Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design

Generation Category (1): Design

Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	17	0	266	283

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	29	2	0	214	0	24	269

Method of Solution:	Newton-Raphson Method
Maximum No. of Iteration:	9999
Precision of Solution:	0.0001000
System Frequency:	50.00 Hz
Unit System:	Metric
Project Filename:	Umm Qasr-ST
Output Filename:	C:\Users\cpra\Desktop\UQ\BGC\Untitled.lfr

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
132kV GRID	132.000	1	100.0	0.0								
200A MDB Agreko	0.415	1	100.0	0.0	0.017	0.008	0.005	0.002				
Area Lighting DB (A8)	0.380	1	100.0	0.0			0.009	0.006				
BUS-MTR1_ENCRYO	11.000	6	100.0	0.0	5.570	3.308						
BUS-MTR2_ENCRYO	11.000	7	100.0	0.0	5.570	3.308						
Bus1	11.000	1	100.0	0.0								
Bus2	0.380	1	100.0	0.0	0.253	0.159						
Bus4	0.380	1	100.0	0.0	0.253	0.159						
Bus7	11.000	1	100.0	0.0								
Bus8	11.000	1	100.0	0.0								
Bus9	0.380	1	100.0	0.0	0.040	0.025						
Bus10	0.380	1	100.0	0.0								
Bus11	0.380	5	100.0	0.0								
Bus12	0.380	1	100.0	0.0	0.040	0.025						
Bus14	0.380	1	100.0	0.0	0.016	0.012						
Bus15	0.380	1	100.0	0.0	0.016	0.012						
Bus17	11.000	1	100.0	0.0								
Bus18	11.000	1	100.0	0.0								
Bus19	11.000	1	100.0	0.0								
Bus20	11.000	1	100.0	0.0								
Bus21	11.000	1	100.0	0.0								
Bus22	3.300	1	100.0	0.0								
Bus23	0.380	1	100.0	0.0								
Bus24	0.380	1	100.0	0.0								
Bus25	3.300	1	100.0	0.0								
Bus27	0.380	1	100.0	0.0								
Bus28	0.380	1	100.0	0.0								
Bus29	3.450	1	100.0	0.0								
Bus30	3.450	1	100.0	0.0								
Bus31	3.450	1	100.0	0.0								
Bus32	3.450	1	100.0	0.0								
Bus33	0.380	1	100.0	0.0								
Bus35	0.380	3	100.0	0.0								

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus36	0.400	1	100.0	0.0	0.512	0.384	0.142	0.106				
Bus37	3.300	1	100.0	0.0								
Bus38	0.400	1	100.0	0.0								
Bus40	0.400	1	100.0	0.0								
Bus41	0.380	4	100.0	0.0								
Bus42	11.000	1	100.0	0.0								
Bus43	11.000	1	100.0	0.0								
Bus45	0.400	1	100.0	0.0								
Bus46	11.000	1	100.0	0.0								
Bus47	3.300	1	100.0	0.0	0.021	0.011						
Bus48	0.400	1	100.0	0.0								
Bus49	3.300	1	100.0	0.0								
Bus50	3.300	1	100.0	0.0								
Bus51	11.000	1	100.0	0.0								
Bus52	0.400	1	100.0	0.0								
Bus53	0.380	1	100.0	0.0	0.040	0.025						
Bus54	0.380	1	100.0	0.0	0.040	0.025						
Bus55	11.000	1	100.0	0.0								
Bus56	11.000	1	100.0	0.0								
Bus57	0.380	1	100.0	0.0	0.253	0.159						
Bus58	0.380	8	100.0	0.0								
Bus59	0.380	1	100.0	0.0	0.253	0.159						
Bus60	0.380	1	100.0	0.0	0.016	0.012						
Bus61	0.380	9	100.0	0.0								
Bus62	0.380	1	100.0	0.0	0.016	0.012						
Bus63	0.400	1	100.0	0.0								
Bus64	0.415	1	100.0	0.0								
Bus65	0.400	1	100.0	0.0								
Bus66	11.000	1	100.0	0.0								
Bus67	0.380	1	100.0	0.0								
Bus68	0.380	1	100.0	0.0	0.253	0.159						
Bus69	0.380	1	100.0	0.0								
Bus70	0.380	1	100.0	0.0	0.253	0.159						
Bus71	0.380	10	100.0	0.0								
Bus72	0.380	1	100.0	0.0	0.016	0.012						

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus73	0.380	1	100.0	0.0	0.016	0.012						
Bus74	0.380	1	100.0	0.0	0.253	0.159						
Bus75	0.380	1	100.0	0.0	0.253	0.159						
Bus76	0.380	1	100.0	0.0	0.016	0.012						
Bus77	0.380	1	100.0	0.0	0.016	0.012						
Bus78	11.000	1	100.0	0.0								
Bus79	132.000	1	100.0	0.0								
Bus80	132.000	1	100.0	0.0								
Bus81	132.000	1	100.0	0.0								
Bus82	0.415	1	100.0	0.0								
Bus83	132.000	1	100.0	0.0								
Bus84	132.000	1	100.0	0.0								
Bus85	0.380	1	100.0	0.0	0.040	0.025						
Bus87	132.000	1	100.0	0.0								
Bus88	132.000	1	100.0	0.0								
Bus89	11.000	1	100.0	0.0	0.053	0.032						
Bus90	11.000	1	100.0	0.0	0.531	0.323						
Bus91	0.380	1	100.0	0.0			0.038	0.018				
Bus92	11.000	1	100.0	0.0	6.532	3.057						
Bus93	0.380	11	109.2	0.0	0.253	0.159						
Bus94	0.380	12	109.2	0.0	0.253	0.159						
Bus95	0.400	1	100.0	0.0	0.016	0.012						
Bus96	0.400	1	100.0	0.0	0.016	0.012						
Bus97	0.380	13	109.2	0.0	0.253	0.159						
Bus98	0.380	14	109.2	0.0	0.253	0.159						
Bus99	0.400	1	100.0	0.0	0.016	0.012						
Bus100	0.400	1	100.0	0.0	0.016	0.012						
Bus101	0.380	15	109.2	0.0	0.253	0.159						
Bus102	0.380	16	109.2	0.0	0.253	0.159						
Bus103	0.400	1	100.0	0.0	0.016	0.012						
Bus104	0.400	1	100.0	0.0	0.016	0.012						
Bus105	0.380	17	109.2	0.0	0.253	0.159						
Bus106	0.380	18	109.2	0.0	0.253	0.159						
Bus107	0.400	1	100.0	0.0	0.016	0.012						
Bus108	0.400	1	100.0	0.0	0.016	0.012						

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus110	11.000	1	100.0	0.0	0.973	0.471						
Bus112	11.000	1	100.0	0.0								
Bus113	11.000	1	100.0	0.0	0.098	0.052						
Bus114	11.000	1	100.0	0.0	6.527	3.054						
Bus115	11.000	1	100.0	0.0	0.978	0.516						
Bus117	11.000	1	100.0	0.0	0.531	0.323						
Bus119	3.300	1	100.0	0.0	0.038	0.022						
Bus120	3.300	1	100.0	0.0	0.871	0.434						
Bus121	3.300	1	100.0	0.0	0.087	0.043						
Bus122	3.300	1	100.0	0.0	0.087	0.043						
Bus123	3.300	1	100.0	0.0	0.020	0.011						
Bus124	3.300	1	100.0	0.0	0.020	0.011						
Bus125	3.300	1	100.0	0.0	0.019	0.011						
Bus126	3.300	1	100.0	0.0	0.019	0.011						
Bus127	3.300	1	100.0	0.0	0.019	0.011						
Bus128	3.300	1	100.0	0.0	0.187	0.108						
Bus129	3.300	1	100.0	0.0	0.018	0.010						
Bus130	3.300	1	100.0	0.0	0.018	0.010						
Bus131	3.300	1	100.0	0.0	0.018	0.010						
Bus132	3.300	1	100.0	0.0	0.181	0.105						
Bus133	3.300	1	100.0	0.0	0.181	0.105						
Bus134	3.300	1	100.0	0.0	0.183	0.105						
Bus135	3.300	1	100.0	0.0	0.018	0.010						
Bus136	3.300	1	100.0	0.0	0.018	0.010						
Bus137	3.300	1	100.0	0.0	0.190	0.109						
Bus138	3.300	1	100.0	0.0	0.190	0.109						
Bus139	3.300	1	100.0	0.0	0.019	0.011						
Bus140	3.300	1	100.0	0.0	0.019	0.011						
Bus141	3.300	1	100.0	0.0	0.198	0.115						
Bus142	3.300	1	100.0	0.0	0.020	0.011						
Bus143	3.300	1	100.0	0.0	0.087	0.043						
Bus144	3.300	1	100.0	0.0	0.871	0.434						
Bus145	3.300	1	100.0	0.0	0.087	0.043						
Bus146	3.300	1	100.0	0.0	0.208	0.106						
Bus149	0.380	1	100.0	0.0	0.040	0.025						

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus151	0.380	1	100.0	0.0	0.040	0.025						
Bus152	0.380	1	100.0	0.0	0.001	0.001						
Bus154	0.380	1	100.0	0.0	0.040	0.025						
Bus157	0.380	1	100.0	0.0			0.003	0.002				
Bus158	0.380	1	100.0	0.0	0.003	0.002						
Bus159	0.380	1	100.0	0.0	0.016	0.007						
Bus160	3.450	1	100.0	0.0	0.901	0.410						
Bus161	0.380	1	100.0	0.0	0.040	0.025						
Bus162	0.380	1	100.0	0.0	0.040	0.025						
Bus163	11.000	1	100.0	0.0								
Bus164	0.380	1	100.0	0.0	0.040	0.025						
Bus165	0.380	1	100.0	0.0	0.040	0.025						
Bus166	0.380	1	100.0	0.0								
Bus167	0.380	1	100.0	0.0								
Bus168	11.000	1	100.0	0.0								
Bus169	0.380	1	100.0	0.0	0.016	0.007						
Bus170	0.380	1	100.0	0.0	0.040	0.025						
Bus171	0.380	1	100.0	0.0	0.040	0.025						
Bus172	0.380	1	100.0	0.0	0.040	0.025						
Bus173	0.380	1	100.0	0.0	0.040	0.025						
Bus174	0.380	1	100.0	0.0	0.040	0.025						
Bus175	0.380	1	100.0	0.0	0.040	0.025						
Bus176	0.380	1	100.0	0.0	0.040	0.025						
Bus177	0.380	1	100.0	0.0	0.040	0.025						
Bus178	0.380	1	100.0	0.0	0.040	0.025						
Bus179	0.380	1	100.0	0.0	0.040	0.025						
Bus180	0.380	1	100.0	0.0	0.040	0.025						
Bus181	0.380	1	100.0	0.0	0.040	0.025						
Bus182	0.380	1	100.0	0.0	0.040	0.025						
Bus183	0.380	1	100.0	0.0	0.040	0.025						
Bus184	0.380	1	100.0	0.0	0.040	0.025						
Bus185	0.380	1	100.0	0.0	0.040	0.025						
Bus186	0.380	1	100.0	0.0	0.040	0.025						
Bus187	0.380	1	100.0	0.0	0.040	0.025						
Bus188	0.380	1	100.0	0.0			0.024	0.015				

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus189	0.380	1	100.0	0.0			0.024	0.015				
Bus190	0.380	1	100.0	0.0			0.024	0.015				
Bus_LVDB-1	0.415	1	100.0	0.0								
Bus_LVDB-2	0.415	1	100.0	0.0								
Bus_MDB-1	0.415	1	100.0	0.0								
Bus_MDB-2	0.415	1	100.0	0.0								
Bus_MDB-3	0.415	1	100.0	0.0								
Bus_MDB-4	0.415	1	100.0	0.0								
Bus_MDB-5	0.415	1	100.0	0.0								
Bus_MDB-6	0.400	1	100.0	0.0								
Bus_MDB-7	0.400	1	100.0	0.0								
Bus_MDB-8	0.400	1	100.0	0.0								
BUS_S002-CMP3	11.000	1	100.0	0.0	3.299	1.691						
CAMP BUS	11.000	1	100.0	0.0	1.800	0.950	0.450	0.238				
LV SWBD_TOFS	0.400	1	100.0	0.0	0.138	0.047	0.550	0.188				
MV RMU-TOFS	11.000	1	100.0	0.0								
POWER SOCKET FEEDER 1 B	0.380	1	100.0	0.0								
S002-ASP1-1 [BUS A]	0.380	1	100.0	0.0			0.092	0.057				
S002-ASP1-1B [BUS B]	0.380	1	100.0	0.0			0.303	0.188				
S002-ASP1-2A	0.380	1	100.0	0.0			0.336	0.243				
S002-ASP1-2B	0.380	1	100.0	0.0			0.015	0.011				
S002-DBASP1-2-22	0.380	1	100.0	0.0	0.014	0.010						
S002-MCC1-1 [BUS]	0.380	1	100.0	0.0	0.229	0.135						
S002-MCC1-2 [BUS]	0.380	1	100.0	0.0								
S002-MCC1-3 [BUS]	0.380	1	100.0	0.0	0.236	0.122						
S002-MCC1-4 [BUS]	0.380	1	100.0	0.0	0.053	0.038						
S002-MCC1-5 [BUS]	0.380	1	100.0	0.0	0.233	0.156						
S002-MCC1-6 [BUS]	0.380	1	100.0	0.0	0.056	0.041						
S002-MCC1-7 [BUS]	0.380	1	100.0	0.0	0.233	0.156						
S002-MCC1-8 [BUS]	0.380	1	100.0	0.0	0.110	0.079						
S002-MCC1-9 [BUS]	0.380	1	100.0	0.0	0.121	0.084						
S002-PC1-1A	0.380	1	100.0	0.0	0.081	0.061						
S002-PC1-1B	0.380	1	100.0	0.0								
S002-PC1-2A	0.380	1	100.0	0.0								
S002-PC1-2B	0.380	1	100.0	0.0	0.055	0.034						

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
S002-SB-1AC001 [BUS A]	11.000	1	100.0	0.0								
S002-SB-1AC001 [BUS B]	11.000	1	100.0	0.0								
S002-SB-1AD001 [BUS A]	3.300	1	100.0	0.0								
S002-SB-1AD001 [BUS B]	3.300	1	100.0	0.0								
S002-SB-1AN001 [BUS A]	0.380	1	100.0	0.0	0.011	0.017	0.043	0.067				
S002-SB-1AN001 [BUS B]	0.380	1	100.0	0.0	0.013	0.008						
S002-SB-01E001 [BUS A]	0.380	1	100.0	0.0	0.179	0.105	0.450	0.255				
S002-SB-01E001 [BUS B]	0.380	1	100.0	0.0	0.106	0.060	0.525	0.289				
S002-SB-02N001 [BUS A]	0.380	1	100.0	0.0	0.244	0.164	0.061	0.041				
S002-SB-02N001 [BUS B]	0.380	1	100.0	0.0	0.148	0.100	0.022	0.015				
S002-SB-03E001 [BUS A]	0.380	1	100.0	0.0	0.397	0.271	0.086	0.058				
S002-SB-03E001 [BUS B]	0.380	1	100.0	0.0	0.373	0.279	0.176	0.132				
S002-SB-03N001 [BUS A]	0.380	1	100.0	0.0	0.621	0.454	0.266	0.194				
S002-SB-03N001 [BUS B]	0.380	1	100.0	0.0	0.356	0.263	0.153	0.113				
S002-SB-04E001 [BUS A]	0.380	1	100.0	0.0	0.142	0.074	0.016	0.006				
S002-SB-04E001 [BUS B]	0.380	1	100.0	0.0	0.031	0.010	0.015	0.011				
S002-SB-05N001A	0.380	1	100.0	0.0	0.078	0.048	0.174	0.064				
S002-SB-05N001B	0.380	1	100.0	0.0	0.395	0.226	0.025	0.010				
S002-SB-PSS1A01 [BUS-A]	132.000	1	100.0	0.0								
S002-SB-PSS01N01 [BUS A]	0.380	1	100.0	0.0	0.094	0.026						
S002-SB-PSS01N01 [BUS B]	0.380	1	100.0	0.0	0.095	0.059						
S002-TR-04DE001_PRI	3.300	1	100.0	0.0								
S002-TR-04DE002_PRI	3.300	1	100.0	0.0								
S003-DB11E001-01	0.380	1	100.0	0.0	0.005	0.004	0.022	0.016				
S003-DB11E001-02	0.380	1	100.0	0.0	0.005	0.004	0.022	0.016				
S003-DB-11N001-03 [BUS A]	0.380	1	100.0	0.0	0.020	0.015						
S003-DB-11N001-03 [BUS B]	0.380	1	100.0	0.0	0.020	0.015	0.005	0.004				
S003-DB11N001-04 [BUS A]	0.380	1	100.0	0.0	0.017	0.013	0.004	0.003				
S003-DB11N001-04 [BUS B]	0.380	1	100.0	0.0	0.019	0.014	0.005	0.003				
S003-DB11N001-11	0.380	1	100.0	0.0			0.089	0.043				
S003-DB11N001-12	0.380	1	100.0	0.0			0.036	0.017				
S003-MDP-B	0.380	1	100.0	0.0			0.308	0.231				
S003-SB01E1	0.380	1	100.0	0.0	0.109	0.082	0.027	0.021				
S003-SB-11C001 [BUS A]	11.000	1	100.0	0.0								
S003-SB-11C001 [BUS B]	11.000	1	100.0	0.0								

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
S003-SB-11D001 [BUS A]	3.300	1	100.0	0.0	0.303	0.161						
S003-SB-11D001 [BUS B]	3.300	1	100.0	0.0	0.276	0.135						
S003-SB-11E001 [BUS A]	0.380	1	100.0	0.0	0.358	0.222	0.089	0.055				
S003-SB-11E001 [BUS B]	0.380	1	100.0	0.0	0.415	0.274	0.065	0.046				
S003-SB-11N001 [BUS A]	0.380	1	100.0	0.0	0.237	0.156	0.055	0.037				
S003-SB-11N001 [BUS B]	0.380	1	100.0	0.0	0.371	0.246	0.069	0.048				
TR-PSS1AC01_11kV BUS	11.000	1	100.0	0.0								
TR-PSS1AC01_132kV BUS	132.000	1	100.0	0.0								
TR-PSS1AC02_11kV BUS	11.000	1	100.0	0.0								
TR-PSS1AC02_132kV BUS	132.000	1	100.0	0.0								
TYPICAL MOTOR FEEDER 1 A	0.380	1	100.0	0.0								
TYPICAL MOTOR FEEDER 1 B	0.380	1	100.0	0.0								
Cable40~	0.434	1	100.0	0.0								
Cable41~	0.434	1	100.0	0.0								
Cable153~	0.434	1	100.0	0.0								
Cable154~	0.434	1	100.0	0.0								
Cable163~	0.434	1	100.0	0.0								
Cable164~	0.434	1	100.0	0.0								
Cable168~	0.434	1	100.0	0.0								
Cable169~	0.434	1	100.0	0.0								
Cable188~	0.434	1	100.0	0.0								
Cable189~	0.434	1	100.0	0.0								
Cable192~	0.434	1	100.0	0.0								
Cable193~	0.434	1	100.0	0.0								
Cable196~	0.434	1	100.0	0.0								
Cable197~	0.434	1	100.0	0.0								
Cable200~	0.434	1	100.0	0.0								
Cable201~	0.434	1	100.0	0.0								
S002-830-1601-N-P-1~	0.418	1	100.0	0.0								
S003-830-2000-E-P~	0.418	1	100.0	0.0								
S003-Cable43~	0.380	8	100.0	0.0								
S003-Cable45~	0.380	9	100.0	0.0								
S003-Cable120~	0.380	10	100.0	0.0								
S002-Cable23~	0.380	4	100.0	0.0								

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
S002-830-1072-E-P~	0.380	3	100.0	0.0								
Cable84~	0.380	5	100.0	0.0								
S002-TR-CMP1_VFD~2	11.500	1	100.0	0.0								
S002-TR-CMP1_VFD~3	11.500	1	100.0	0.0								
S002-TR-CMP2_VFD~2	11.500	1	100.0	0.0								
S002-TR-CMP2_VFD~3	11.500	1	100.0	0.0								
Total Number of Buses: 283					50.343	27.926	4.822	2.899	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
132kV GRID	132.000	Swing	1	100.0	0.0					
BUS-MTR1_ENCRYO	11.000	Swing	6	100.0	0.0					
BUS-MTR2_ENCRYO	11.000	Swing	7	100.0	0.0					
Bus11	0.380	Swing	5	100.0	0.0					
Bus35	0.380	Swing	3	100.0	0.0					
Bus41	0.380	Swing	4	100.0	0.0					
Bus58	0.380	Swing	8	100.0	0.0					
Bus61	0.380	Swing	9	100.0	0.0					
Bus71	0.380	Swing	10	100.0	0.0					
Bus93	0.380	Swing	11	109.2	0.0					
Bus94	0.380	Swing	12	109.2	0.0					
Bus97	0.380	Swing	13	109.2	0.0					
Bus98	0.380	Swing	14	109.2	0.0					
Bus101	0.380	Swing	15	109.2	0.0					
Bus102	0.380	Swing	16	109.2	0.0					
Bus105	0.380	Swing	17	109.2	0.0					
Bus106	0.380	Swing	18	109.2	0.0					
						0.000	0.000			

Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)

Line/Cable/Busway			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable2	1.0MCUN3	70	384.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable3	1.0MCUN3	70	378.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable4	1.0MCUN3	70	372.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable5	1.0MCUN3	70	372.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable6	1.0MCUN3	70	366.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable7	1.0MCUN3	70	360.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable10	1.0MCUN3	70	366.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable11	1.0MCUN3	70	360.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable12	1.0MCUN3	70	354.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable13	1.0MCUN3	70	360.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable19	1.0MCUN3	70	354.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable20	1.0MCUN3	35	360.0	0.0	1	75	0.638076	0.080000	0.0001015
Cable21	1.0MCUN3	70	348.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable22	1.0MCUN3	70	354.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable23	1.0MCUN3	16	100.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable24	1.0MCUN3	16	100.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable25	1.0MCUN3	70	348.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable26	1.0MCUN3	35	360.0	0.0	1	75	0.638076	0.080000	0.0001015
Cable27	1.0MCUN3	70	342.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable28	1.0MCUN3	70	324.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable29	1.0MCUN3	70	330.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable30	1.0MCUN3	70	336.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable31	1.0MCUN3	70	330.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable32	1.0MCUN3	70	336.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable33	1.0MCUN3	70	342.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable34	1.0MCUN3	70	336.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable35	1.0MCUN3	70	342.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable36	1.0MCUN3	70	348.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable37	1.0MCUN3	70	342.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable38	1.0MCUN3	70	348.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable39	1.0MCUN3	70	354.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable40	1.0NCUN1	240	45.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable41	1.0NCUN1	240	50.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable42	1.0MCUN3	70	348.0	0.0	1	75	0.326191	0.075000	0.0001307

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable43	1.0MCUN3	70	354.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable44	1.0MCUN3	70	360.0	0.0	1	75	0.326191	0.075000	0.0001307
Cable45	1.0MCUN3	150	276.0	0.0	1	75	0.151650	0.073000	0.0001442
Cable48	1.0MCUN3	150	130.0	0.0	1	75	0.151650	0.073000	0.0001442
Cable49	1.0MCUN3	150	160.0	0.0	1	75	0.151650	0.073000	0.0001442
Cable83	1.0NCUN4	16	50.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable118	11NCUS3	95	1000.0	0.0	1	75	0.236536	0.102000	0.0001049
Cable123	1.0NCUN2	16	55.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable128	1.0NCUN1	240	25.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable134	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable136	1.0NCUN1	240	40.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable137	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable144	1.0NCUN1	240	30.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable145	1.0NCUN1	240	30.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable146	1.0NCUN1	240	30.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable147	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable148	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable149	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable150	1.0NCUN1	240	40.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable151	1.0NCUN1	240	25.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable152	1.0NCUN1	240	8.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable153	1.0NCUN1	240	30.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable154	1.0NCUN1	240	35.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable155	1.0NCUN3	16	35.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable156	1.0NCUN3	16	40.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable157	1.0NCUN1	240	40.0	0.0	7	75	0.092516	0.092000	0.0001703
Cable158	1.0NCUN1	240	26.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable159	1.0NCUN1	240	24.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable160	1.0NCUN1	240	23.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable161	1.0NCUN1	240	15.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable162	1.0NCUN1	240	25.0	0.0	4	75	0.092516	0.092000	0.0001703
Cable163	1.0NCUN1	240	12.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable164	1.0NCUN1	240	18.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable165	1.0NCUN3	16	27.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable166	1.0NCUN3	16	32.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable167	1.0NCUN1	240	25.0	0.0	4	75	0.092516	0.092000	0.0001703

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable168	1.0NCUN1	240	25.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable169	1.0NCUN1	240	36.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable170	1.0NCUN3	16	35.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable171	1.0NCUN3	16	46.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable176	1.0NCUN3	95	25.0	0.0	4	75	0.235582	0.073000	0.0001433
Cable188	1.0NCUN1	240	10.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable189	1.0NCUN1	240	15.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable190	1.0NCUN3	16	25.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable191	1.0NCUN3	16	30.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable192	1.0NCUN1	240	25.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable193	1.0NCUN1	240	35.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable194	1.0NCUN3	16	35.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable195	1.0NCUN3	16	45.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable196	1.0NCUN1	240	43.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable197	1.0NCUN1	240	26.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable198	1.0NCUN3	16	50.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable199	1.0NCUN3	16	35.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable200	1.0NCUN1	240	17.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable201	1.0NCUN1	240	23.0	0.0	2	75	0.092516	0.092000	0.0001703
Cable202	1.0NCUN3	16	25.0	0.0	1	75	1.392512	0.080000	0.0000971
Cable203	1.0NCUN3	16	32.0	0.0	1	75	1.392512	0.080000	0.0000971
C_688-KM-01	3.3NCUN3	95	500.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-1063-N-P	1.0MCUN1	630	30.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-830-1066-N-P	1.0MCUN1	630	30.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-830-1069-N-P	1.0MCUN1	630	35.0	0.0	3	75	0.040059	0.086000	0.0002023
S002-830-1072-N-P	1.0MCUN1	630	35.0	0.0	3	75	0.040059	0.086000	0.0002023
S002-830-1420-N-P-1	1.0MCUN1	630	25.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-830-1500-N-P-1	1.0MCUN1	630	30.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-830-9001-A-P	11MCUS1	400	380.0	0.0	1	75	0.061042	0.101000	0.0001762
S002-830-9001-C-P	11NCUS1	1000	81.0	0.0	8	75	0.032428	0.093000	0.0002560
S002-830-9001-D-P-1	3.3NCUN1	630	100.0	0.0	4	75	0.040059	0.086000	0.0002023
S002-830-9004-A-P	11MCUS1	400	380.0	0.0	1	75	0.061042	0.101000	0.0001762
S002-830-9004-D-P	3.3NCUN3	240	90.0	0.0	1	75	0.094424	0.073000	0.0001464
S002-830-9005-D-P	3.3NCUN3	240	90.0	0.0	1	75	0.094424	0.073000	0.0001464
S002-830-9006-D-P	3.3NCUN3	25	580.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9007-D-P	3.3NCUN3	25	580.0	0.0	1	75	0.884149	0.096000	0.0000591

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
S002-830-9008-C-P-1	11NCUS1	1000	105.0	0.0	8	75	0.032428	0.093000	0.0002560
S002-830-9008-D-P	3.3NCUN3	25	400.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9009-D-P	3.3NCUN3	25	400.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9010-D-P	3.3NCUN3	25	690.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9011-D-P	3.3NCUN3	25	690.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9012-D-P	3.3NCUN3	25	870.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9013-D-P	3.3NCUN3	25	870.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9014-D-P	3.3NCUN3	25	550.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9015-D-P	3.3NCUN3	25	560.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9016-D-P	3.3NCUN3	95	440.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9017-D-P	3.3NCUN3	95	450.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9018-D-P	3.3NCUN3	95	470.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9019-D-P	3.3NCUN3	95	160.0	0.0	1	75	0.235582	0.080000	0.0001037
S002-830-9020-D-P	3.3NCUN3	35	270.0	0.0	1	75	0.638076	0.092000	0.0000669
S002-830-9021-D-P-1	3.3NCUN1	630	100.0	0.0	4	75	0.040059	0.086000	0.0002023
S002-830-9024-D-P	3.3NCUN3	240	90.0	0.0	1	75	0.094424	0.073000	0.0001464
S002-830-9025-D-P	3.3NCUN3	240	90.0	0.0	1	75	0.094424	0.073000	0.0001464
S002-830-9026-D-P	3.3NCUN3	25	580.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9027-D-P	3.3NCUN3	25	580.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9028-D-P	3.3NCUN3	25	400.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9029-D-P	3.3NCUN3	25	400.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9030-D-P	3.3NCUN3	25	690.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9031-D-P	3.3NCUN3	25	690.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9032-D-P	3.3NCUN3	25	870.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9033-D-P	3.3NCUN3	25	870.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9034-D-P	3.3NCUN3	25	550.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9035-D-P	3.3NCUN3	25	560.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9036-D-P	3.3NCUN3	95	440.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9037-D-P	3.3NCUN3	95	450.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9038-D-P	3.3NCUN3	95	470.0	0.0	2	75	0.235582	0.080000	0.0001037
S002-830-9039-D-P	3.3NCUN3	25	270.0	0.0	1	75	0.884149	0.096000	0.0000591
S002-830-9051-C-P	11NCUS3	185	120.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9053-C-P	11NCUS3	185	850.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9054-C-P	11NCUS3	185	960.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9055-C-P	11NCUS3	185	960.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9056-C-P-1	11NCUS1	630	60.0	0.0	2	75	0.040059	0.096500	0.0002193

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
S002-830-9059-C-P	11NCUS3	185	475.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9061-C-P-1	11NCUS3	400	330.0	0.0	2	75	0.061042	0.084900	0.0001822
S002-830-9062-C-P	11NCUS3	185	600.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9063-C-P	11NCUS3	185	590.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9064-C-P	11NCUS3	185	90.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9066-C-P	11NCUS3	185	850.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9067-C-P	11NCUS3	185	960.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9068-C-P	11NCUS3	185	960.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9069-C-P-1	11NCUS1	630	60.0	0.0	2	75	0.040059	0.096500	0.0002193
S002-830-9072-C-P	11NCUS3	185	485.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9073-C-P-1	11NCUS3	400	320.0	0.0	2	75	0.061042	0.084900	0.0001822
S002-830-9074-C-P	11NCUS3	185	480.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9076-C-P	11NCUS3	185	590.0	0.0	1	75	0.122083	0.093100	0.0001351
S002-830-9077-C-P-1	11NCUS3	150	7000.0	0.0	2	75	0.151650	0.096200	0.0001232
S002-830-9078-C-P-1	11MCUS3	150	7000.0	0.0	2	75	0.151650	0.096200	0.0001232
S002-830-9079-C-P	11NCUS3	95	550.0	0.0	1	75	0.236536	0.102000	0.0001049
S002-Cable20	1.0NCUN1	630	505.0	0.0	3	75	0.040059	0.086000	0.0002023
S002-Cable21	1.0NCUN4	185	200.0	0.0	1	75	0.122083	0.073000	0.0001442
S002-Cable22	1.0NCUN1	630	505.0	0.0	3	75	0.040059	0.086000	0.0002023
S002-Cable26	1.0NCUN4	95	170.0	0.0	1	75	0.235582	0.073000	0.0001433
S002-Cable29	1.0NCUN1	800	25.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable34	1.0MCUN4	185	90.0	0.0	1	75	0.122083	0.073000	0.0001442
S002-Cable35	1.0MCUN1	630	15.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-Cable85	1.0NCUN1	800	30.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable87	1.0NCUN1	800	25.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable88	1.0NCUN1	800	35.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable89	1.0MCUN1	630	15.0	0.0	5	75	0.040059	0.086000	0.0002023
S002-Cable90	1.0MCUN4	185	90.0	0.0	1	75	0.122083	0.073000	0.0001442
S002-Cable95	1.0NCUN1	800	36.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable115	1.0NCUN1	630	28.0	0.0	1	75	0.040059	0.086000	0.0002023
S002-Cable119	1.0NCUN1	630	30.0	0.0	1	75	0.040059	0.086000	0.0002023
S002-Cable124	1.0NCUN1	800	29.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable125	1.0NCUN1	800	29.0	0.0	1	75	0.033287	0.087000	
S002-Cable126	1.0NCUN1	800	24.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable127	1.0NCUN1	800	24.0	0.0	1	75	0.031475	0.091000	0.0002061
S002-Cable131	1.0NCUN3	95	67.0	0.0	1	75	0.235582	0.073000	0.0001433

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
S002-Cable132	1.0NCUN3	95	95.0	0.0	1	75	0.235582	0.073000	0.0001433
S002-Cable138	1.0NCUN4	16	100.0	0.0	1	75	1.392512	0.080000	0.0000971
S002-Cable181	1.0NCUN4	35	100.0	0.0	1	75	0.638076	0.080000	0.0001015
S002-C_TR-04DE001	3.3NCUN3	185	500.0	0.0	1	75	0.121129	0.075000	0.0001329
S002-C_TR-04DE002	3.3NCUN3	185	500.0	0.0	1	75	0.121129	0.075000	0.0001329
S003-830-1000-E-P	1.0MCUN1	630	50.0	0.0	4	75	0.040059	0.086000	0.0002023
S003-830-1004-E-P	1.0MCUN1	630	50.0	0.0	4	75	0.040059	0.086000	0.0002023
S003-830-9001-C-P	1.0MCUN3	185	75.0	0.0	1	75	0.122083	0.073000	0.0001442
S003-830-9001-D-P	3.3NCUN1	630	53.0	0.0	1	75	0.040059	0.086000	0.0002023
S003-830-9002-C-P	1.0MCUN3	185	50.0	0.0	1	75	0.122083	0.073000	0.0001442
S003-830-9003-C-P	3.3NCUN3	185	50.0	0.0	1	75	0.121129	0.075000	0.0001329
S003-830-9004-C-P	3.3NCUN3	185	45.0	0.0	1	75	0.121129	0.075000	0.0001329
S003-830-9005-C-P	11MCUS3	185	50.0	0.0	1	75	0.122083	0.093100	0.0001351
S003-830-9006-C-P	11MCUS3	185	67.0	0.0	1	75	0.122083	0.093100	0.0001351
S003-830-9019-D-P	3.3NCUN1	630	53.0	0.0	1	75	0.040059	0.086000	0.0002023
S003-830-9021-C-P	11NCUS3	150	605.0	0.0	1	75	0.151650	0.096200	0.0001232
S003-830-9022-C-P	11NCUS3	150	605.0	0.0	1	75	0.151650	0.096200	0.0001232
S003-Cable19	1.0NCUN4	70	50.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable42	1.0NCUN1	630	780.0	0.0	4	75	0.040059	0.086000	0.0002023
S003-Cable44	1.0NCUN4	70	50.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable93	1.0NCUN4	70	55.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable94	1.0NCUN4	70	50.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable96	1.0NCUN4	70	50.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable105	1.0NCUN1	630	300.0	0.0	5	75	0.040059	0.086000	0.0002023
S003-Cable113	11NCUS3	150	850.0	0.0	3	75	0.151650	0.096200	0.0001232
S003-Cable121	1.0NCUN4	70	50.0	0.0	1	75	0.326191	0.075000	0.0001307
S003-Cable142	1.0MCUN4	70	160.0	0.0	2	75	0.326191	0.075000	0.0001307
S003-Cable173	1.0MCUN4	70	160.0	0.0	2	75	0.326191	0.075000	0.0001307
S002-830-1601-N-P-1	1.0MCUN1	630	15.0	0.0	5	75	0.040059	0.086000	0.0002023
S003-830-2000-E-P	1.0NCUN3	120	600.0	0.0	1	75	0.186940	0.073000	0.0001499
S003-Cable43	1.0NCUN4	240	200.0	0.0	1	75	0.092516	0.072000	0.0001565
S003-Cable45	1.0NCUN4	150	60.0	0.0	1	75	0.151650	0.073000	0.0001442
S003-Cable120	1.0MCUN1	630	15.0	0.0	8	75	0.040059	0.086000	0.0002023
S002-Cable23	1.0MCUN1	630	15.0	0.0	10	75	0.040059	0.086000	0.0002023
S002-830-1072-E-P	1.0MCUN1	630	42.0	0.0	3	75	0.040059	0.086000	0.0002023
Cable84	1.0NCUN4	150	50.0	0.0	1	75	0.151650	0.073000	0.0001442

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
FD Tranfo	3-Phase	0.750	3.300	0.400	5.00	3.50	0	0	0	0	0	5.0000	Dyn	0.000
S002-TML1 [EXISTING]	3-Phase	1.000	11.000	0.416	4.42	3.50	0	0	0	5.000	0	4.4200	Dyn	0.000
S002-TML1A-1 [EXISTING]	3-Phase	1.600	3.300	0.400	6.21	6.00	0	0	0	5.000	0	6.2100	Dyn	0.000
S002-TML1A-2 [EXISTING]	3-Phase	1.600	3.300	0.400	6.19	6.00	0	0	0	5.000	0	6.1900	Dyn	0.000
S002-TML1B-1 [EXISTING]	3-Phase	1.600	3.300	0.400	6.18	6.00	0	0	0	5.000	0	6.1800	Dyn	0.000
S002-TML1B-2 [EXISTING]	3-Phase	1.600	3.300	0.400	6.16	6.00	0	0	0	5.000	0	6.1600	Dyn	0.000
S002-TML2 [EXISTING]	3-Phase	1.000	11.000	0.416	4.27	3.50	0	0	0	5.000	0	4.2700	Dyn	0.000
S002-TMM-1A [EXISTING]	3-Phase	12.000	11.000	3.450	8.88	20.00	0	0	0	0	0	8.8800	Dyn	0.000
S002-TMM-1B [EXISTING]	3-Phase	12.000	11.000	3.450	8.68	20.00	0	0	0	0	0	8.6800	Dyn	0.000
S002-TR-01CE001	3-Phase	2.000	11.000	0.400	6.00	6.00	0	0	7.5	2.500	0	6.4500	Dyn	0.000
S002-TR-01CE002	3-Phase	2.000	11.000	0.400	6.00	6.00	0	0	7.5	2.500	0	6.4500	Dyn	0.000
S002-TR-03CE001	3-Phase	1.250	11.000	0.400	4.89	5.24	0	0	0	0	0	4.8900	Dyn	0.000
S002-TR-03CE002	3-Phase	1.250	11.000	0.400	4.89	5.25	0	0	0	0	0	4.8900	Dyn	0.000
S002-TR-03CN001	3-Phase	2.000	11.000	0.400	6.08	6.35	0	0	0	2.500	0	6.0800	Dyn	0.000
S002-TR-03CN002	3-Phase	2.000	11.000	0.400	6.03	6.35	0	0	0	2.500	0	6.0300	Dyn	0.000
S002-TR-04DE001	3-Phase	0.400	3.300	0.400	4.05	1.50	0	0	10.0	2.500	0	4.4550	Dyn	0.000
S002-TR-PSS1AC01	3-Phase	50.000	132.000	11.500	10.30	32.50	0	0	0	0	0	10.3000	YNd	0.000
S002-TR-PSS1AC02	3-Phase	50.000	132.000	11.500	10.30	32.50	0	0	0	0	0	10.3000	YNd	0.000
S003-TMM1A [EXISTING]	3-Phase	3.500	11.000	3.600	6.88	8.50	0	0	0	2.500	0	6.8800	Dyn	0.000
S003-TMM1B [EXISTING]	3-Phase	3.500	11.000	3.600	6.94	8.50	0	0	0	2.500	0	6.9400	Dyn	0.000
S003-TR-11CE001	3-Phase	1.600	11.000	0.400	6.00	3.50	0	0	7.5	0	0	6.4500	Dyn	0.000
S003-TR-11CE002	3-Phase	1.600	11.000	0.400	6.00	6.00	0	0	7.5	0	0	6.4500	Dyn	0.000
S003-TR-11CN001	3-Phase	2.500	11.000	0.400	6.00	6.00	0	0	7.5	0	0	6.4500	Dyn	0.000
S003-TR-11CN002	3-Phase	2.500	11.000	0.400	6.00	6.00	0	0	7.5	0	0	6.4500	Dyn	0.000
T2	3-Phase	1.000	11.000	0.416	5.00	3.50	0	0	0	0	0	5.0000	Dyn	0.000
TR-05CE001A	3-Phase	2.500	11.000	0.400	8.50	6.00	0	0	10.0	0	0	9.3500	Dyn	0.000
TR-05CE001B	3-Phase	2.500	11.000	0.400	8.50	6.00	0	0	10.0	0	0	9.3500	Dyn	0.000
TX-A	3-Phase	3.150	11.000	0.415	6.50	6.00	0	0	10.0	-5.000	0	7.1500	Dyn	0.000
TX-B	3-Phase	3.150	11.000	0.415	6.50	6.00	0	0	10.0	-5.000	0	7.1500	Dyn	0.000

2-Winding Transformer Load Tap Changer (LTC) Settings

Transformer	Connected Buses ("*" LTC Side)		Transformer Load Tap Changer Setting					
ID	Primary Bus ID	Secondary Bus ID	% Min. Tap	% Max. Tap	% Step	Regulated Bus ID	% V	kV
S002-TR-PSS1AC01	* TR-PSS1AC01_132kV BUS	TR-PSS1AC01_11kV BUS	-10.00	10.00	1.250	TR-PSS1AC01_11kV BUS	100.80	11.088
S002-TR-PSS1AC02	* TR-PSS1AC02_132kV BUS	TR-PSS1AC02_11kV BUS	-10.00	10.00	1.250	TR-PSS1AC02_11kV BUS	100.80	11.088

3-Winding Transformer Input Data

Transformer	Rating			Tap	Impedance					Z Variation		Phase Shift	
ID	Winding	MVA	kV	%		% Z1	X1/R1	MVAb	% Tol.	+ 5%	- 5%	Type	Angle
S002-TR-CMP1_VFD	Primary:	7.500	11.000	0	Zps =	10.00	45.00	7.500	10.0	0	0		
	Secondary:	3.250	11.000	0	Zpt =	10.00	45.00	7.500	10.0			Std Pos. Seq.	0.000
	Tertiary:	3.250	11.000	0	Zst =	10.00	45.00	7.500	10.0			Std Pos. Seq.	0.000
S002-TR-CMP2_VFD	Primary:	7.500	11.000	0	Zps =	10.00	45.00	7.500	10.0	0	0		
	Secondary:	3.250	11.000	0	Zpt =	10.00	45.00	7.500	10.0			Std Pos. Seq.	0.000
	Tertiary:	3.250	11.000	0	Zst =	10.00	45.00	7.500	10.0			Std Pos. Seq.	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
FD Tranfo	2W XFMR	Bus37	Bus36	140.80	492.81	512.53	
S002-TML1 [EXISTING]	2W XFMR	Bus7	Bus27	116.65	408.28	424.62	
S002-TML1A-1 [EXISTING]	2W XFMR	Bus30	S002-PC1-1A	56.08	336.50	341.15	
S002-TML1A-2 [EXISTING]	2W XFMR	Bus31	S002-PC1-2A	55.90	335.42	340.05	
S002-TML1B-1 [EXISTING]	2W XFMR	Bus29	S002-PC1-1B	55.81	334.88	339.50	
S002-TML1B-2 [EXISTING]	2W XFMR	Bus32	S002-PC1-2B	55.63	333.79	338.40	
S002-TML2 [EXISTING]	2W XFMR	Bus46	Bus24	112.69	394.43	410.21	
S002-TMM-1A [EXISTING]	2W XFMR	Bus1	Bus22	3.38	67.62	67.71	
S002-TMM-1B [EXISTING]	2W XFMR	Bus21	Bus25	3.30	66.10	66.18	
S002-TR-01CE001	2W XFMR	Bus43	Bus10	49.72	298.33	302.44	
S002-TR-01CE002	2W XFMR	Bus42	Bus23	49.72	298.33	302.44	
S002-TR-03CE001	2W XFMR	Bus17	Bus28	67.13	351.57	357.92	
S002-TR-03CE002	2W XFMR	Bus18	Bus33	67.02	351.59	357.92	
S002-TR-03CN001	2W XFMR	Bus112	Bus67	44.35	281.62	285.09	
S002-TR-03CN002	2W XFMR	Bus8	Bus69	43.99	279.31	282.75	
S002-TR-04DE001	2W XFMR	S002-TR-04DE001_PRI	S002-SB-04E001 [BUS A]	530.09	795.13	955.63	
S002-TR-PSS1AC01	2W XFMR	TR-PSS1AC01_132kV BUS	TR-PSS1AC01_11kV BUS	0.63	20.33	20.34	
S002-TR-PSS1AC02	2W XFMR	TR-PSS1AC02_132kV BUS	TR-PSS1AC02_11kV BUS	0.65	21.11	21.11	
S003-TMM1A [EXISTING]	2W XFMR	Bus168	Bus50	21.54	183.08	184.35	
S003-TMM1B [EXISTING]	2W XFMR	Bus163	Bus49	21.73	184.68	185.95	
S003-TR-11CE001	2W XFMR	Bus19	Bus166	101.33	354.64	368.83	
S003-TR-11CE002	2W XFMR	Bus20	Bus167	60.64	363.81	368.83	
S003-TR-11CN001	2W XFMR	Bus56	S003-SB-11N001 [BUS A]	38.81	232.84	236.05	
S003-TR-11CN002	2W XFMR	Bus55	S003-SB-11N001 [BUS B]	38.81	232.84	236.05	
T2	2W XFMR	Bus51	LV SWBD_TOFS	125.68	439.87	457.47	
TR-05CE001A	2W XFMR	S002-SB-1AC001 [BUS A]	S002-SB-05N001A	56.25	337.53	342.19	
TR-05CE001B	2W XFMR	S002-SB-1AC001 [BUS B]	S002-SB-05N001B	56.25	337.53	342.19	
TX-A	2W XFMR	Bus66	Bus38	32.43	194.61	197.29	
TX-B	2W XFMR	Bus78	Bus65	32.43	194.61	197.29	
S002-TR-CMP1_VFD	3W Xfmr	S002-SB-1AC001 [BUS A]	S002-TR-CMP1_VFD~2	4.47	201.24	201.29	
	3W Xfmr	S002-SB-1AC001 [BUS A]	S002-TR-CMP1_VFD~3	4.47	201.24	201.29	
	3W Xfmr	S002-TR-CMP1_VFD~2	S002-TR-CMP1_VFD~3	4.47	201.24	201.29	
S002-TR-CMP2_VFD	3W Xfmr	S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~2	4.47	201.24	201.29	
	3W Xfmr	S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~3	4.47	201.24	201.29	
	3W Xfmr	S002-TR-CMP2_VFD~2	S002-TR-CMP2_VFD~3	4.47	201.24	201.29	
Cable2	Cable	TYPICAL MOTOR FEEDER 1 A	Bus9	7162.64	1646.88	7349.53	0.0000088

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable3	Cable	TYPICAL MOTOR FEEDER 1 A	Bus12	7050.72	1621.15	7234.69	0.0000086
Cable4	Cable	TYPICAL MOTOR FEEDER 1 A	Bus53	6938.81	1595.42	7119.86	0.0000085
Cable5	Cable	TYPICAL MOTOR FEEDER 1 A	Bus54	6938.81	1595.42	7119.86	0.0000085
Cable6	Cable	TYPICAL MOTOR FEEDER 1 A	Bus85	6826.89	1569.68	7005.02	0.0000084
Cable7	Cable	TYPICAL MOTOR FEEDER 1 A	Bus149	6714.97	1543.95	6890.19	0.0000082
Cable10	Cable	TYPICAL MOTOR FEEDER 1 A	Bus151	6826.89	1569.68	7005.02	0.0000084
Cable11	Cable	TYPICAL MOTOR FEEDER 1 A	Bus154	6714.97	1543.95	6890.19	0.0000082
Cable12	Cable	TYPICAL MOTOR FEEDER 1 A	Bus161	6603.06	1518.22	6775.35	0.0000081
Cable13	Cable	TYPICAL MOTOR FEEDER 1 A	Bus162	6714.97	1543.95	6890.19	0.0000082
Cable19	Cable	TYPICAL MOTOR FEEDER 1 A	Bus164	6603.06	1518.22	6775.35	0.0000081
Cable20	Cable	S002-MCC1-8 [BUS]	Bus159	12018.05	1506.79	12112.14	0.0000070
Cable21	Cable	TYPICAL MOTOR FEEDER 1 A	Bus165	6491.14	1492.49	6660.51	0.0000080
Cable22	Cable	TYPICAL MOTOR FEEDER 1 A	Bus170	6603.06	1518.22	6775.35	0.0000081
Cable23	Cable	S002-SB-1AN001 [BUS B]	Bus152	7285.48	418.55	7297.49	0.0000019
Cable24	Cable	S002-SB-1AN001 [BUS A]	Bus158	7285.48	418.55	7297.49	0.0000019
Cable25	Cable	TYPICAL MOTOR FEEDER 1 A	Bus171	6491.14	1492.49	6660.51	0.0000080
Cable26	Cable	S002-MCC1-9 [BUS]	Bus169	12018.05	1506.79	12112.14	0.0000070
Cable27	Cable	TYPICAL MOTOR FEEDER 1 A	Bus172	6379.22	1466.75	6545.68	0.0000078
Cable28	Cable	TYPICAL MOTOR FEEDER 1 B	Bus173	6043.48	1389.56	6201.17	0.0000074
Cable29	Cable	TYPICAL MOTOR FEEDER 1 B	Bus174	6155.39	1415.29	6316.00	0.0000075
Cable30	Cable	TYPICAL MOTOR FEEDER 1 B	Bus175	6267.31	1441.02	6430.84	0.0000077
Cable31	Cable	TYPICAL MOTOR FEEDER 1 B	Bus176	6155.39	1415.29	6316.00	0.0000075
Cable32	Cable	TYPICAL MOTOR FEEDER 1 B	Bus177	6267.31	1441.02	6430.84	0.0000077
Cable33	Cable	TYPICAL MOTOR FEEDER 1 B	Bus178	6379.22	1466.75	6545.68	0.0000078
Cable34	Cable	TYPICAL MOTOR FEEDER 1 B	Bus179	6267.31	1441.02	6430.84	0.0000077
Cable35	Cable	TYPICAL MOTOR FEEDER 1 B	Bus180	6379.22	1466.75	6545.68	0.0000078
Cable36	Cable	TYPICAL MOTOR FEEDER 1 B	Bus181	6491.14	1492.49	6660.51	0.0000080
Cable37	Cable	TYPICAL MOTOR FEEDER 1 B	Bus182	6379.22	1466.75	6545.68	0.0000078
Cable38	Cable	TYPICAL MOTOR FEEDER 1 B	Bus183	6491.14	1492.49	6660.51	0.0000080

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable39	Cable	TYPICAL MOTOR FEEDER 1 B	Bus184	6603.06	1518.22	6775.35	0.0000081
Cable40	Cable	Bus_MDB-8	Cable40~	110.58	109.97	155.95	0.0000029
Cable41	Cable	Bus_MDB-8	Cable41~	122.87	122.19	173.28	0.0000032
Cable42	Cable	TYPICAL MOTOR FEEDER 1 B	Bus185	6491.14	1492.49	6660.51	0.0000080
Cable43	Cable	TYPICAL MOTOR FEEDER 1 B	Bus186	6603.06	1518.22	6775.35	0.0000081
Cable44	Cable	TYPICAL MOTOR FEEDER 1 B	Bus187	6714.97	1543.95	6890.19	0.0000082
Cable45	Cable	POWER SOCKET FEEDER 1 B	Bus188	2393.44	1152.13	2656.30	0.0000070
Cable48	Cable	POWER SOCKET FEEDER 1 B	Bus189	1127.34	542.67	1251.16	0.0000033
Cable49	Cable	POWER SOCKET FEEDER 1 B	Bus190	1387.50	667.90	1539.89	0.0000040
Cable83	Cable	Bus_MDB-8	Bus14	3698.81	212.50	3704.91	0.0000009
Cable118	Cable	MV RMU-TOFS	CAMP BUS	17.89	7.71	19.48	0.0138730
Cable123	Cable	Bus_MDB-8	Bus15	4068.69	233.75	4075.40	0.0000010
Cable128	Cable	Bus_LVDB-2	Bus_MDB-8	30.72	30.55	43.32	0.0000032
Cable134	Cable	Bus52	Bus_LVDB-1	9.83	9.77	13.86	0.0000010
Cable136	Cable	Bus65	Bus52	28.08	27.93	39.61	0.0000090
Cable137	Cable	Bus63	Bus_LVDB-1	9.83	9.77	13.86	0.0000010
Cable144	Cable	Bus38	Bus40	21.06	20.95	29.71	0.0000067
Cable145	Cable	Bus38	Bus45	21.06	20.95	29.71	0.0000067
Cable146	Cable	Bus38	Bus48	21.06	20.95	29.71	0.0000067
Cable147	Cable	Bus40	Bus_LVDB-2	9.83	9.77	13.86	0.0000010
Cable148	Cable	Bus45	Bus_LVDB-2	9.83	9.77	13.86	0.0000010
Cable149	Cable	Bus48	Bus_LVDB-2	9.83	9.77	13.86	0.0000010
Cable150	Cable	Bus65	Bus63	28.08	27.93	39.61	0.0000090
Cable151	Cable	Bus_LVDB-2	Bus_MDB-5	30.72	30.55	43.32	0.0000032
Cable152	Cable	Bus64	Bus_LVDB-1	9.83	9.77	13.86	0.0000010
Cable153	Cable	Bus_MDB-5	Cable153~	73.72	73.31	103.97	0.0000019
Cable154	Cable	Bus_MDB-5	Cable154~	86.01	85.53	121.30	0.0000022
Cable155	Cable	Bus_MDB-5	Bus60	2589.17	148.75	2593.44	0.0000006
Cable156	Cable	Bus_MDB-5	Bus62	2959.05	170.00	2963.93	0.0000007
Cable157	Cable	Bus65	Bus64	28.08	27.93	39.61	0.0000090
Cable158	Cable	Bus_LVDB-1	Bus_MDB-1	31.95	31.77	45.05	0.0000033
Cable159	Cable	Bus_LVDB-1	Bus_MDB-2	29.49	29.32	41.59	0.0000031
Cable160	Cable	Bus_LVDB-1	Bus_MDB-3	28.26	28.10	39.85	0.0000029
Cable161	Cable	Bus_LVDB-1	Bus_MDB-4	18.43	18.33	25.99	0.0000019
Cable162	Cable	Bus_LVDB-2	Bus_MDB-6	30.72	30.55	43.32	0.0000032
Cable163	Cable	Bus_MDB-6	Cable163~	29.49	29.32	41.59	0.0000008
Cable164	Cable	Bus_MDB-6	Cable164~	44.23	43.99	62.38	0.0000012

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable165	Cable	Bus_MDB-6	Bus72	1997.36	114.75	2000.65	0.0000005
Cable166	Cable	Bus_MDB-6	Bus73	2367.24	136.00	2371.14	0.0000006
Cable167	Cable	Bus_LVDB-2	Bus_MDB-7	30.72	30.55	43.32	0.0000032
Cable168	Cable	Bus_MDB-7	Cable168~	61.44	61.09	86.64	0.0000016
Cable169	Cable	Bus_MDB-7	Cable169~	88.47	87.97	124.76	0.0000023
Cable170	Cable	Bus_MDB-7	Bus76	2589.17	148.75	2593.44	0.0000006
Cable171	Cable	Bus_MDB-7	Bus77	3402.91	195.50	3408.52	0.0000008
Cable176	Cable	Bus64	Bus82	78.22	24.24	81.89	0.0000027
Cable188	Cable	Bus_MDB-1	Cable188~	24.57	24.44	34.66	0.0000006
Cable189	Cable	Bus_MDB-1	Cable189~	36.86	36.66	51.98	0.0000010
Cable190	Cable	Bus_MDB-1	Bus95	1849.41	106.25	1852.46	0.0000005
Cable191	Cable	Bus_MDB-1	Bus96	2219.29	127.50	2222.95	0.0000005
Cable192	Cable	Bus_MDB-2	Cable192~	61.44	61.09	86.64	0.0000016
Cable193	Cable	Bus_MDB-2	Cable193~	86.01	85.53	121.30	0.0000022
Cable194	Cable	Bus_MDB-2	Bus99	2589.17	148.75	2593.44	0.0000006
Cable195	Cable	Bus_MDB-2	Bus100	3328.93	191.25	3334.42	0.0000008
Cable196	Cable	Bus_MDB-3	Cable196~	105.67	105.08	149.02	0.0000028
Cable197	Cable	Bus_MDB-3	Cable197~	63.89	63.54	90.11	0.0000017
Cable198	Cable	Bus_MDB-3	Bus103	3698.81	212.50	3704.91	0.0000009
Cable199	Cable	Bus_MDB-3	Bus104	2589.17	148.75	2593.44	0.0000006
Cable200	Cable	Bus_MDB-4	Cable200~	41.78	41.54	58.92	0.0000011
Cable201	Cable	Bus_MDB-4	Cable201~	56.52	56.21	79.71	0.0000015
Cable202	Cable	Bus_MDB-4	Bus107	1849.41	106.25	1852.46	0.0000005
Cable203	Cable	Bus_MDB-4	Bus108	2367.24	136.00	2371.14	0.0000006
C_688-KM-01	Cable	S002-SB-1AD001 [BUS A]	Bus160	45.27	15.37	47.81	0.0013490
S002-830-1063-N-P	Cable	Bus67	S002-SB-03N001 [BUS A]	13.74	29.51	32.55	0.0000053
S002-830-1066-N-P	Cable	Bus69	S002-SB-03N001 [BUS B]	13.74	29.51	32.55	0.0000053
S002-830-1069-N-P	Cable	Bus28	S002-SB-03E001 [BUS A]	26.72	57.37	63.29	0.0000037
S002-830-1072-N-P	Cable	Bus33	S002-SB-03E001 [BUS B]	26.72	57.37	63.29	0.0000037
S002-830-1420-N-P-1	Cable	Bus10	S002-SB-01E001 [BUS A]	11.45	24.59	27.13	0.0000044
S002-830-1500-N-P-1	Cable	Bus23	S002-SB-01E001 [BUS B]	13.74	29.51	32.55	0.0000053
S002-830-9001-A-P	Cable	Bus81	TR-PSS1AC02_132kV BUS	0.01	0.02	0.03	1.1666414
S002-830-9001-C-P	Cable	TR-PSS1AC01_11kV BUS	S002-SB-1AC001 [BUS A]	0.02	0.07	0.08	0.0219387
S002-830-9001-D-P-1	Cable	Bus22	S002-SB-1AD001 [BUS A]	0.77	1.65	1.82	0.0010527
S002-830-9004-A-P	Cable	Bus79	TR-PSS1AC01_132kV BUS	0.01	0.02	0.03	1.1666414
S002-830-9004-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus30	6.53	5.05	8.26	0.0001714
S002-830-9005-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus31	6.53	5.05	8.26	0.0001714
S002-830-9006-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus132	394.19	42.80	396.51	0.0004459
S002-830-9007-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus131	394.19	42.80	396.51	0.0004459

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
S002-830-9008-C-P-1	Cable	TR-PSS1AC02_11kV BUS	S002-SB-1AC001 [BUS B]	0.03	0.09	0.10	0.0284390
S002-830-9008-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus130	271.85	29.52	273.45	0.0003075
S002-830-9009-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus129	271.85	29.52	273.45	0.0003075
S002-830-9010-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus128	468.95	50.92	471.71	0.0005305
S002-830-9011-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus127	468.95	50.92	471.71	0.0005305
S002-830-9012-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus126	591.28	64.20	594.76	0.0006689
S002-830-9013-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus125	591.28	64.20	594.76	0.0006689
S002-830-9014-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus124	373.80	40.59	376.00	0.0004229
S002-830-9015-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus123	380.60	41.32	382.83	0.0004306
S002-830-9016-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus122	39.84	13.53	42.07	0.0011872
S002-830-9017-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus121	40.75	13.84	43.03	0.0012141
S002-830-9018-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus120	42.56	14.45	44.94	0.0012681
S002-830-9019-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus119	28.97	9.84	30.60	0.0002158
S002-830-9020-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus47	132.43	19.09	133.80	0.0002350
S002-830-9021-D-P-1	Cable	Bus25	S002-SB-1AD001 [BUS B]	0.77	1.65	1.82	0.0010527
S002-830-9024-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus29	6.53	5.05	8.26	0.0001714
S002-830-9025-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus32	6.53	5.05	8.26	0.0001714
S002-830-9026-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus133	394.19	42.80	396.51	0.0004459
S002-830-9027-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus134	394.19	42.80	396.51	0.0004459
S002-830-9028-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus135	271.85	29.52	273.45	0.0003075
S002-830-9029-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus136	271.85	29.52	273.45	0.0003075
S002-830-9030-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus137	468.95	50.92	471.71	0.0005305
S002-830-9031-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus138	468.95	50.92	471.71	0.0005305
S002-830-9032-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus139	591.28	64.20	594.76	0.0006689
S002-830-9033-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus140	591.28	64.20	594.76	0.0006689
S002-830-9034-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus141	373.80	40.59	376.00	0.0004229
S002-830-9035-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus142	380.60	41.32	382.83	0.0004306
S002-830-9036-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus143	39.84	13.53	42.07	0.0011872
S002-830-9037-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus144	40.75	13.84	43.03	0.0012141
S002-830-9038-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus145	42.56	14.45	44.94	0.0012681
S002-830-9039-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus146	183.50	19.92	184.58	0.0002076
S002-830-9051-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus43	1.11	0.84	1.39	0.0021440
S002-830-9053-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus7	7.85	5.98	9.87	0.0151869
S002-830-9054-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus112	8.86	6.76	11.14	0.0171523
S002-830-9055-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus17	8.86	6.76	11.14	0.0171523
S002-830-9056-C-P-1	Cable	S002-SB-1AC001 [BUS A]	Bus1	0.09	0.22	0.24	0.0034803
S002-830-9059-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus110	4.38	3.34	5.51	0.0084868
S002-830-9061-C-P-1	Cable	S002-SB-1AC001 [BUS A]	Bus92	0.76	1.06	1.30	0.0159033
S002-830-9062-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus90	5.54	4.22	6.97	0.0107202

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
S002-830-9063-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus89	5.45	4.15	6.85	0.0105415
S002-830-9064-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus42	0.83	0.63	1.04	0.0016080
S002-830-9066-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus46	7.85	5.98	9.87	0.0151869
S002-830-9067-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus8	8.86	6.76	11.14	0.0171523
S002-830-9068-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus18	8.86	6.76	11.14	0.0171523
S002-830-9069-C-P-1	Cable	S002-SB-1AC001 [BUS B]	Bus21	0.09	0.22	0.24	0.0034803
S002-830-9072-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus113	4.48	3.41	5.63	0.0086655
S002-830-9073-C-P-1	Cable	S002-SB-1AC001 [BUS B]	Bus114	0.74	1.03	1.27	0.0154214
S002-830-9074-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus115	4.43	3.38	5.57	0.0085761
S002-830-9076-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus117	5.45	4.15	6.85	0.0105415
S002-830-9077-C-P-1	Cable	S002-SB-1AC001 [BUS A]	S003-SB-11C001 [BUS A]	40.13	25.46	47.53	0.2281048
S002-830-9078-C-P-1	Cable	S002-SB-1AC001 [BUS B]	S003-SB-11C001 [BUS B]	40.13	25.46	47.53	0.2281048
S002-830-9079-C-P	Cable	S002-SB-1AC001 [BUS A]	MV RMU-TOFS	9.84	4.24	10.71	0.0076302
S002-Cable20	Cable	S002-SB-01E001 [BUS A]	S002-SB-PSS01N01 [BUS A]	385.60	827.82	913.22	0.0000536
S002-Cable21	Cable	S002-ASP1-2B	S002-DBASP1-2-22	1277.45	763.86	1488.41	0.0000055
S002-Cable22	Cable	S002-SB-01E001 [BUS B]	S002-SB-PSS01N01 [BUS B]	385.60	827.82	913.22	0.0000536
S002-Cable26	Cable	S002-SB-04E001 [BUS A]	Bus157	2095.32	649.28	2193.61	0.0000047
S002-Cable29	Cable	S002-PC1-1B	S002-MCC1-2 [BUS]	41.17	119.03	125.94	0.0000010
S002-Cable34	Cable	S002-PC1-1A	S002-SB-1AN001 [BUS A]	574.85	343.74	669.78	0.0000025
S002-Cable35	Cable	Bus27	S002-SB-02N001 [BUS A]	6.35	13.64	15.05	0.0000029
S002-Cable85	Cable	S002-PC1-1A	S002-MCC1-1 [BUS]	49.40	142.83	151.13	0.0000012
S002-Cable87	Cable	S002-PC1-1A	S002-MCC1-3 [BUS]	41.17	119.03	125.94	0.0000010
S002-Cable88	Cable	S002-PC1-2A	S002-MCC1-5 [BUS]	57.64	166.64	176.32	0.0000014
S002-Cable89	Cable	Bus24	S002-SB-02N001 [BUS B]	6.35	13.64	15.05	0.0000029
S002-Cable90	Cable	S002-PC1-1B	S002-SB-1AN001 [BUS B]	574.85	343.74	669.78	0.0000025
S002-Cable95	Cable	S002-PC1-2A	S002-MCC1-7 [BUS]	59.28	171.40	181.36	0.0000014
S002-Cable115	Cable	S002-PC1-2B	S002-MCC1-8 [BUS]	58.68	125.98	138.98	0.0000011
S002-Cable119	Cable	S002-PC1-2A	S002-MCC1-9 [BUS]	62.87	134.98	148.91	0.0000012
S002-Cable124	Cable	S002-PC1-1A	S002-ASP1-1 [BUS A]	47.75	138.07	146.10	0.0000011
S002-Cable125	Cable	S002-PC1-1B	S002-ASP1-1B [BUS B]	50.50	132.00	141.33	
S002-Cable126	Cable	S002-PC1-2A	S002-ASP1-2A	39.52	114.26	120.91	0.0000009
S002-Cable127	Cable	S002-PC1-2B	S002-ASP1-2B	39.52	114.26	120.91	0.0000009
S002-Cable131	Cable	S002-SB-01E001 [BUS A]	S002-MCC1-4 [BUS]	902.58	279.68	944.92	0.0000017
S002-Cable132	Cable	S002-SB-01E001 [BUS B]	S002-MCC1-6 [BUS]	1279.78	396.57	1339.82	0.0000024
S002-Cable138	Cable	S002-SB-02N001 [BUS A]	Area Lighting DB (A8)	7362.10	422.95	7374.24	0.0000018
S002-Cable181	Cable	S002-SB-1AN001 [BUS A]	Bus91	3338.35	418.55	3364.48	0.0000019
S002-C_TR-04DE001	Cable	S002-SB-1AD001 [BUS A]	S002-TR-04DE001_PRI	46.56	28.83	54.76	0.0008645
S002-C_TR-04DE002	Cable	S002-SB-1AD001 [BUS B]	S002-TR-04DE002_PRI	46.56	28.83	54.76	0.0008645
S003-830-1000-E-P	Cable	Bus166	S003-SB-11E001 [BUS A]	28.63	61.47	67.81	0.0000071

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
S003-830-1004-E-P	Cable	Bus167	S003-SB-11E001 [BUS B]	28.63	61.47	67.81	0.0000071
S003-830-9001-C-P	Cable	S003-SB-11C001 [BUS A]	Bus56	0.69	0.41	0.81	0.0014303
S003-830-9001-D-P	Cable	Bus50	S003-SB-11D001 [BUS A]	1.50	3.22	3.55	0.0001519
S003-830-9002-C-P	Cable	S003-SB-11C001 [BUS B]	Bus55	0.46	0.28	0.54	0.0009535
S003-830-9003-C-P	Cable	S003-SB-11C001 [BUS A]	Bus19	0.46	0.28	0.54	0.0008788
S003-830-9004-C-P	Cable	S003-SB-11C001 [BUS B]	Bus20	0.41	0.26	0.48	0.0007909
S003-830-9005-C-P	Cable	S003-SB-11C001 [BUS A]	Bus168	0.46	0.35	0.58	0.0008933
S003-830-9006-C-P	Cable	S003-SB-11C001 [BUS B]	Bus163	0.62	0.47	0.78	0.0011971
S003-830-9019-D-P	Cable	Bus49	S003-SB-11D001 [BUS B]	1.50	3.22	3.55	0.0001519
S003-830-9021-C-P	Cable	S003-SB-11C001 [BUS B]	Bus78	6.94	4.40	8.22	0.0098574
S003-830-9022-C-P	Cable	S003-SB-11C001 [BUS A]	Bus66	6.94	4.40	8.22	0.0098574
S003-Cable19	Cable	S003-SB-11N001 [BUS A]	S003-DB-11N001-03 [BUS A]	932.64	214.44	956.97	0.0000011
S003-Cable42	Cable	S003-SB-11N001 [BUS B]	S003-SB01E1	446.68	958.97	1057.89	0.0001104
S003-Cable44	Cable	S003-SB-11N001 [BUS A]	S003-DB11N001-04 [BUS A]	932.64	214.44	956.97	0.0000011
S003-Cable93	Cable	S003-SB-11E001 [BUS A]	S003-DB11E001-01	1025.90	235.88	1052.67	0.0000013
S003-Cable94	Cable	S003-SB-11E001 [BUS B]	S003-DB11E001-02	932.64	214.44	956.97	0.0000011
S003-Cable96	Cable	S003-SB-11N001 [BUS B]	S003-DB-11N001-03 [BUS B]	932.64	214.44	956.97	0.0000011
S003-Cable105	Cable	S003-SB-11N001 [BUS A]	S003-MDP-B	137.44	295.07	325.51	0.0000531
S003-Cable113	Cable	S003-SB-11D001 [BUS B]	Bus37	30.33	19.24	35.92	0.0044501
S003-Cable121	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-04 [BUS B]	932.64	214.44	956.97	0.0000011
S003-Cable142	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-11	1492.22	343.10	1531.15	0.0000073
S003-Cable173	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-12	1492.22	343.10	1531.15	0.0000073
S002-830-1601-N-P-1	Cable	S002-SB-01E001 [BUS B]	S002-830-1601-N-P-1~	6.87	14.75	16.28	0.0000027
S003-830-2000-E-P	Cable	S003-830-2000-E-P~	S003-SB-11E001 [BUS B]	6413.91	2504.63	6885.60	0.0000157
S003-Cable43	Cable	Bus58	S003-Cable43~	1281.39	997.23	1623.71	0.0000045
S003-Cable45	Cable	Bus61	S003-Cable45~	630.13	303.32	699.33	0.0000012
S003-Cable120	Cable	Bus71	S003-Cable120~	5.20	11.17	12.32	0.0000035
S002-Cable23	Cable	Bus41	S002-Cable23~	4.16	8.93	9.86	0.0000044
S002-830-1072-E-P	Cable	Bus35	S002-830-1072-E-P~	38.84	83.38	91.98	0.0000037
Cable84	Cable	Bus11	Cable84~	525.10	252.77	582.78	0.0000010
=E02-Q0	Tie Breakr	132kV GRID	Bus83				
=E06-Q0	Tie Breakr	132kV GRID	Bus84				
CB1	Tie Breakr	S002-SB-05N001A	TYPICAL MOTOR FEEDER 1 A				
CB8	Tie Breakr	S002-SB-05N001B	TYPICAL MOTOR FEEDER 1 B				
CB19	Tie Breakr	S002-SB-05N001B	POWER SOCKET FEEDER 1 B				
IAC001=R17	Tie Breakr	S002-SB-1AC001 [BUS B]	BUS_S002-CMP3				
QM	Tie Breakr	Bus82	200A MDB Agreko				
SB-04E001=CB3	Tie Breakr	S002-SB-04E001 [BUS B]	S002-SB-04E001 [BUS A]				

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
RMU TOFS	Tie Fuse	MV RMU-TOFS	Bus51				
SW3	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus79				
SW5	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus80				
SW7	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus81				
SW9	Tie Switch	Bus83	S002-SB-PSS1A01 [BUS-A]				
SW11	Tie Switch	Bus84	S002-SB-PSS1A01 [BUS-A]				
SW13	Tie Switch	Bus87	S002-SB-PSS1A01 [BUS-A]				
SW15	Tie Switch	Bus88	S002-SB-PSS1A01 [BUS-A]				
VFD1	Tie Switch	Cable153~	Bus57				
VFD2	Tie Switch	Cable154~	Bus59				
VFD9	Tie Switch	Cable163~	Bus68				
VFD10	Tie Switch	Cable164~	Bus70				
VFD13	Tie Switch	Cable168~	Bus74				
VFD14	Tie Switch	Cable169~	Bus75				
VFD15	Tie Switch	Cable40~	Bus2				
VFD16	Tie Switch	Cable41~	Bus4				

Equipment Cable Input Data

Equipment Cable	Equipment		ohms or siemens/1000 m per Conductor									O/L Heater	
					Length							Resistance	
ID	ID	Type	Library	Size	Adj. (m)	% Tol	#/ph	T (°C)	R	X	Y	Adj.(ohm)	% Tol
Cable16	614-KM-01A	Ind. Motor	11NCUS3	400	300.0	0.0	2	75	.06104	.08490	.0001822	.0000	0.0
Cable17	614-KM-01B	Ind. Motor	11NCUS3	400	300.0	0.0	2	75	.06104	.08490	.0001822	.0000	0.0
Cable9	614-KM-11	Ind. Motor	11NCUS3	400	300.0	0.0	2	75	.06104	.08490	.0001822	.0000	0.0
Cable86	S002-40-MP-04B	Ind. Motor	3.3NCUN3	25	270.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable46	S002-40-MP-05A	Ind. Motor	11NCUS3	185	600.0	0.0	1	75	.12208	.09310	.0001351	.0000	0.0
Cable50	S002-40-MP-05B	Ind. Motor	11NCUS3	185	590.0	0.0	1	75	.12208	.09310	.0001351	.0000	0.0
Cable47	S002-40-MP-5C	Ind. Motor	11NCUS3	185	590.0	0.0	1	75	.12208	.09310	.0001621	.0000	0.0
MSB1B-MK1C-P	S002-6100-MK-1C	Ind. Motor	11NCUS3	400	320.0	0.0	2	75	.06104	.08490	.0001822	.0000	0.0
MSB1A-MK1D-P	S002-6100-MK-1D	Ind. Motor	11NCUS3	400	330.0	0.0	2	75	.06104	.08490	.0001822	.0000	0.0
Cable130	S002-6100-MP-1B	Ind. Motor	1.0NCUN3	95	150.0	0.0	1	75	.23568	.07400	.0000000	.0000	0.0
Cable54	S002-6100-MP-3D	Ind. Motor	3.3NCUN3	25	580.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable55	S002-6100-MP-4A	Ind. Motor	3.3NCUN3	25	400.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable57	S002-6100-MP-4C	Ind. Motor	3.3NCUN3	25	400.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable63	S002-6100-MP-6A	Ind. Motor	3.3NCUN3	25	700.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable65	S002-6100-MP-6C	Ind. Motor	3.3NCUN3	25	700.0	0.0	1	75	.88320	.09800	.0000000	.0000	0.0
Cable108	S002-6300-MK-1A	Ind. Motor	1.0NCUN3	120	210.0	0.0	2	75	.18732	.07200	.0000000	.0000	0.0
Cable135	S002-MCC1-4_M	Ind. Motor	1.0NCUN3	16	220.0	0.0	1	75	1.39251	.08000	.0000971	.0000	0.0
Cable99	S003-7100-MK-1A	Ind. Motor	3.3NCUN3	50	375.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable100	S003-7100-MK-1B	Ind. Motor	3.3NCUN3	50	330.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable101	S003-7100-MK-1C	Ind. Motor	3.3NCUN3	50	360.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable98	S003-7100-MK-2A	Ind. Motor	3.3NCUN3	50	360.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable102	S003-7100-MK-2B	Ind. Motor	3.3NCUN3	50	375.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable103	S003-7100-MK-2C	Ind. Motor	3.3NCUN3	50	320.0	0.0	1	75	.47116	.08900	.0000789	.0000	0.0
Cable104	S003-7200-MP-02B	Ind. Motor	3.3NCUN3	150	190.0	0.0	1	75	.15070	.07600	.0001219	.0000	0.0
Cable97	S003-7200-MP-03A	Ind. Motor	3.3NCUN3	150	590.0	0.0	1	75	.15070	.07600	.0001219	.0000	0.0
Cable129	S003-7400-MP-4A	Ind. Motor	1.0NCUN3	50	150.0	0.0	1	75	.47116	.07900	.0001188	.0000	0.0
Cable133	S003-7400-P-21B	Ind. Motor	1.0NCUN3	120	150.0	0.0	1	75	.18694	.07300	.0001499	.0000	0.0
Cable110	S003-7500-K-1A	Ind. Motor	1.0NCUN3	240	210.0	0.0	2	75	.09433	.07200	.0000000	.0000	0.0

VFD Input Data

	AC Rating							Operating			Output		
	MVA	kV		FLA		Input							Bypass
ID	Output	Input	Output	Input	Output	% EFF	% PF	V/Hz	% Frequency	Hz	% Imax	Switch
S002-CMP1_VFD	6.000	11.000	11.000	331.49	314.92	95.00	100.00	100.00	100.0	50.0	150.0	No
S002-CMP2_VFD	6.000	11.000	11.000	331.49	314.92	95.00	100.00	100.00	100.0	50.0	150.0	No
VFD25	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD26	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD27	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD28	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD29	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD30	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD31	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No
VFD32	0.300	0.415	0.415	410.41	410.41	100.00	100.00	100.00	100.0	50.0	150.0	No

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
* 132kV GRID	132.000	100.000	0.0	56.580	26.246	0.000	0.000	Bus83	0.000	0.000	0.0	0.0	
								Bus84	56.580	26.246	272.8	90.7	
200A MDB Agreko	0.415	101.359	-6.6	0.000	0.000	0.022	0.011	Bus82	-0.022	-0.011	33.6	90.0	
Area Lighting DB (A8)	0.380	103.060	-3.8	0.000	0.000	0.009	0.007	S002-SB-02N001 [BUS A]	-0.009	-0.007	16.7	80.0	
Bus1	11.000	101.236	-3.5	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-4.455	-2.794	272.6	84.7	
								Bus22	4.455	2.794	272.6	84.7	
Bus2	0.380	107.790	-6.0	0.000	0.000	0.253	0.159	Cable40~	-0.253	-0.159	421.6	84.7	
Bus4	0.380	107.729	-6.0	0.000	0.000	0.253	0.159	Cable41~	-0.253	-0.159	421.8	84.7	
Bus7	11.000	101.206	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-0.321	-0.222	20.3	82.2	
								Bus27	0.321	0.222	20.3	82.2	5.000
Bus8	11.000	100.219	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-0.515	-0.392	33.9	79.6	
								Bus69	0.515	0.392	33.9	79.6	2.500
Bus9	0.380	100.027	-4.4	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.5	85.0	
Bus10	0.380	101.966	-4.8	0.000	0.000	0.000	0.000	S002-SB-01E001 [BUS A]	0.795	0.436	1350.7	87.7	
								Bus43	-0.795	-0.436	1350.7	87.7	
* Bus11	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Cable84~	0.000	0.000	0.0	0.0	
Bus12	0.380	100.091	-4.4	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.4	85.0	
Bus14	0.380	107.600	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-8	-0.016	-0.012	27.9	80.0	
Bus15	0.380	107.525	-5.6	0.000	0.000	0.016	0.012	Bus_MDB-8	-0.016	-0.012	27.9	80.0	
Bus17	11.000	101.175	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-0.494	-0.349	31.4	81.6	
								Bus28	0.494	0.349	31.4	81.6	
Bus18	11.000	100.210	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-0.564	-0.442	37.5	78.7	
								Bus33	0.564	0.442	37.5	78.7	
Bus19	11.000	98.955	-3.4	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS A]	-0.483	-0.315	30.6	83.7	
								Bus166	0.483	0.315	30.6	83.7	
Bus20	11.000	97.829	-3.7	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS B]	-0.513	-0.360	33.6	81.9	
								Bus167	0.513	0.360	33.6	81.9	
Bus21	11.000	100.290	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-2.853	-1.654	172.6	86.5	
								Bus25	2.853	1.654	172.6	86.5	
Bus22	3.300	103.588	-5.3	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS A]	4.445	2.595	869.2	86.4	
								Bus1	-4.445	-2.595	869.2	86.4	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Bus23	0.380	100.902	-4.8	0.000	0.000	0.000	0.000	S002-SB-01E001 [BUS B]	0.791	0.457	1376.0	86.6	
								Bus42	-0.791	-0.457	1376.0	86.6	
Bus24	0.380	103.763	-3.8	0.000	0.000	0.000	0.000	S002-SB-02N001 [BUS B]	0.172	0.116	303.7	83.0	
								Bus46	-0.172	-0.116	303.7	83.0	
Bus25	3.300	103.517	-4.6	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS B]	2.849	1.576	550.2	87.5	
								Bus21	-2.849	-1.576	550.2	87.5	
Bus27	0.380	104.008	-4.1	0.000	0.000	0.000	0.000	S002-SB-02N001 [BUS A]	0.320	0.216	563.0	82.9	
								Bus7	-0.320	-0.216	563.0	82.9	
Bus28	0.380	104.740	-4.4	0.000	0.000	0.000	0.000	S002-SB-03E001 [BUS A]	0.491	0.336	862.9	82.6	
								Bus17	-0.491	-0.336	862.9	82.6	
Bus29	3.450	98.927	-4.6	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS B]	-0.331	-0.212	66.6	84.2	
								S002-PC1-1B	0.331	0.212	66.6	84.2	5.000
Bus30	3.450	98.916	-5.3	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS A]	-0.742	-0.515	152.9	82.1	
								S002-PC1-1A	0.742	0.515	152.9	82.1	5.000
Bus31	3.450	98.891	-5.3	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS A]	-0.949	-0.706	200.1	80.2	
								S002-PC1-2A	0.949	0.706	200.1	80.2	5.000
Bus32	3.450	98.940	-4.6	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS B]	-0.212	-0.145	43.4	82.6	
								S002-PC1-2B	0.212	0.145	43.4	82.6	5.000
Bus33	0.380	103.284	-4.5	0.000	0.000	0.000	0.000	S002-SB-03E001 [BUS B]	0.560	0.422	1032.2	79.8	
								Bus18	-0.560	-0.422	1032.2	79.8	
* Bus35	0.380	100.000	0.0	0.000	0.000	0.000	0.000	S002-830-1072-E-P~	0.000	0.000	0.0	0.0	
Bus36	0.400	97.433	-6.7	0.000	0.000	0.647	0.485	Bus37	-0.647	-0.485	1197.4	80.0	
Bus37	3.300	101.894	-4.8	0.000	0.000	0.000	0.000	S003-SB-11D001 [BUS B]	-0.659	-0.529	145.1	78.0	
								Bus36	0.659	0.529	145.1	78.0	
Bus38	0.400	103.648	-5.8	0.000	0.000	0.000	0.000	Bus40	0.724	0.462	1196.5	84.3	
								Bus45	0.724	0.462	1196.5	84.3	
								Bus48	0.724	0.462	1196.5	84.3	
								Bus66	-2.173	-1.386	3589.5	84.3	
Bus40	0.400	103.365	-5.9	0.000	0.000	0.000	0.000	Bus38	-0.723	-0.460	1196.5	84.4	
								Bus_LVDB-2	0.723	0.460	1196.5	84.4	
* Bus41	0.380	100.000	0.0	0.000	0.000	0.000	0.000	S002-Cable23~	0.000	0.000	0.0	0.0	
Bus42	11.000	100.286	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-0.796	-0.486	48.8	85.4	
								Bus23	0.796	0.486	48.8	85.4	2.500

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Bus43	11.000	101.233	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-0.799	-0.464	47.9	86.5	
								Bus10	0.799	0.464	47.9	86.5	2.500
Bus45	0.400	103.365	-5.9	0.000	0.000	0.000	0.000	Bus38	-0.723	-0.460	1196.5	84.4	
								Bus_LVDB-2	0.723	0.460	1196.5	84.4	
Bus46	11.000	100.275	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-0.173	-0.118	10.9	82.6	
								Bus24	0.173	0.118	10.9	82.6	5.000
Bus47	3.300	103.465	-5.3	0.000	0.000	0.021	0.011	S002-SB-1AD001 [BUS A]	-0.021	-0.011	3.9	89.0	
Bus48	0.400	103.365	-5.9	0.000	0.000	0.000	0.000	Bus38	-0.723	-0.460	1196.5	84.4	
								Bus_LVDB-2	0.723	0.460	1196.5	84.4	
Bus49	3.300	102.324	-4.8	0.000	0.000	0.000	0.000	S003-SB-11D001 [BUS B]	0.938	0.662	196.4	81.7	
								Bus163	-0.938	-0.662	196.4	81.7	
Bus50	3.300	104.882	-3.8	0.000	0.000	0.000	0.000	S003-SB-11D001 [BUS A]	0.304	0.161	57.3	88.3	
								Bus168	-0.304	-0.161	57.3	88.3	
Bus51	11.000	100.862	-3.4	0.000	0.000	0.000	0.000	LV SWBD_TOFS	0.725	0.273	40.3	93.6	
								MV RMU-TOFS	-0.725	-0.273	40.3	93.6	
Bus52	0.400	105.185	-6.6	0.000	0.000	0.000	0.000	Bus_LVDB-1	0.726	0.037	997.8	99.9	
								Bus65	-0.726	-0.037	997.8	99.9	
Bus53	0.380	100.156	-4.4	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.4	85.0	
Bus54	0.380	100.156	-4.4	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.4	85.0	
Bus55	11.000	97.827	-3.7	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS B]	-0.758	-0.520	49.3	82.5	
								S003-SB-11N001 [BUS B]	0.758	0.520	49.3	82.5	
Bus56	11.000	98.951	-3.4	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS A]	-0.657	-0.485	43.3	80.5	
								S003-SB-11N001 [BUS A]	0.657	0.485	43.3	80.5	
Bus57	0.380	107.975	-6.0	0.000	0.000	0.253	0.159	Cable153~	-0.253	-0.159	420.8	84.7	
* Bus58	0.380	100.000	0.0	0.000	0.000	0.000	0.000	S003-Cable43~	0.000	0.000	0.0	0.0	
Bus59	0.380	107.913	-6.0	0.000	0.000	0.253	0.159	Cable154~	-0.253	-0.159	421.1	84.7	
Bus60	0.380	107.824	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-5	-0.016	-0.012	27.8	80.0	
* Bus61	0.380	100.000	0.0	0.000	0.000	0.000	0.000	S003-Cable45~	0.000	0.000	0.0	0.0	
Bus62	0.380	107.750	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-5	-0.016	-0.012	27.9	80.0	
Bus63	0.400	105.185	-6.6	0.000	0.000	0.000	0.000	Bus_LVDB-1	0.726	0.037	997.8	99.9	
								Bus65	-0.726	-0.037	997.8	99.9	
Bus64	0.415	101.380	-6.6	0.000	0.000	0.000	0.000	Bus_LVDB-1	0.710	0.029	974.9	99.9	
								Bus65	-0.732	-0.040	1005.9	99.9	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus82	0.022	0.011	33.6	90.0	
Bus65	0.400	105.425	-6.5	0.000	0.000	0.000	0.000	Bus52	0.728	0.039	997.8	99.9	
								Bus63	0.728	0.039	997.8	99.9	
								Bus64	0.734	0.041	1005.9	99.8	
								Bus78	-2.189	-0.118	3001.6	99.9	
Bus66	11.000	98.715	-3.4	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS A]	-2.198	-1.535	142.5	82.0	
								Bus38	2.198	1.535	142.5	82.0	-5.000
Bus67	0.380	101.186	-4.9	0.000	0.000	0.000	0.000	S002-SB-03N001 [BUS A]	0.894	0.655	1664.3	80.7	
								Bus112	-0.894	-0.655	1664.3	80.7	
Bus68	0.380	108.195	-5.9	0.000	0.000	0.253	0.159	Cable163~	-0.253	-0.159	420.0	84.7	
Bus69	0.380	101.412	-4.3	0.000	0.000	0.000	0.000	S002-SB-03N001 [BUS B]	0.513	0.379	956.4	80.4	
								Bus8	-0.513	-0.379	956.4	80.4	
Bus70	0.380	108.122	-5.9	0.000	0.000	0.253	0.159	Cable164~	-0.253	-0.159	420.3	84.7	
* Bus71	0.380	100.000	0.0	0.000	0.000	0.000	0.000	S003-Cable120~	0.000	0.000	0.0	0.0	
Bus72	0.380	107.943	-5.8	0.000	0.000	0.016	0.012	Bus_MDB-6	-0.016	-0.012	27.8	80.0	
Bus73	0.380	107.869	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-6	-0.016	-0.012	27.8	80.0	
Bus74	0.380	108.036	-5.9	0.000	0.000	0.253	0.159	Cable168~	-0.253	-0.159	420.6	84.7	
Bus75	0.380	107.901	-6.0	0.000	0.000	0.253	0.159	Cable169~	-0.253	-0.159	421.1	84.7	
Bus76	0.380	107.824	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-7	-0.016	-0.012	27.8	80.0	
Bus77	0.380	107.661	-5.7	0.000	0.000	0.016	0.012	Bus_MDB-7	-0.016	-0.012	27.9	80.0	
Bus78	11.000	97.651	-3.8	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS B]	-2.207	-0.223	119.2	99.5	
								Bus65	2.207	0.223	119.2	99.5	-5.000
Bus79	132.000	100.000	0.0	0.000	0.000	0.000	0.000	TR-PSS1AC01_132kV BUS	28.642	14.004	139.4	89.8	
								S002-SB-PSS1A01 [BUS-A]	-28.642	-14.004	139.4	89.8	
Bus80	132.000	100.000	0.0	0.000	0.000	0.000	0.000	S002-SB-PSS1A01 [BUS-A]	0.000	0.000	0.0	0.0	
Bus81	132.000	100.000	0.0	0.000	0.000	0.000	0.000	TR-PSS1AC02_132kV BUS	27.937	12.242	133.4	91.6	
								S002-SB-PSS1A01 [BUS-A]	-27.937	-12.242	133.4	91.6	
Bus82	0.415	101.359	-6.6	0.000	0.000	0.000	0.000	Bus64	-0.022	-0.011	33.6	90.0	
								200A MDB Agreko	0.022	0.011	33.6	90.0	
Bus83	132.000	100.000	0.0	0.000	0.000	0.000	0.000	132kV GRID	0.000	0.000	0.0	0.0	
								S002-SB-PSS1A01 [BUS-A]	0.000	0.000	0.0	0.0	
Bus84	132.000	100.000	0.0	0.000	0.000	0.000	0.000	132kV GRID	-56.580	-26.246	272.8	90.7	
								S002-SB-PSS1A01 [BUS-A]	56.580	26.246	272.8	90.7	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Bus85	0.380	100.221	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.3	85.0	
Bus87	132.000	100.000	0.0	0.000	0.000	0.000	0.000	S002-SB-PSS1A01 [BUS-A]	0.000	0.000	0.0	0.0	
Bus88	132.000	100.000	0.0	0.000	0.000	0.000	0.000	S002-SB-PSS1A01 [BUS-A]	0.000	0.000	0.0	0.0	
Bus89	11.000	101.243	-3.4	0.000	0.000	0.053	0.032	S002-SB-1AC001 [BUS A]	-0.053	-0.032	3.2	85.4	
Bus90	11.000	101.201	-3.4	0.000	0.000	0.531	0.323	S002-SB-1AC001 [BUS A]	-0.531	-0.323	32.2	85.4	
Bus91	0.380	98.170	-6.3	0.000	0.000	0.036	0.018	S002-SB-1AN001 [BUS A]	-0.036	-0.018	62.6	90.0	
Bus92	11.000	101.159	-3.5	0.000	0.000	6.532	3.057	S002-SB-1AC001 [BUS A]	-6.532	-3.057	374.2	90.6	
Bus95	0.400	104.556	-6.6	0.000	0.000	0.016	0.012	Bus_MDB-1	-0.016	-0.012	27.3	80.0	
Bus96	0.400	104.487	-6.6	0.000	0.000	0.016	0.012	Bus_MDB-1	-0.016	-0.012	27.3	80.0	
Bus99	0.400	104.433	-6.5	0.000	0.000	0.016	0.012	Bus_MDB-2	-0.016	-0.012	27.3	80.0	
Bus100	0.400	104.294	-6.5	0.000	0.000	0.016	0.012	Bus_MDB-2	-0.016	-0.012	27.3	80.0	
Bus103	0.400	104.232	-6.4	0.000	0.000	0.016	0.012	Bus_MDB-3	-0.016	-0.012	27.4	80.0	
Bus104	0.400	104.440	-6.5	0.000	0.000	0.016	0.012	Bus_MDB-3	-0.016	-0.012	27.3	80.0	
Bus107	0.400	104.641	-6.5	0.000	0.000	0.016	0.012	Bus_MDB-4	-0.016	-0.012	27.2	80.0	
Bus108	0.400	104.545	-6.5	0.000	0.000	0.016	0.012	Bus_MDB-4	-0.016	-0.012	27.3	80.0	
Bus110	11.000	101.184	-3.5	0.000	0.000	0.973	0.471	S002-SB-1AC001 [BUS A]	-0.973	-0.471	56.1	90.0	
Bus112	11.000	101.111	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-0.900	-0.695	59.0	79.1	
								Bus67	0.900	0.695	59.0	79.1	2.500
Bus113	11.000	100.290	-3.4	0.000	0.000	0.098	0.052	S002-SB-1AC001 [BUS B]	-0.098	-0.052	5.8	88.4	
Bus114	11.000	100.210	-3.5	0.000	0.000	6.527	3.054	S002-SB-1AC001 [BUS B]	-6.527	-3.054	377.4	90.6	
Bus115	11.000	100.231	-3.5	0.000	0.000	0.978	0.516	S002-SB-1AC001 [BUS B]	-0.978	-0.516	57.9	88.4	
Bus117	11.000	100.251	-3.4	0.000	0.000	0.531	0.323	S002-SB-1AC001 [BUS B]	-0.531	-0.323	32.5	85.4	
Bus119	3.300	103.483	-5.3	0.000	0.000	0.038	0.022	S002-SB-1AD001 [BUS A]	-0.038	-0.022	7.5	87.0	
Bus120	3.300	102.996	-5.3	0.000	0.000	0.871	0.434	S002-SB-1AD001 [BUS A]	-0.871	-0.434	165.3	89.5	
Bus121	3.300	103.451	-5.3	0.000	0.000	0.087	0.043	S002-SB-1AD001 [BUS A]	-0.087	-0.043	16.5	89.5	
Bus122	3.300	103.452	-5.3	0.000	0.000	0.087	0.043	S002-SB-1AD001 [BUS A]	-0.087	-0.043	16.5	89.5	
Bus123	3.300	103.406	-5.3	0.000	0.000	0.020	0.011	S002-SB-1AD001 [BUS A]	-0.020	-0.011	3.9	86.5	
Bus124	3.300	103.408	-5.3	0.000	0.000	0.020	0.011	S002-SB-1AD001 [BUS A]	-0.020	-0.011	3.9	86.5	
Bus125	3.300	103.363	-5.3	0.000	0.000	0.019	0.011	S002-SB-1AD001 [BUS A]	-0.019	-0.011	3.7	86.5	
Bus126	3.300	103.363	-5.3	0.000	0.000	0.019	0.011	S002-SB-1AD001 [BUS A]	-0.019	-0.011	3.7	86.5	
Bus127	3.300	103.391	-5.3	0.000	0.000	0.019	0.011	S002-SB-1AD001 [BUS A]	-0.019	-0.011	3.7	86.5	
Bus128	3.300	102.410	-5.0	0.000	0.000	0.187	0.108	S002-SB-1AD001 [BUS A]	-0.187	-0.108	36.9	86.5	
Bus129	3.300	103.438	-5.3	0.000	0.000	0.018	0.010	S002-SB-1AD001 [BUS A]	-0.018	-0.010	3.5	86.5	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Bus130	3.300	103.438	-5.3	0.000	0.000	0.018	0.010	S002-SB-1AD001 [BUS A]	-0.018	-0.010	3.5	86.5	
Bus131	3.300	103.411	-5.3	0.000	0.000	0.018	0.010	S002-SB-1AD001 [BUS A]	-0.018	-0.010	3.5	86.5	
Bus132	3.300	102.615	-5.1	0.000	0.000	0.181	0.105	S002-SB-1AD001 [BUS A]	-0.181	-0.105	35.7	86.5	
Bus133	3.300	102.578	-4.4	0.000	0.000	0.181	0.105	S002-SB-1AD001 [BUS B]	-0.181	-0.105	35.7	86.5	
Bus134	3.300	102.568	-4.4	0.000	0.000	0.183	0.105	S002-SB-1AD001 [BUS B]	-0.183	-0.105	36.0	86.7	
Bus135	3.300	103.401	-4.6	0.000	0.000	0.018	0.010	S002-SB-1AD001 [BUS B]	-0.018	-0.010	3.5	86.5	
Bus136	3.300	103.401	-4.6	0.000	0.000	0.018	0.010	S002-SB-1AD001 [BUS B]	-0.018	-0.010	3.5	86.5	
Bus137	3.300	102.357	-4.4	0.000	0.000	0.190	0.109	S002-SB-1AD001 [BUS B]	-0.190	-0.109	37.4	86.8	
Bus138	3.300	102.357	-4.4	0.000	0.000	0.190	0.109	S002-SB-1AD001 [BUS B]	-0.190	-0.109	37.4	86.8	
Bus139	3.300	103.326	-4.6	0.000	0.000	0.019	0.011	S002-SB-1AD001 [BUS B]	-0.019	-0.011	3.7	86.5	
Bus140	3.300	103.326	-4.6	0.000	0.000	0.019	0.011	S002-SB-1AD001 [BUS B]	-0.019	-0.011	3.7	86.5	
Bus141	3.300	102.545	-4.4	0.000	0.000	0.198	0.115	S002-SB-1AD001 [BUS B]	-0.198	-0.115	39.0	86.5	
Bus142	3.300	103.369	-4.6	0.000	0.000	0.020	0.011	S002-SB-1AD001 [BUS B]	-0.020	-0.011	3.9	86.5	
Bus143	3.300	103.415	-4.6	0.000	0.000	0.087	0.043	S002-SB-1AD001 [BUS B]	-0.087	-0.043	16.5	89.5	
Bus144	3.300	102.980	-4.6	0.000	0.000	0.871	0.434	S002-SB-1AD001 [BUS B]	-0.871	-0.434	165.3	89.5	
Bus145	3.300	103.411	-4.6	0.000	0.000	0.087	0.043	S002-SB-1AD001 [BUS B]	-0.087	-0.043	16.5	89.5	
Bus146	3.300	102.995	-4.5	0.000	0.000	0.208	0.106	S002-SB-1AD001 [BUS B]	-0.208	-0.106	39.6	89.1	
Bus149	0.380	100.285	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.3	85.0	
Bus151	0.380	100.221	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.3	85.0	
Bus152	0.380	102.352	-5.3	0.000	0.000	0.001	0.001	S002-SB-1AN001 [BUS B]	-0.001	-0.001	1.8	90.0	
Bus154	0.380	100.285	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.3	85.0	
Bus157	0.380	103.594	-6.0	0.000	0.000	0.003	0.002	S002-SB-04E001 [BUS A]	-0.003	-0.002	5.8	80.0	
Bus158	0.380	99.586	-6.6	0.000	0.000	0.003	0.002	S002-SB-1AN001 [BUS A]	-0.003	-0.002	5.5	90.0	
Bus159	0.380	100.010	-4.7	0.000	0.000	0.016	0.007	S002-MCC1-8 [BUS]	-0.016	-0.007	26.8	92.0	
Bus160	3.450	98.476	-5.3	0.000	0.000	0.901	0.410	S002-SB-1AD001 [BUS A]	-0.901	-0.410	168.2	91.0	
Bus161	0.380	100.349	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus162	0.380	100.285	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.3	85.0	
Bus163	11.000	97.823	-3.7	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS B]	-0.942	-0.692	62.7	80.6	
								Bus49	0.942	0.692	62.7	80.6	2.500
Bus164	0.380	100.349	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus165	0.380	100.414	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus166	0.380	102.307	-4.3	0.000	0.000	0.000	0.000	S003-SB-11E001 [BUS A]	0.479	0.302	841.4	84.6	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus19	-0.479	-0.302	841.4	84.6	
Bus167	0.380	101.092	-4.8	0.000	0.000	0.000	0.000	S003-SB-11E001 [BUS B]	0.510	0.343	924.1	83.0	
								Bus20	-0.510	-0.343	924.1	83.0	
Bus168	11.000	98.956	-3.4	0.000	0.000	0.000	0.000	S003-SB-11C001 [BUS A]	-0.304	-0.164	18.3	88.0	
								Bus50	0.304	0.164	18.3	88.0	2.500
Bus169	0.380	97.096	-6.9	0.000	0.000	0.016	0.007	S002-MCC1-9 [BUS]	-0.016	-0.007	27.6	92.0	
Bus170	0.380	100.349	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus171	0.380	100.414	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus172	0.380	100.478	-4.5	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 A	-0.040	-0.025	71.2	85.0	
Bus173	0.380	98.713	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.4	85.0	
Bus174	0.380	98.648	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.5	85.0	
Bus175	0.380	98.582	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.5	85.0	
Bus176	0.380	98.648	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.5	85.0	
Bus177	0.380	98.582	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.5	85.0	
Bus178	0.380	98.517	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus179	0.380	98.582	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.5	85.0	
Bus180	0.380	98.517	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus181	0.380	98.451	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus182	0.380	98.517	-5.0	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus183	0.380	98.451	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus184	0.380	98.385	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.7	85.0	
Bus185	0.380	98.451	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.6	85.0	
Bus186	0.380	98.385	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.7	85.0	
Bus187	0.380	98.320	-4.9	0.000	0.000	0.040	0.025	TYPICAL MOTOR FEEDER 1 B	-0.040	-0.025	72.7	85.0	
Bus188	0.380	101.194	-5.6	0.000	0.000	0.025	0.015	POWER SOCKET FEEDER 1 B	-0.025	-0.015	43.4	85.0	
Bus189	0.380	101.675	-5.6	0.000	0.000	0.025	0.015	POWER SOCKET FEEDER 1 B	-0.025	-0.015	43.6	85.0	
Bus190	0.380	101.576	-5.6	0.000	0.000	0.025	0.015	POWER SOCKET FEEDER 1 B	-0.025	-0.015	43.6	85.0	
Bus_LVDB-1	0.415	101.302	-6.6	0.000	0.000	0.000	0.000	Bus52	-0.726	-0.036	997.8	99.9	
								Bus63	-0.726	-0.036	997.8	99.9	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus64	-0.709	-0.029	974.9	99.9	
								Bus_MDB-1	0.540	0.025	742.2	99.9	
								Bus_MDB-2	0.540	0.026	743.1	99.9	
								Bus_MDB-3	0.541	0.026	743.3	99.9	
								Bus_MDB-4	0.540	0.025	742.0	99.9	
Bus_LVDB-2	0.415	99.502	-5.9	0.000	0.000	0.000	0.000	Bus_MDB-8	0.542	0.345	899.0	84.4	
								Bus40	-0.722	-0.459	1196.5	84.4	
								Bus45	-0.722	-0.459	1196.5	84.4	
								Bus48	-0.722	-0.459	1196.5	84.4	
								Bus_MDB-5	0.542	0.345	897.5	84.4	
								Bus_MDB-6	0.541	0.344	895.7	84.4	
								Bus_MDB-7	0.541	0.345	897.3	84.4	
Bus_MDB-1	0.415	101.108	-6.7	0.000	0.000	0.000	0.000	Bus_LVDB-1	-0.539	-0.024	742.2	99.9	
								Cable188~	0.254	0.000	348.8	100.0	
								Cable189~	0.254	0.000	348.9	100.0	
								Bus95	0.016	0.012	27.3	80.2	
								Bus96	0.016	0.012	27.3	80.2	
Bus_MDB-2	0.415	101.122	-6.7	0.000	0.000	0.000	0.000	Bus_LVDB-1	-0.540	-0.025	743.1	99.9	
								Cable192~	0.254	0.000	349.1	100.0	
								Cable193~	0.254	0.001	349.3	100.0	
								Bus99	0.016	0.012	27.3	80.2	
								Bus100	0.016	0.012	27.3	80.3	
Bus_MDB-3	0.415	101.130	-6.7	0.000	0.000	0.000	0.000	Bus_LVDB-1	-0.540	-0.025	743.3	99.9	
								Cable196~	0.254	0.001	349.5	100.0	
								Cable197~	0.254	0.000	349.1	100.0	
								Bus103	0.016	0.012	27.4	80.3	
								Bus104	0.016	0.012	27.3	80.2	
Bus_MDB-4	0.415	101.190	-6.7	0.000	0.000	0.000	0.000	Bus_LVDB-1	-0.539	-0.024	742.0	99.9	
								Cable200~	0.254	0.000	348.7	100.0	
								Cable201~	0.254	0.000	348.8	100.0	
								Bus107	0.016	0.012	27.2	80.2	
								Bus108	0.016	0.012	27.3	80.2	
Bus_MDB-5	0.415	99.204	-5.9	0.000	0.000	0.000	0.000	Bus_LVDB-2	-0.540	-0.343	897.5	84.4	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Cable153~	0.254	0.160	420.8	84.7	
								Cable154~	0.254	0.160	421.1	84.7	
								Bus60	0.016	0.012	27.8	80.2	
								Bus62	0.016	0.012	27.9	80.3	
Bus_MDB-6	0.400	102.925	-5.9	0.000	0.000	0.000	0.000	Bus_LVDB-2	-0.539	-0.342	895.7	84.4	
								Cable163~	0.254	0.159	420.0	84.7	
								Cable164~	0.254	0.159	420.3	84.7	
								Bus72	0.016	0.012	27.8	80.2	
								Bus73	0.016	0.012	27.8	80.2	
Bus_MDB-7	0.400	102.924	-5.9	0.000	0.000	0.000	0.000	Bus_LVDB-2	-0.540	-0.343	897.3	84.4	
								Cable168~	0.254	0.160	420.6	84.7	
								Cable169~	0.254	0.160	421.1	84.7	
								Bus76	0.016	0.012	27.8	80.2	
								Bus77	0.016	0.012	27.9	80.3	
Bus_MDB-8	0.400	102.923	-5.9	0.000	0.000	0.000	0.000	Cable40~	0.254	0.160	421.6	84.6	
								Cable41~	0.255	0.160	421.8	84.6	
								Bus14	0.016	0.012	27.9	80.3	
								Bus15	0.016	0.012	27.9	80.3	
								Bus_LVDB-2	-0.541	-0.344	899.0	84.4	
BUS_S002-CMP3	11.000	100.297	-3.4	0.000	0.000	3.299	1.691	S002-SB-1AC001 [BUS B]	-3.299	-1.691	194.0	89.0	
CAMP BUS	11.000	100.323	-3.4	0.000	0.000	2.253	1.190	MV RMU-TOFS	-2.253	-1.190	133.3	88.4	
LV SWBD_TOFS	0.400	102.566	-5.2	0.000	0.000	0.717	0.245	Bus51	-0.717	-0.245	1065.7	94.6	
MV RMU-TOFS	11.000	100.862	-3.4	0.000	0.000	0.000	0.000	CAMP BUS	2.265	1.182	133.0	88.7	
								S002-SB-1AC001 [BUS A]	-2.990	-1.455	173.1	89.9	
								Bus51	0.725	0.273	40.3	93.6	
POWER SOCKET FEEDER 1 B	0.380	102.108	-5.6	0.000	0.000	0.000	0.000	Bus188	0.025	0.015	43.4	85.1	
								Bus189	0.025	0.015	43.6	85.0	
								Bus190	0.025	0.015	43.6	85.0	
								S002-SB-05N001B	-0.075	-0.046	130.6	85.0	
S002-ASP1-1 [BUS A]	0.380	100.929	-6.9	0.000	0.000	0.094	0.058	S002-PC1-1A	-0.094	-0.058	166.4	85.0	
S002-ASP1-1B [BUS B]	0.380	102.058	-5.5	0.000	0.000	0.315	0.195	S002-PC1-1B	-0.315	-0.195	552.4	85.0	
S002-ASP1-2A	0.380	99.635	-7.5	0.000	0.000	0.334	0.241	S002-PC1-2A	-0.334	-0.241	627.9	81.0	
S002-ASP1-2B	0.380	102.921	-5.1	0.000	0.000	0.016	0.012	S002-DBASP1-2-22	0.014	0.010	25.1	80.0	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								S002-PC1-2B	-0.030	-0.022	54.7	80.6	
S002-DBASP1-2-22	0.380	102.596	-5.0	0.000	0.000	0.014	0.010	S002-ASP1-2B	-0.014	-0.010	25.1	80.0	
S002-MCC1-1 [BUS]	0.380	100.689	-7.0	0.000	0.000	0.229	0.135	S002-PC1-1A	-0.229	-0.135	401.7	86.1	
S002-MCC1-2 [BUS]	0.380	102.600	-5.3	0.000	0.000	0.000	0.000	S002-PC1-1B	0.000	0.000	0.0	0.0	
S002-MCC1-3 [BUS]	0.380	100.774	-7.0	0.000	0.000	0.236	0.122	S002-PC1-1A	-0.236	-0.122	401.3	88.8	
S002-MCC1-4 [BUS]	0.380	101.024	-4.8	0.000	0.000	0.053	0.038	S002-SB-01E001 [BUS A]	-0.053	-0.038	98.9	81.3	
S002-MCC1-5 [BUS]	0.380	99.653	-7.5	0.000	0.000	0.233	0.156	S002-PC1-2A	-0.233	-0.156	427.2	83.1	
S002-MCC1-6 [BUS]	0.380	99.534	-4.7	0.000	0.000	0.056	0.041	S002-SB-01E001 [BUS B]	-0.056	-0.041	106.4	81.0	
S002-MCC1-7 [BUS]	0.380	99.638	-7.5	0.000	0.000	0.233	0.156	S002-PC1-2A	-0.233	-0.156	427.3	83.0	
S002-MCC1-8 [BUS]	0.380	102.733	-5.1	0.000	0.000	0.110	0.079	Bus159	0.017	0.007	26.8	92.3	
								S002-PC1-2B	-0.127	-0.086	226.3	82.8	
S002-MCC1-9 [BUS]	0.380	99.900	-7.3	0.000	0.000	0.121	0.084	Bus169	0.017	0.007	27.6	92.3	
								S002-PC1-2A	-0.138	-0.090	251.2	83.7	
S002-PC1-1A	0.380	101.093	-6.8	0.000	0.000	0.081	0.061	S002-SB-1AN001 [BUS A]	0.095	0.103	211.4	67.8	
								S002-MCC1-1 [BUS]	0.230	0.137	401.7	85.9	
								S002-MCC1-3 [BUS]	0.237	0.123	401.3	88.7	
								S002-ASP1-1 [BUS A]	0.094	0.058	166.4	84.9	
								Bus30	-0.737	-0.483	1324.2	83.6	
S002-PC1-1B	0.380	102.600	-5.3	0.000	0.000	0.000	0.000	S002-MCC1-2 [BUS]	0.000	0.000	0.0	0.0	
								S002-SB-1AN001 [BUS B]	0.014	0.009	24.5	85.4	
								S002-ASP1-1B [BUS B]	0.316	0.198	552.4	84.8	
								Bus29	-0.330	-0.206	576.8	84.8	
S002-PC1-2A	0.380	100.177	-7.2	0.000	0.000	0.000	0.000	S002-MCC1-5 [BUS]	0.233	0.158	427.2	82.8	
								S002-MCC1-7 [BUS]	0.233	0.158	427.3	82.8	
								S002-MCC1-9 [BUS]	0.138	0.091	251.2	83.6	
								S002-ASP1-2A	0.334	0.244	627.9	80.8	
								Bus31	-0.939	-0.651	1733.3	82.2	
S002-PC1-2B	0.380	102.968	-5.1	0.000	0.000	0.055	0.034	S002-MCC1-8 [BUS]	0.127	0.086	226.3	82.7	
								S002-ASP1-2B	0.030	0.022	54.7	80.5	
								Bus32	-0.212	-0.142	376.3	83.0	
S002-SB-1AC001 [BUS A]	11.000	101.247	-3.4	0.000	0.000	0.000	0.000	TR-PSS1AC01_11kV BUS	-28.572	-13.018	1627.6	91.0	
								Bus43	0.799	0.462	47.9	86.6	
								Bus7	0.322	0.208	19.9	83.9	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus112	0.901	0.680	58.5	79.8	
								Bus17	0.494	0.334	30.9	82.9	
								Bus1	4.455	2.792	272.5	84.7	
								Bus110	0.974	0.464	55.9	90.3	
								Bus92	6.536	3.048	373.9	90.6	
								Bus90	0.531	0.314	32.0	86.1	
								Bus89	0.053	0.022	3.0	92.1	
								S003-SB-11C001 [BUS A]	3.733	2.335	228.3	84.8	
								MV RMU-TOFS	3.002	1.453	172.9	90.0	
								S002-SB-05N001A	0.900	0.534	54.2	86.0	
								S002-TR-CMP1_VFD~2	5.872	0.371	305.0	99.8	
								&S002-TR-CMP1_VFD~3					
S002-SB-1AC001 [BUS B]	11.000	100.297	-3.4	0.000	0.000	0.000	0.000	TR-PSS1AC02_11kV BUS	-27.871	-11.397	1575.7	92.6	
								Bus42	0.796	0.484	48.8	85.4	
								Bus46	0.173	0.104	10.5	85.7	
								Bus8	0.516	0.377	33.4	80.7	
								Bus18	0.565	0.427	37.0	79.8	
								Bus21	2.853	1.651	172.5	86.6	
								Bus113	0.098	0.044	5.6	91.3	
								Bus114	6.531	3.046	377.1	90.6	
								Bus115	0.978	0.508	57.7	88.7	
								Bus117	0.531	0.314	32.3	86.1	
								S003-SB-11C001 [BUS B]	4.526	1.646	252.0	94.0	
								S002-SB-05N001B	1.133	0.726	70.4	84.2	
								BUS_S002-CMP3	3.299	1.691	194.0	89.0	
								S002-TR-CMP2_VFD~2	5.872	0.378	307.9	99.8	
								&S002-TR-CMP2_VFD~3					
S002-SB-1AD001 [BUS A]	3.300	103.499	-5.3	0.000	0.000	0.000	0.000	Bus160	0.906	0.411	168.1	91.1	
								Bus22	-4.442	-2.591	869.3	86.4	
								Bus30	0.743	0.516	152.9	82.2	
								Bus31	0.950	0.707	200.1	80.2	
								Bus132	0.183	0.105	35.6	86.8	
								Bus131	0.018	0.010	3.5	87.3	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus130	0.018	0.010	3.5	87.1	
								Bus129	0.018	0.010	3.5	87.1	
								Bus128	0.189	0.108	36.9	86.8	
								Bus127	0.019	0.010	3.6	87.5	
								Bus126	0.019	0.010	3.6	87.7	
								Bus125	0.019	0.010	3.6	87.7	
								Bus124	0.020	0.011	3.8	87.2	
								Bus123	0.020	0.011	3.8	87.2	
								Bus122	0.087	0.042	16.4	89.9	
								Bus121	0.087	0.042	16.4	89.9	
								Bus120	0.875	0.434	165.2	89.6	
								Bus119	0.039	0.022	7.5	87.2	
								Bus47	0.021	0.010	3.9	89.4	
								S002-TR-04DE001_PRI	0.213	0.110	40.5	88.8	
S002-SB-1AD001 [BUS B]	3.300	103.461	-4.6	0.000	0.000	0.000	0.000	Bus25	-2.848	-1.575	550.3	87.5	
								Bus29	0.332	0.212	66.6	84.2	
								Bus32	0.212	0.145	43.4	82.6	
								Bus133	0.183	0.105	35.6	86.8	
								Bus134	0.185	0.105	36.0	87.0	
								Bus135	0.018	0.010	3.5	87.1	
								Bus136	0.018	0.010	3.5	87.1	
								Bus137	0.192	0.109	37.3	87.1	
								Bus138	0.192	0.109	37.3	87.1	
								Bus139	0.019	0.010	3.6	87.7	
								Bus140	0.019	0.010	3.6	87.7	
								Bus141	0.200	0.115	39.0	86.8	
								Bus142	0.020	0.011	3.8	87.2	
								Bus143	0.087	0.042	16.4	89.9	
								Bus144	0.875	0.434	165.2	89.6	
								Bus145	0.087	0.042	16.4	90.0	
								Bus146	0.209	0.106	39.6	89.2	
								S002-TR-04DE002_PRI	0.000	-0.001	0.1	0.0	
S002-SB-1AN001 [BUS A]	0.380	99.911	-6.6	0.000	0.000	0.053	0.083	Bus158	0.003	0.002	5.5	90.1	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								S002-PC1-1A	-0.094	-0.103	211.4	67.5	
								Bus91	0.037	0.018	62.6	90.3	
S002-SB-1AN001 [BUS B]	0.380	102.458	-5.3	0.000	0.000	0.013	0.008	Bus152	0.001	0.001	1.8	90.1	
								S002-PC1-1B	-0.014	-0.009	24.5	85.4	
S002-SB-01E001 [BUS A]	0.380	101.731	-4.9	0.000	0.000	0.645	0.368	Bus10	-0.793	-0.434	1350.8	87.7	
								S002-SB-PSS01N01 [BUS A]	0.094	0.027	146.7	96.1	
								S002-MCC1-4 [BUS]	0.054	0.038	98.9	81.4	
S002-SB-01E001 [BUS B]	0.380	100.610	-5.0	0.000	0.000	0.637	0.353	Bus23	-0.790	-0.454	1376.0	86.7	
								S002-SB-PSS01N01 [BUS B]	0.095	0.060	170.0	84.7	
								S002-MCC1-6 [BUS]	0.057	0.041	106.4	81.2	
								S002-830-1601-N-P-1~	0.000	0.000	0.0	0.0	
S002-SB-02N001 [BUS A]	0.380	103.946	-4.2	0.000	0.000	0.310	0.208	Bus27	-0.319	-0.215	563.0	82.9	
								Area Lighting DB (A8)	0.009	0.007	16.7	80.3	
S002-SB-02N001 [BUS B]	0.380	103.729	-3.8	0.000	0.000	0.172	0.116	Bus24	-0.172	-0.116	303.7	83.0	
S002-SB-03E001 [BUS A]	0.380	104.366	-4.5	0.000	0.000	0.490	0.333	Bus28	-0.490	-0.333	862.9	82.7	
S002-SB-03E001 [BUS B]	0.380	102.825	-4.7	0.000	0.000	0.559	0.419	Bus33	-0.559	-0.419	1032.2	80.0	
S002-SB-03N001 [BUS A]	0.380	100.808	-5.0	0.000	0.000	0.892	0.651	Bus67	-0.892	-0.651	1664.3	80.8	
S002-SB-03N001 [BUS B]	0.380	101.194	-4.3	0.000	0.000	0.513	0.378	Bus69	-0.513	-0.378	956.4	80.5	
S002-SB-04E001 [BUS A]	0.380	103.699	-6.1	0.000	0.000	0.160	0.081	Bus157	0.003	0.002	5.8	80.1	
								S002-TR-04DE001_PRI	-0.209	-0.105	343.3	89.3	
								S002-SB-04E001 [BUS B]	0.046	0.022	75.0	90.4	
S002-SB-04E001 [BUS B]	0.380	103.699	-6.1	0.000	0.000	0.046	0.022	S002-SB-04E001 [BUS A]	-0.046	-0.022	75.0	90.4	
S002-SB-05N001A	0.380	103.999	-5.2	0.000	0.000	0.266	0.117	S002-SB-1AC001 [BUS A]	-0.893	-0.495	1491.6	87.5	
								TYPICAL MOTOR FEEDER 1 A	0.627	0.378	1069.5	85.6	
S002-SB-05N001B	0.380	102.108	-5.6	0.000	0.000	0.421	0.236	S002-SB-1AC001 [BUS B]	-1.122	-0.660	1937.1	86.2	
								TYPICAL MOTOR FEEDER 1 B	0.626	0.378	1088.6	85.6	
								POWER SOCKET FEEDER 1 B	0.075	0.046	130.6	85.0	
S002-SB-PSS1A01 [BUS-A]	132.000	100.000	0.0	0.000	0.000	0.000	0.000	Bus79	28.642	14.004	139.4	89.8	
								Bus80	0.000	0.000	0.0	0.0	
								Bus81	27.937	12.242	133.4	91.6	
								Bus83	0.000	0.000	0.0	0.0	
								Bus84	-56.580	-26.246	272.8	90.7	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus87	0.000	0.000	0.0	0.0	
								Bus88	0.000	0.000	0.0	0.0	
S002-SB-PSS01N01 [BUS A]	0.380	101.033	-5.4	0.000	0.000	0.094	0.026	S002-SB-01E001 [BUS A]	-0.094	-0.026	146.8	96.3	
S002-SB-PSS01N01 [BUS B]	0.380	99.573	-5.3	0.000	0.000	0.095	0.059	S002-SB-01E001 [BUS B]	-0.095	-0.059	170.1	85.0	
S002-TR-04DE001_PRI	3.300	103.348	-5.3	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS A]	-0.213	-0.111	40.6	88.7	
								S002-SB-04E001 [BUS A]	0.213	0.111	40.6	88.7	2.500
S002-TR-04DE002_PRI	3.300	103.462	-4.6	0.000	0.000	0.000	0.000	S002-SB-1AD001 [BUS B]	0.000	0.000	0.0	0.0	
S003-DB11E001-01	0.380	101.529	-4.4	0.000	0.000	0.028	0.021	S003-SB-11E001 [BUS A]	-0.028	-0.021	51.6	80.0	
S003-DB11E001-02	0.380	100.301	-4.9	0.000	0.000	0.028	0.020	S003-SB-11E001 [BUS B]	-0.028	-0.020	52.0	80.4	
S003-DB-11N001-03 [BUS A]	0.380	102.300	-4.2	0.000	0.000	0.020	0.015	S003-SB-11N001 [BUS A]	-0.020	-0.015	37.5	80.0	
S003-DB-11N001-03 [BUS B]	0.380	100.891	-4.7	0.000	0.000	0.025	0.019	S003-SB-11N001 [BUS B]	-0.025	-0.019	47.7	80.0	
S003-DB11N001-04 [BUS A]	0.380	102.287	-4.2	0.000	0.000	0.021	0.016	S003-SB-11N001 [BUS A]	-0.021	-0.016	39.4	80.0	
S003-DB11N001-04 [BUS B]	0.380	100.917	-4.7	0.000	0.000	0.023	0.017	S003-SB-11N001 [BUS B]	-0.023	-0.017	43.8	80.0	
S003-DB11N001-11	0.380	99.445	-4.5	0.000	0.000	0.088	0.043	S003-SB-11N001 [BUS B]	-0.088	-0.043	149.4	90.0	
S003-DB11N001-12	0.380	100.506	-4.7	0.000	0.000	0.036	0.017	S003-SB-11N001 [BUS B]	-0.036	-0.017	60.2	90.0	
S003-MDP-B	0.380	101.205	-4.7	0.000	0.000	0.315	0.237	S003-SB-11N001 [BUS A]	-0.315	-0.237	592.0	80.0	
S003-SB01E1	0.380	99.278	-5.4	0.000	0.000	0.136	0.102	S003-SB-11N001 [BUS B]	-0.136	-0.102	260.9	80.0	
S003-SB-11C001 [BUS A]	11.000	98.959	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	-3.648	-2.491	234.3	82.6	
								Bus56	0.657	0.484	43.3	80.5	
								Bus19	0.483	0.315	30.6	83.8	
								Bus168	0.304	0.163	18.3	88.1	
								Bus66	2.204	1.529	142.3	82.2	
S003-SB-11C001 [BUS B]	11.000	97.833	-3.7	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	-4.423	-1.786	255.9	92.7	
								Bus55	0.758	0.519	49.3	82.5	
								Bus20	0.513	0.359	33.6	81.9	
								Bus163	0.942	0.691	62.7	80.6	
								Bus78	2.210	0.216	119.2	99.5	
S003-SB-11D001 [BUS A]	3.300	104.870	-3.8	0.000	0.000	0.303	0.161	Bus50	-0.303	-0.161	57.3	88.3	
S003-SB-11D001 [BUS B]	3.300	102.279	-4.8	0.000	0.000	0.276	0.135	Bus49	-0.938	-0.662	196.4	81.7	
								Bus37	0.662	0.527	144.8	78.2	
S003-SB-11E001 [BUS A]	0.380	101.925	-4.5	0.000	0.000	0.450	0.279	Bus166	-0.478	-0.300	841.4	84.7	
								S003-DB11E001-01	0.028	0.021	51.6	80.1	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
S003-SB-11E001 [BUS B]	0.380	100.665	-5.0	0.000	0.000	0.481	0.320	Bus167	-0.509	-0.341	924.1	83.1	
								S003-DB11E001-02	0.028	0.020	52.0	80.5	
								S003-830-2000-E-P~	0.000	0.000	0.0	0.0	
S003-SB-11N001 [BUS A]	0.380	102.562	-4.3	0.000	0.000	0.295	0.195	S003-DB-11N001-03 [BUS A]	0.020	0.015	37.5	80.1	
								S003-DB11N001-04 [BUS A]	0.021	0.016	39.4	80.1	
								S003-MDP-B	0.318	0.242	592.0	79.6	
								Bus56	-0.654	-0.468	1191.4	81.4	
S003-SB-11N001 [BUS B]	0.380	101.223	-4.8	0.000	0.000	0.441	0.295	S003-SB01E1	0.138	0.106	260.9	79.4	
								S003-DB-11N001-03 [BUS B]	0.025	0.019	47.7	80.1	
								S003-DB11N001-04 [BUS B]	0.023	0.017	43.8	80.1	
								S003-DB11N001-11	0.090	0.043	149.4	90.2	
								S003-DB11N001-12	0.036	0.017	60.2	90.1	
								Bus55	-0.754	-0.498	1356.1	83.5	
TR-PSS1AC01_11kV BUS	11.000	101.265	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS A]	28.574	13.005	1627.2	91.0	
								TR-PSS1AC01_132kV BUS	-28.574	-13.005	1627.2	91.0	
TR-PSS1AC01_132kV BUS	132.000	99.993	0.0	0.000	0.000	0.000	0.000	Bus79	-28.641	-15.168	141.8	88.4	
								TR-PSS1AC01_11kV BUS	28.641	15.168	141.8	88.4	
TR-PSS1AC02_11kV BUS	11.000	100.318	-3.4	0.000	0.000	0.000	0.000	S002-SB-1AC001 [BUS B]	27.874	11.380	1575.2	92.6	
								TR-PSS1AC02_132kV BUS	-27.874	-11.380	1575.2	92.6	
TR-PSS1AC02_132kV BUS	132.000	99.993	0.0	0.000	0.000	0.000	0.000	Bus81	-27.936	-13.407	135.5	90.2	
								TR-PSS1AC02_11kV BUS	27.936	13.407	135.5	90.2	1.250
TYPICAL MOTOR FEEDER 1 A	0.380	103.999	-5.2	0.000	0.000	0.000	0.000	Bus9	0.042	0.025	71.5	85.7	
								Bus12	0.042	0.025	71.4	85.7	
								Bus53	0.042	0.025	71.4	85.7	
								Bus54	0.042	0.025	71.4	85.7	
								Bus85	0.042	0.025	71.3	85.7	
								Bus149	0.042	0.025	71.3	85.6	
								Bus151	0.042	0.025	71.3	85.7	
								Bus154	0.042	0.025	71.3	85.6	
								Bus161	0.042	0.025	71.2	85.6	
								Bus162	0.042	0.025	71.3	85.6	
								Bus164	0.042	0.025	71.2	85.6	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus165	0.042	0.025	71.2	85.6	
								Bus170	0.042	0.025	71.2	85.6	
								Bus171	0.042	0.025	71.2	85.6	
								Bus172	0.042	0.025	71.2	85.6	
								S002-SB-05N001A	-0.627	-0.378	1069.5	85.6	
TYPICAL MOTOR FEEDER 1 B	0.380	102.108	-5.6	0.000	0.000	0.000	0.000	Bus173	0.042	0.025	72.4	85.6	
								Bus174	0.042	0.025	72.5	85.6	
								Bus175	0.042	0.025	72.5	85.6	
								Bus176	0.042	0.025	72.5	85.6	
								Bus177	0.042	0.025	72.5	85.6	
								Bus178	0.042	0.025	72.6	85.6	
								Bus179	0.042	0.025	72.5	85.6	
								Bus180	0.042	0.025	72.6	85.6	
								Bus181	0.042	0.025	72.6	85.6	
								Bus182	0.042	0.025	72.6	85.6	
								Bus183	0.042	0.025	72.6	85.6	
								Bus184	0.042	0.025	72.7	85.7	
								Bus185	0.042	0.025	72.6	85.6	
								Bus186	0.042	0.025	72.7	85.7	
								Bus187	0.042	0.025	72.7	85.7	
								S002-SB-05N001B	-0.626	-0.378	1088.6	85.6	
Cable40~	0.434	94.408	-6.0	0.000	0.000	0.000	0.000	Bus_MDB-8	-0.253	-0.159	421.6	84.7	
								Bus2	0.253	0.159	421.6	84.7	
Cable41~	0.434	94.354	-6.0	0.000	0.000	0.000	0.000	Bus_MDB-8	-0.253	-0.159	421.8	84.7	
								Bus4	0.253	0.159	421.8	84.7	
Cable153~	0.434	94.570	-6.0	0.000	0.000	0.000	0.000	Bus_MDB-5	-0.253	-0.159	420.8	84.7	
								Bus57	0.253	0.159	420.8	84.7	
Cable154~	0.434	94.516	-6.0	0.000	0.000	0.000	0.000	Bus_MDB-5	-0.253	-0.159	421.1	84.7	
								Bus59	0.253	0.159	421.1	84.7	
Cable163~	0.434	94.763	-5.9	0.000	0.000	0.000	0.000	Bus_MDB-6	-0.253	-0.159	420.0	84.7	
								Bus68	0.253	0.159	420.0	84.7	
Cable164~	0.434	94.699	-5.9	0.000	0.000	0.000	0.000	Bus_MDB-6	-0.253	-0.159	420.3	84.7	
								Bus70	0.253	0.159	420.3	84.7	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
Cable168~	0.434	94.623	-5.9	0.000	0.000	0.000	0.000	Bus_MDB-7	-0.253	-0.159	420.6	84.7	
								Bus74	0.253	0.159	420.6	84.7	
Cable169~	0.434	94.505	-6.0	0.000	0.000	0.000	0.000	Bus_MDB-7	-0.253	-0.159	421.1	84.7	
								Bus75	0.253	0.159	421.1	84.7	
Cable188~	0.434	96.647	-6.7	0.000	0.000	0.253	0.000	Bus_MDB-1	-0.253	0.000	348.8	100.0	
								VFD25	0.253	0.000	348.8	100.0	
Cable189~	0.434	96.615	-6.8	0.000	0.000	0.253	0.000	Bus_MDB-1	-0.253	0.000	348.9	100.0	
								VFD26	0.253	0.000	348.9	100.0	
Cable192~	0.434	96.564	-6.8	0.000	0.000	0.253	0.000	Bus_MDB-2	-0.253	0.000	349.1	100.0	
								VFD27	0.253	0.000	349.1	100.0	
Cable193~	0.434	96.500	-6.8	0.000	0.000	0.253	0.000	Bus_MDB-2	-0.253	0.000	349.3	100.0	
								VFD28	0.253	0.000	349.3	100.0	
Cable196~	0.434	96.455	-6.9	0.000	0.000	0.253	0.000	Bus_MDB-3	-0.253	0.000	349.5	100.0	
								VFD29	0.253	0.000	349.5	100.0	
Cable197~	0.434	96.565	-6.8	0.000	0.000	0.253	0.000	Bus_MDB-3	-0.253	0.000	349.1	100.0	
								VFD30	0.253	0.000	349.1	100.0	
Cable200~	0.434	96.681	-6.7	0.000	0.000	0.253	0.000	Bus_MDB-4	-0.253	0.000	348.7	100.0	
								VFD31	0.253	0.000	348.7	100.0	
Cable201~	0.434	96.642	-6.8	0.000	0.000	0.253	0.000	Bus_MDB-4	-0.253	0.000	348.8	100.0	
								VFD32	0.253	0.000	348.8	100.0	
S002-830-1601-N-P-1~	0.418	91.424	-5.0	0.000	0.000	0.000	0.000	S002-SB-01E001 [BUS B]	0.000	0.000	0.0	0.0	
S003-830-2000-E-P~	0.418	91.474	-5.0	0.000	0.000	0.000	0.000	S003-SB-11E001 [BUS B]	0.000	0.000	0.0	0.0	
S003-Cable43~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus58	0.000	0.000	0.0	0.0	
S003-Cable45~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus61	0.000	0.000	0.0	0.0	
S003-Cable120~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus71	0.000	0.000	0.0	0.0	
S002-Cable23~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus41	0.000	0.000	0.0	0.0	
S002-830-1072-E-P~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus35	0.000	0.000	0.0	0.0	
Cable84~	0.380	100.000	0.0	0.000	0.000	0.000	0.000	Bus11	0.000	0.000	0.0	0.0	
S002-TR-CMP1_VFD~2	11.500	96.516	-7.1	0.000	0.000	2.932	0.000	S002-CMP1_VFD	2.932	0.000	152.5	100.0	
								S002-TR-CMP1_VFD~3	-2.932	0.000	152.5	100.0	
								&S002-SB-1AC001 [BUS A]					
S002-TR-CMP1_VFD~3	11.500	96.516	-7.1	0.000	0.000	2.932	0.000	S002-CMP1_VFD	2.932	0.000	152.5	100.0	
								S002-SB-1AC001 [BUS A]	-2.932	0.000	152.5	100.0	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								&S002-TR-CMP1_VFD~2					
S002-TR-CMP2_VFD~2	11.500	95.600	-7.1	0.000	0.000	2.932	0.000	S002-CMP2_VFD	2.932	0.000	154.0	100.0	
								S002-TR-CMP2_VFD~3	-2.932	0.000	154.0	100.0	
								&S002-SB-1AC001 [BUS B]					
S002-TR-CMP2_VFD~3	11.500	95.600	-7.1	0.000	0.000	2.932	0.000	S002-CMP2_VFD	2.932	0.000	154.0	100.0	
								S002-SB-1AC001 [BUS B]	-2.932	0.000	154.0	100.0	
								&S002-TR-CMP2_VFD~2					
S002-CMP1_VFD	11.000	100.000	0.0	5.570	3.308	0.000	0.000	BUS-MTR1_ENCRYO	5.570	3.308	340.0	86.0	
S002-CMP2_VFD	11.000	100.000	0.0	5.570	3.308	0.000	0.000	BUS-MTR2_ENCRYO	5.570	3.308	340.0	86.0	
VFD25	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus93	0.253	0.159	416.1	84.7	
VFD26	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus94	0.253	0.159	416.1	84.7	
VFD27	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus97	0.253	0.159	416.1	84.7	
VFD28	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus98	0.253	0.159	416.1	84.7	
VFD29	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus101	0.253	0.159	416.1	84.7	
VFD30	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus102	0.253	0.159	416.1	84.7	
VFD31	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus105	0.253	0.159	416.1	84.7	
VFD32	0.380	109.211	0.0	0.253	0.159	0.000	0.000	Bus106	0.253	0.159	416.1	84.7	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

\# Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
BUS-MTR1_ENCRYO	11.000										6.478	86.0	340.0	
BUS-MTR2_ENCRYO	11.000										6.478	86.0	340.0	
Bus93	0.380										0.299	84.7	416.1	
Bus94	0.380										0.299	84.7	416.1	
Bus97	0.380										0.299	84.7	416.1	
Bus98	0.380										0.299	84.7	416.1	
Bus101	0.380										0.299	84.7	416.1	
Bus102	0.380										0.299	84.7	416.1	
Bus105	0.380										0.299	84.7	416.1	
Bus106	0.380										0.299	84.7	416.1	
132kV GRID	132.000										62.371	90.7	272.8	
200A MDB Agreko	0.415		0.017	0.008	0.005	0.002					0.025	90.0	33.6	
Area Lighting DB (A8)	0.380				0.009	0.007					0.011	80.0	16.7	
Bus1	11.000										5.258	84.7	272.6	
Bus2	0.380		0.253	0.159							0.299	84.7	421.6	
Bus4	0.380		0.253	0.159							0.299	84.7	421.8	
Bus7	11.000										0.391	82.2	20.3	
Bus8	11.000										0.648	79.6	33.9	
Bus9	0.380		0.040	0.025							0.047	85.0	71.5	
Bus10	0.380										0.907	87.7	1350.7	
Bus11	0.380											-	-	
Bus12	0.380		0.040	0.025							0.047	85.0	71.4	
Bus14	0.380		0.016	0.012							0.020	80.0	27.9	
Bus15	0.380		0.016	0.012							0.020	80.0	27.9	
Bus17	11.000										0.605	81.6	31.4	
Bus18	11.000										0.717	78.7	37.5	
Bus19	11.000										0.577	83.7	30.6	
Bus20	11.000										0.626	81.9	33.6	
Bus21	11.000										3.297	86.5	172.6	
Bus22	3.300										5.147	86.4	869.2	
Bus23	0.380										0.914	86.6	1376.0	
Bus24	0.380										0.207	83.0	303.7	
Bus25	3.300										3.256	87.5	550.2	
Bus27	0.380										0.385	82.9	563.0	
Bus28	0.380										0.595	82.6	862.9	
Bus29	3.450										0.394	84.2	66.6	
Bus30	3.450										0.904	82.1	152.9	
Bus31	3.450										1.182	80.2	200.1	
Bus32	3.450										0.257	82.6	43.4	

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
Bus33	0.380										0.702	79.8	1032.2	
Bus35	0.380											-	-	
Bus36	0.400		0.512	0.384	0.135	0.101					0.808	80.0	1197.4	
Bus37	3.300										0.845	78.0	145.1	
Bus38	0.400										2.578	84.3	3589.5	
Bus40	0.400										0.857	84.4	1196.5	
Bus41	0.380												-	
Bus42	11.000										0.933	85.4	48.8	
Bus43	11.000										0.924	86.5	47.9	
Bus45	0.400										0.857	84.4	1196.5	
Bus46	11.000										0.209	82.6	10.9	
Bus47	3.300		0.021	0.011							0.023	89.0	3.9	
Bus48	0.400										0.857	84.4	1196.5	
Bus49	3.300										1.149	81.7	196.4	
Bus50	3.300										0.344	88.3	57.3	
Bus51	11.000										0.774	93.6	40.3	
Bus52	0.400										0.727	99.9	997.8	
Bus53	0.380		0.040	0.025							0.047	85.0	71.4	
Bus54	0.380		0.040	0.025							0.047	85.0	71.4	
Bus55	11.000										0.919	82.5	49.3	
Bus56	11.000										0.817	80.5	43.3	
Bus57	0.380		0.253	0.159							0.299	84.7	420.8	
Bus58	0.380											-	-	
Bus59	0.380		0.253	0.159							0.299	84.7	421.1	
Bus60	0.380		0.016	0.012							0.020	80.0	27.8	
Bus61	0.380											-	-	
Bus62	0.380		0.016	0.012							0.020	80.0	27.9	
Bus63	0.400										0.727	99.9	997.8	
Bus64	0.415										0.733	99.9	1005.9	
Bus65	0.400										2.192	99.9	3001.6	
Bus66	11.000										2.681	82.0	142.5	
Bus67	0.380										1.108	80.7	1664.3	
Bus68	0.380		0.253	0.159							0.299	84.7	420.0	
Bus69	0.380										0.638	80.4	956.4	
Bus70	0.380		0.253	0.159							0.299	84.7	420.3	
Bus71	0.380												-	
Bus72	0.380		0.016	0.012							0.020	80.0	27.8	
Bus73	0.380		0.016	0.012							0.020	80.0	27.8	
Bus74	0.380		0.253	0.159							0.299	84.7	420.6	
Bus75	0.380		0.253	0.159							0.299	84.7	421.1	
Bus76	0.380		0.016	0.012							0.020	80.0	27.8	
Bus77	0.380		0.016	0.012							0.020	80.0	27.9	

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Percent Loading
Bus78	11.000										2.218	99.5	119.2	
Bus79	132.000										31.882	89.8	139.4	
Bus80	132.000													
Bus81	132.000										30.502	91.6	133.4	
Bus82	0.415										0.025	90.0	33.6	
Bus83	132.000													
Bus84	132.000										62.371	90.7	272.8	
Bus85	0.380		0.040	0.025							0.047	85.0	71.3	
Bus87	132.000													
Bus88	132.000													
Bus89	11.000		0.053	0.032							0.062	85.4	3.2	
Bus90	11.000		0.531	0.323							0.622	85.4	32.2	
Bus91	0.380				0.036	0.018					0.040	90.0	62.6	
Bus92	11.000		6.532	3.057							7.212	90.6	374.2	
Bus95	0.400		0.016	0.012							0.020	80.0	27.3	
Bus96	0.400		0.016	0.012							0.020	80.0	27.3	
Bus99	0.400		0.016	0.012							0.020	80.0	27.3	
Bus100	0.400		0.016	0.012							0.020	80.0	27.3	
Bus103	0.400		0.016	0.012							0.020	80.0	27.4	
Bus104	0.400		0.016	0.012							0.020	80.0	27.3	
Bus107	0.400		0.016	0.012							0.020	80.0	27.2	
Bus108	0.400		0.016	0.012							0.020	80.0	27.3	
Bus110	11.000		0.973	0.471							1.082	90.0	56.1	
Bus112	11.000										1.137	79.1	59.0	
Bus113	11.000		0.098	0.052							0.111	88.4	5.8	
Bus114	11.000		6.527	3.054							7.206	90.6	377.4	
Bus115	11.000		0.978	0.516							1.105	88.4	57.9	
Bus117	11.000		0.531	0.323							0.622	85.4	32.5	
Bus119	3.300		0.038	0.022							0.044	87.0	7.5	
Bus120	3.300		0.871	0.434							0.973	89.5	165.3	
Bus121	3.300		0.087	0.043							0.097	89.5	16.5	
Bus122	3.300		0.087	0.043							0.097	89.5	16.5	
Bus123	3.300		0.020	0.011							0.023	86.5	3.9	
Bus124	3.300		0.020	0.011							0.023	86.5	3.9	
Bus125	3.300		0.019	0.011							0.022	86.5	3.7	
Bus126	3.300		0.019	0.011							0.022	86.5	3.7	
Bus127	3.300		0.019	0.011							0.022	86.5	3.7	
Bus128	3.300		0.187	0.108							0.216	86.5	36.9	
Bus129	3.300		0.018	0.010							0.021	86.5	3.5	
Bus130	3.300		0.018	0.010							0.021	86.5	3.5	
Bus131	3.300		0.018	0.010							0.021	86.5	3.5	
Bus132	3.300		0.181	0.105							0.209	86.5	35.7	

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
Bus133	3.300		0.181	0.105							0.209	86.5	35.7	
Bus134	3.300		0.183	0.105							0.211	86.7	36.0	
Bus135	3.300		0.018	0.010							0.021	86.5	3.5	
Bus136	3.300		0.018	0.010							0.021	86.5	3.5	
Bus137	3.300		0.190	0.109							0.219	86.8	37.4	
Bus138	3.300		0.190	0.109							0.219	86.8	37.4	
Bus139	3.300		0.019	0.011							0.022	86.5	3.7	
Bus140	3.300		0.019	0.011							0.022	86.5	3.7	
Bus141	3.300		0.198	0.115							0.229	86.5	39.0	
Bus142	3.300		0.020	0.011							0.023	86.5	3.9	
Bus143	3.300		0.087	0.043							0.097	89.5	16.5	
Bus144	3.300		0.871	0.434							0.973	89.5	165.3	
Bus145	3.300		0.087	0.043							0.097	89.5	16.5	
Bus146	3.300		0.208	0.106							0.233	89.1	39.6	
Bus149	0.380		0.040	0.025							0.047	85.0	71.3	
Bus151	0.380		0.040	0.025							0.047	85.0	71.3	
Bus152	0.380		0.001	0.001							0.001	90.0	1.8	
Bus154	0.380		0.040	0.025							0.047	85.0	71.3	
Bus157	0.380				0.003	0.002					0.004	80.0	5.8	
Bus158	0.380		0.003	0.002							0.004	90.0	5.5	
Bus159	0.380		0.016	0.007							0.018	92.0	26.8	
Bus160	3.450		0.901	0.410							0.990	91.0	168.2	
Bus161	0.380		0.040	0.025							0.047	85.0	71.2	
Bus162	0.380		0.040	0.025							0.047	85.0	71.3	
Bus163	11.000										1.169	80.6	62.7	
Bus164	0.380		0.040	0.025							0.047	85.0	71.2	
Bus165	0.380		0.040	0.025							0.047	85.0	71.2	
Bus166	0.380										0.567	84.6	841.4	
Bus167	0.380										0.615	83.0	924.1	
Bus168	11.000										0.345	88.0	18.3	
Bus169	0.380		0.016	0.007							0.018	92.0	27.6	
Bus170	0.380		0.040	0.025							0.047	85.0	71.2	
Bus171	0.380		0.040	0.025							0.047	85.0	71.2	
Bus172	0.380		0.040	0.025							0.047	85.0	71.2	
Bus173	0.380		0.040	0.025							0.047	85.0	72.4	
Bus174	0.380		0.040	0.025							0.047	85.0	72.5	
Bus175	0.380		0.040	0.025							0.047	85.0	72.5	
Bus176	0.380		0.040	0.025							0.047	85.0	72.5	
Bus177	0.380		0.040	0.025							0.047	85.0	72.5	
Bus178	0.380		0.040	0.025							0.047	85.0	72.6	
Bus179	0.380		0.040	0.025							0.047	85.0	72.5	
Bus180	0.380		0.040	0.025							0.047	85.0	72.6	

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
Bus181	0.380		0.040	0.025							0.047	85.0	72.6	
Bus182	0.380		0.040	0.025							0.047	85.0	72.6	
Bus183	0.380		0.040	0.025							0.047	85.0	72.6	
Bus184	0.380		0.040	0.025							0.047	85.0	72.7	
Bus185	0.380		0.040	0.025							0.047	85.0	72.6	
Bus186	0.380		0.040	0.025							0.047	85.0	72.7	
Bus187	0.380		0.040	0.025							0.047	85.0	72.7	
Bus188	0.380				0.025	0.015					0.029	85.0	43.4	
Bus189	0.380				0.025	0.015					0.029	85.0	43.6	
Bus190	0.380				0.025	0.015					0.029	85.0	43.6	
Bus_LVDB-1	0.415	4200.0									2.163	99.9	2970.6	70.7
Bus_LVDB-2	0.415	4200.0									2.567	84.4	3589.5	85.5
Bus_MDB-1	0.415	2000.0									0.539	99.9	742.2	37.1
Bus_MDB-2	0.415	2000.0									0.540	99.9	743.1	37.2
Bus_MDB-3	0.415	2000.0									0.540	99.9	743.3	37.2
Bus_MDB-4	0.415	2000.0									0.540	99.9	742.0	37.1
Bus_MDB-5	0.415	2000.0									0.640	84.4	897.5	44.9
Bus_MDB-6	0.400	2000.0									0.639	84.4	895.7	44.8
Bus_MDB-7	0.400	2000.0									0.640	84.4	897.3	44.9
Bus_MDB-8	0.400	2000.0									0.641	84.4	899.0	45.0
BUS_S002-CMP3	11.000		3.299	1.691							3.708	89.0	194.0	
CAMP BUS	11.000		1.800	0.950	0.453	0.239					2.548	88.4	133.3	
LV SWBD_TOFS	0.400	2000.0	0.138	0.047	0.579	0.198					0.757	94.6	1065.7	53.3
MV RMU-TOFS	11.000	630.0									3.326	89.9	173.1	27.5
POWER SOCKET FEEDER 1 B	0.380										0.088	85.0	130.6	
S002-ASP1-1 [BUS A]	0.380	1200.0			0.094	0.058					0.111	85.0	166.4	13.9
S002-ASP1-1B [BUS B]	0.380	1200.0			0.315	0.195					0.371	85.0	552.4	46.0
S002-ASP1-2A	0.380	1200.0			0.334	0.241					0.412	81.0	627.9	52.3
S002-ASP1-2B	0.380	1200.0			0.016	0.012					0.037	80.6	54.7	4.6
S002-DBASP1-2-22	0.380	80.0	0.014	0.010							0.017	80.0	25.1	31.4
S002-MCC1-1 [BUS]	0.380	800.0	0.229	0.135							0.266	86.1	401.7	50.2
S002-MCC1-2 [BUS]	0.380	800.0											-	-
S002-MCC1-3 [BUS]	0.380	800.0	0.236	0.122							0.266	88.8	401.3	50.2
S002-MCC1-4 [BUS]	0.380	800.0	0.053	0.038							0.066	81.3	98.9	12.4
S002-MCC1-5 [BUS]	0.380	800.0	0.233	0.156							0.280	83.1	427.2	53.4
S002-MCC1-6 [BUS]	0.380	800.0	0.056	0.041							0.070	81.0	106.4	13.3
S002-MCC1-7 [BUS]	0.380	800.0	0.233	0.156							0.280	83.0	427.3	53.4
S002-MCC1-8 [BUS]	0.380	800.0	0.110	0.079							0.153	82.8	226.3	28.3
S002-MCC1-9 [BUS]	0.380	800.0	0.121	0.084							0.165	83.7	251.2	31.4
S002-PC1-1A	0.380	3000.0	0.081	0.061							0.881	83.6	1324.2	44.1
S002-PC1-1B	0.380	3000.0									0.390	84.8	576.8	19.2
S002-PC1-2A	0.380	3000.0									1.143	82.2	1733.3	57.8

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
S002-PC1-2B	0.380	3000.0	0.055	0.034							0.255	83.0	376.3	12.5
S002-SB-1AC001 [BUS A]	11.000	3150.0									31.398	91.0	1627.6	51.7
S002-SB-1AC001 [BUS B]	11.000	3150.0									30.111	92.6	1575.7	50.0
S002-SB-1AD001 [BUS A]	3.300	2500.0									5.143	86.4	869.3	34.8
S002-SB-1AD001 [BUS B]	3.300	2500.0									3.255	87.5	550.4	22.0
S002-SB-1AN001 [BUS A]	0.380	800.0	0.011	0.017	0.043	0.067					0.139	67.5	211.4	26.4
S002-SB-1AN001 [BUS B]	0.380	800.0	0.013	0.008							0.016	85.4	24.5	3.1
S002-SB-01E001 [BUS A]	0.380	3150.0	0.179	0.105	0.466	0.264					0.904	87.7	1350.8	42.9
S002-SB-01E001 [BUS B]	0.380	3150.0	0.106	0.060	0.531	0.293					0.911	86.7	1376.0	43.7
S002-SB-02N001 [BUS A]	0.380	2000.0	0.244	0.164	0.066	0.044					0.385	82.9	563.0	28.1
S002-SB-02N001 [BUS B]	0.380	2000.0	0.148	0.100	0.024	0.016					0.207	83.0	303.7	15.2
S002-SB-03E001 [BUS A]	0.380	2500.0	0.397	0.271	0.093	0.063					0.593	82.7	862.9	34.5
S002-SB-03E001 [BUS B]	0.380	2500.0	0.373	0.279	0.186	0.140					0.699	80.0	1032.2	41.3
S002-SB-03N001 [BUS A]	0.380	3150.0	0.621	0.454	0.271	0.198					1.104	80.8	1664.3	52.8
S002-SB-03N001 [BUS B]	0.380	3150.0	0.356	0.263	0.156	0.115					0.637	80.5	956.4	30.4
S002-SB-04E001 [BUS A]	0.380	800.0	0.142	0.074	0.017	0.007					0.234	89.3	343.3	42.9
S002-SB-04E001 [BUS B]	0.380	400.0	0.031	0.010	0.016	0.012					0.051	90.4	75.0	18.8
S002-SB-05N001A	0.380	4000.0	0.078	0.048	0.188	0.069					1.021	87.5	1491.6	37.3
S002-SB-05N001B	0.380	4000.0	0.395	0.226	0.026	0.010					1.302	86.2	1937.1	48.4
S002-SB-PSS1A01 [BUS-A]	132.000	2000.0									62.371	90.7	272.8	13.6
S002-SB-PSS01N01 [BUS A]	0.380	630.0	0.094	0.026							0.098	96.3	146.8	23.3
S002-SB-PSS01N01 [BUS B]	0.380	630.0	0.095	0.059							0.111	85.0	170.1	27.0
S002-TR-04DE001_PRI	3.300										0.240	88.7	40.6	
S002-TR-04DE002_PRI	3.300												-	
S003-DB11E001-01	0.380	100.0	0.005	0.004	0.022	0.017					0.034	80.0	51.6	51.6
S003-DB11E001-02	0.380	100.0	0.005	0.004	0.022	0.016					0.034	80.4	52.0	52.0
S003-DB-11N001-03 [BUS A]	0.380	125.0	0.020	0.015							0.025	80.0	37.5	30.0
S003-DB-11N001-03 [BUS B]	0.380	125.0	0.020	0.015	0.005	0.004					0.032	80.0	47.7	38.1
S003-DB11N001-04 [BUS A]	0.380	125.0	0.017	0.013	0.004	0.003					0.026	80.0	39.4	31.5
S003-DB11N001-04 [BUS B]	0.380	125.0	0.019	0.014	0.005	0.004					0.029	80.0	43.8	35.1
S003-DB11N001-11	0.380				0.088	0.043					0.098	90.0	149.4	
S003-DB11N001-12	0.380				0.036	0.017					0.040	90.0	60.2	
S003-MDP-B	0.380	1000.0			0.315	0.237					0.394	80.0	592.0	59.2
S003-SB01E1	0.380	630.0	0.109	0.082	0.027	0.020					0.171	80.0	260.9	41.4
S003-SB-11C001 [BUS A]	11.000	1250.0									4.417	82.6	234.3	18.7
S003-SB-11C001 [BUS B]	11.000	1250.0									4.770	92.7	255.9	20.5
S003-SB-11D001 [BUS A]	3.300	1250.0	0.303	0.161							0.344	88.3	57.3	4.6
S003-SB-11D001 [BUS B]	3.300	1250.0	0.276	0.135							1.148	81.7	196.4	15.7
S003-SB-11E001 [BUS A]	0.380	2500.0	0.358	0.222	0.093	0.058					0.564	84.7	841.4	33.7
S003-SB-11E001 [BUS B]	0.380	2500.0	0.415	0.274	0.066	0.046					0.612	83.1	924.1	37.0
S003-SB-11N001 [BUS A]	0.380	4000.0	0.237	0.156	0.058	0.039					0.804	81.4	1191.4	29.8
S003-SB-11N001 [BUS B]	0.380	4000.0	0.371	0.246	0.070	0.049					0.903	83.5	1356.1	33.9

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Percent Loading
TR-PSS1AC01_11kV BUS	11.000										31.395	91.0	1627.2	
TR-PSS1AC01_132kV BUS	132.000										32.409	88.4	141.8	
TR-PSS1AC02_11kV BUS	11.000										30.107	92.6	1575.2	
TR-PSS1AC02_132kV BUS	132.000										30.986	90.2	135.5	
TYPICAL MOTOR FEEDER 1 A	0.380										0.732	85.6	1069.5	
TYPICAL MOTOR FEEDER 1 B	0.380										0.732	85.6	1088.6	

* Indicates operating load of a bus exceeds the bus critical limit (106.0% of the Continuous Ampere rating).
\# Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
					Capability (MVA)	Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%		MVA	%	MVA	%
Cable2	Cable	165.52	71.48	43.18					
Cable3	Cable	165.52	71.43	43.16					
Cable4	Cable	165.52	71.39	43.13					
Cable5	Cable	165.52	71.39	43.13					
Cable6	Cable	165.52	71.34	43.10					
Cable7	Cable	165.52	71.30	43.07					
Cable10	Cable	165.52	71.34	43.10					
Cable11	Cable	165.52	71.30	43.07					
Cable12	Cable	165.52	71.25	43.05					
Cable13	Cable	165.52	71.30	43.07					
Cable19	Cable	165.52	71.25	43.05					
Cable20	Cable	110.80	26.81	24.19					
Cable21	Cable	165.52	71.20	43.02					
Cable22	Cable	165.52	71.25	43.05					
Cable23	Cable	68.91	1.79	2.60					
Cable24	Cable	68.91	5.53	8.03					
Cable25	Cable	165.52	71.20	43.02					
Cable26	Cable	110.80	27.61	24.92					
Cable27	Cable	165.52	71.16	42.99					
Cable28	Cable	165.52	72.43	43.76					
Cable29	Cable	165.52	72.48	43.79					
Cable30	Cable	165.52	72.53	43.82					
Cable31	Cable	165.52	72.48	43.79					
Cable32	Cable	165.52	72.53	43.82					
Cable33	Cable	165.52	72.58	43.85					
Cable34	Cable	165.52	72.53	43.82					
Cable35	Cable	165.52	72.58	43.85					
Cable36	Cable	165.52	72.62	43.88					
Cable37	Cable	165.52	72.58	43.85					
Cable38	Cable	165.52	72.62	43.88					
Cable39	Cable	165.52	72.67	43.91					
Cable40	Cable	776.79	421.57	54.27					
Cable41	Cable	776.79	421.81	54.30					

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
Cable42	Cable	165.52	72.62	43.88					
Cable43	Cable	165.52	72.67	43.91					
Cable44	Cable	165.52	72.72	43.93					
Cable45	Cable	269.32	43.41	16.12					
Cable48	Cable	269.32	43.62	16.20					
Cable49	Cable	269.32	43.58	16.18					
Cable83	Cable	36.76	27.90	75.88					
Cable118	Cable	230.46	133.29	57.84					
Cable123	Cable	45.49	27.91	61.36					
Cable128	Cable	1553.58	899.03	57.87					
Cable134	Cable	1553.58	997.85	64.23					
Cable136	Cable	2718.76	997.85	36.70					
Cable137	Cable	1553.58	997.85	64.23					
Cable144	Cable	2718.76	1196.50	44.01					
Cable145	Cable	2718.76	1196.50	44.01					
Cable146	Cable	2718.76	1196.50	44.01					
Cable147	Cable	1553.58	1196.50	77.02					
Cable148	Cable	1553.58	1196.50	77.02					
Cable149	Cable	1553.58	1196.50	77.02					
Cable150	Cable	2718.76	997.85	36.70					
Cable151	Cable	1553.58	897.47	57.77					
Cable152	Cable	1553.58	974.94	62.75					
Cable153	Cable	776.79	420.85	54.18					
Cable154	Cable	776.79	421.09	54.21					
Cable155	Cable	42.45	27.84	65.58					
Cable156	Cable	42.45	27.86	65.62					
Cable157	Cable	2718.76	1005.89	37.00					
Cable158	Cable	1553.58	742.24	47.78					
Cable159	Cable	1553.58	743.09	47.83					
Cable160	Cable	1553.58	743.28	47.84					
Cable161	Cable	1553.58	742.00	47.76					
Cable162	Cable	1553.58	895.73	57.66					
Cable163	Cable	776.79	419.99	54.07					
Cable164	Cable	776.79	420.27	54.10					
Cable165	Cable	42.45	27.81	65.50					

CKT / Branch		Busway / Cable & Reactor			Transformer				
					Capability	Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	(MVA)	MVA	%	MVA	%
Cable166	Cable	42.45	27.83	65.55					
Cable167	Cable	1553.58	897.30	57.76					
Cable168	Cable	776.79	420.61	54.15					
Cable169	Cable	776.79	421.13	54.21					
Cable170	Cable	42.45	27.84	65.58					
Cable171	Cable	42.45	27.88	65.68					
Cable176	Cable	702.16	33.64	4.79					
Cable188	Cable	776.79	348.81	44.90					
Cable189	Cable	776.79	348.93	44.92					
Cable190	Cable	42.45	27.27	64.25					
Cable191	Cable	42.45	27.29	64.29					
Cable192	Cable	776.79	349.11	44.94					
Cable193	Cable	776.79	349.35	44.97					
Cable194	Cable	42.45	27.30	64.32					
Cable195	Cable	42.45	27.34	64.41					
Cable196	Cable	776.79	349.51	44.99					
Cable197	Cable	776.79	349.11	44.94					
Cable198	Cable	42.45	27.36	64.45					
Cable199	Cable	42.45	27.30	64.32					
Cable200	Cable	776.79	348.69	44.89					
Cable201	Cable	776.79	348.83	44.91					
Cable202	Cable	42.45	27.25	64.19					
Cable203	Cable	42.45	27.27	64.25					
C_688-KM-01	Cable	390.70	168.19	43.05					
S002-830-1063-N-P	Cable	4383.32	1664.27	37.97					
S002-830-1066-N-P	Cable	4383.32	956.37	21.82					
S002-830-1069-N-P	Cable	2629.99	862.86	32.81					
S002-830-1072-N-P	Cable	2629.99	1032.19	39.25					
S002-830-1420-N-P-1	Cable	4383.32	1350.75	30.82					
S002-830-1500-N-P-1	Cable	4383.32	1375.97	31.39					
S002-830-9001-A-P	Cable	618.47	135.54	21.92					
S002-830-9001-D-P-1	Cable	3506.65	869.32	24.79					
S002-830-9004-A-P	Cable	708.61	141.76	20.01					
S002-830-9004-D-P	Cable	355.42	152.87	43.01					
S002-830-9005-D-P	Cable	355.42	200.09	56.30					

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
S002-830-9006-D-P	Cable	74.44	35.67	47.92					
S002-830-9007-D-P	Cable	74.44	3.54	4.75					
S002-830-9008-C-P-1	Cable	8699.24	1575.72	18.11					
S002-830-9008-D-P	Cable	74.44	3.54	4.75					
S002-830-9009-D-P	Cable	74.44	3.54	4.75					
S002-830-9010-D-P	Cable	74.44	36.93	49.61					
S002-830-9011-D-P	Cable	74.44	3.66	4.91					
S002-830-9012-D-P	Cable	74.44	3.66	4.92					
S002-830-9013-D-P	Cable	74.44	3.66	4.92					
S002-830-9014-D-P	Cable	74.44	3.87	5.20					
S002-830-9015-D-P	Cable	74.44	3.87	5.20					
S002-830-9016-D-P	Cable	390.70	16.46	4.21					
S002-830-9017-D-P	Cable	390.70	16.46	4.21					
S002-830-9018-D-P	Cable	390.70	165.28	42.30					
S002-830-9019-D-P	Cable	195.35	7.48	3.83					
S002-830-9020-D-P	Cable	94.90	3.92	4.13					
S002-830-9021-D-P-1	Cable	3506.65	550.31	15.69					
S002-830-9024-D-P	Cable	355.42	66.59	18.74					
S002-830-9025-D-P	Cable	355.42	43.44	12.22					
S002-830-9026-D-P	Cable	74.44	35.68	47.94					
S002-830-9027-D-P	Cable	74.44	36.02	48.38					
S002-830-9028-D-P	Cable	74.44	3.54	4.76					
S002-830-9029-D-P	Cable	74.44	3.54	4.76					
S002-830-9030-D-P	Cable	74.44	37.38	50.21					
S002-830-9031-D-P	Cable	74.44	37.38	50.21					
S002-830-9032-D-P	Cable	74.44	3.66	4.92					
S002-830-9033-D-P	Cable	74.44	3.66	4.92					
S002-830-9034-D-P	Cable	74.44	39.02	52.42					
S002-830-9035-D-P	Cable	74.44	3.87	5.20					
S002-830-9036-D-P	Cable	390.70	16.46	4.21					
S002-830-9037-D-P	Cable	390.70	165.30	42.31					
S002-830-9038-D-P	Cable	390.70	16.46	4.21					
S002-830-9039-D-P	Cable	74.44	39.57	53.16					
S002-830-9051-C-P	Cable	221.77	47.92	21.61					
S002-830-9053-C-P	Cable	336.70	20.28	6.02					

CKT / Branch		Busway / Cable & Reactor			Transformer				
		Ampacity	Loading		Capability	Loading (input)		Loading (output)	
ID	Type	(Amp)	Amp	%	(MVA)	MVA	%	MVA	%
S002-830-9054-C-P	Cable	336.70	59.04	17.54					
S002-830-9055-C-P	Cable	221.77	31.38	14.15					
S002-830-9056-C-P-1	Cable	1859.31	272.63	14.66					
S002-830-9059-C-P	Cable	221.77	56.10	25.30					
S002-830-9061-C-P-1	Cable	608.59	374.18	61.48					
S002-830-9062-C-P	Cable	221.77	32.24	14.54					
S002-830-9063-C-P	Cable	221.77	3.22	1.45					
S002-830-9064-C-P	Cable	221.77	48.81	22.01					
S002-830-9066-C-P	Cable	336.70	10.94	3.25					
S002-830-9067-C-P	Cable	336.70	33.93	10.08					
S002-830-9068-C-P	Cable	336.70	37.53	11.15					
S002-830-9069-C-P-1	Cable	1859.31	172.57	9.28					
S002-830-9072-C-P	Cable	221.77	5.78	2.61					
S002-830-9073-C-P-1	Cable	608.59	377.42	62.02					
S002-830-9074-C-P	Cable	221.77	57.88	26.10					
S002-830-9076-C-P	Cable	221.77	32.55	14.68					
S002-830-9077-C-P-1	Cable	723.14	234.28	32.40					
S002-830-9078-C-P-1	Cable	582.84	255.91	43.91					
S002-830-9079-C-P	Cable	230.46	173.05	75.09					
S002-Cable20	Cable	2629.99	146.75	5.58					
S002-Cable21	Cable	269.35	25.15	9.34					
S002-Cable22	Cable	2629.99	170.07	6.47					
S002-Cable26	Cable	166.39	5.80	3.49					
S002-Cable29	Cable	1057.94	0.00	0.00					
S002-Cable35	Cable	4383.32	562.97	12.84					
S002-Cable85	Cable	1057.94	401.66	37.97					
S002-Cable87	Cable	1057.94	401.33	37.93					
S002-Cable88	Cable	1057.94	427.20	40.38					
S002-Cable89	Cable	4383.32	303.68	6.93					
S002-Cable90	Cable	269.35	24.46	9.08					
S002-Cable95	Cable	1057.94	427.27	40.39					
S002-Cable115	Cable	876.66	226.30	25.81					
S002-Cable119	Cable	876.66	251.19	28.65					
S002-Cable124	Cable	1083.69	166.37	15.35					
S002-Cable125	Cable	1032.23	552.38	53.51					

CKT / Branch		Busway / Cable & Reactor			Transformer				
					Capability	Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	(MVA)	MVA	%	MVA	%
S002-Cable126	Cable	1057.94	627.95	59.36					
S002-Cable127	Cable	1057.94	54.68	5.17					
S002-Cable131	Cable	175.54	98.87	56.32					
S002-Cable132	Cable	175.54	106.37	60.59					
S002-Cable138	Cable	36.76	16.69	45.40					
S002-Cable181	Cable	73.22	62.65	85.56					
S002-C_TR-04DE001	Cable	305.52	40.60	13.29					
S002-C_TR-04DE002	Cable	305.52	0.13	0.04					
S003-830-1000-E-P	Cable	3506.65	841.38	23.99					
S003-830-1004-E-P	Cable	3506.65	924.11	26.35					
S003-830-9001-C-P	Cable	300.89	43.33	14.40					
S003-830-9001-D-P	Cable	876.66	57.33	6.54					
S003-830-9002-C-P	Cable	300.89	49.31	16.39					
S003-830-9003-C-P	Cable	305.52	30.60	10.01					
S003-830-9004-C-P	Cable	305.52	33.60	11.00					
S003-830-9005-C-P	Cable	336.70	18.30	5.44					
S003-830-9006-C-P	Cable	336.70	62.71	18.62					
S003-830-9019-D-P	Cable	876.66	196.42	22.40					
S003-830-9021-C-P	Cable	296.87	119.20	40.15					
S003-830-9022-C-P	Cable	296.87	142.55	48.02					
S003-Cable19	Cable	128.85	37.48	29.09					
S003-Cable42	Cable	2389.72	260.94	10.92					
S003-Cable44	Cable	128.85	39.35	30.54					
S003-Cable93	Cable	128.85	51.56	40.01					
S003-Cable94	Cable	128.85	51.97	40.33					
S003-Cable96	Cable	128.85	47.65	36.98					
S003-Cable105	Cable	4383.32	591.99	13.51					
S003-Cable113	Cable	890.61	145.14	16.30					
S003-Cable121	Cable	128.85	43.82	34.01					
S003-Cable142	Cable	257.71	149.41	57.98					
S003-Cable173	Cable	257.71	60.23	23.37					
S002-830-1601-N-P-1	Cable	4383.32	0.00	0.00					
S003-830-2000-E-P	Cable	214.13	0.02	0.01					
S003-Cable120	Cable	7013.31	0.01	0.00					
S002-Cable23	Cable	8766.63	0.01	0.00					

CKT / Branch		Busway / Cable & Reactor			Transformer				
					Capability (MVA)	Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%		MVA	%	MVA	%
S002-830-1072-E-P	Cable	2629.99	0.01	0.00					
* FD Tranfo	Transformer				0.750	0.845	112.7	0.808	107.8
S002-TML1 [EXISTING]	Transformer				1.000	0.391	39.1	0.385	38.5
S002-TML1A-1 [EXISTING]	Transformer				1.600	0.904	56.5	0.881	55.1
S002-TML1A-2 [EXISTING]	Transformer				1.600	1.182	73.9	1.143	71.4
S002-TML1B-1 [EXISTING]	Transformer				1.600	0.394	24.6	0.390	24.3
S002-TML1B-2 [EXISTING]	Transformer				1.600	0.257	16.1	0.255	15.9
S002-TML2 [EXISTING]	Transformer				1.000	0.209	20.9	0.207	20.7
S002-TMM-1A [EXISTING]	Transformer				12.000	5.258	43.8	5.147	42.9
S002-TMM-1B [EXISTING]	Transformer				12.000	3.297	27.5	3.256	27.1
S002-TR-01CE001	Transformer				2.000	0.924	46.2	0.907	45.3
S002-TR-01CE002	Transformer				2.000	0.933	46.6	0.914	45.7
S002-TR-03CE001	Transformer				0.785	0.605	77.0	0.595	75.8
S002-TR-03CE002	Transformer				1.250	0.717	57.3	0.702	56.1
S002-TR-03CN001	Transformer				2.000	1.137	56.9	1.108	55.4
S002-TR-03CN002	Transformer				2.000	0.648	32.4	0.638	31.9
S002-TR-04DE001	Transformer				0.400	0.240	60.0	0.234	58.6
S002-TR-PSS1AC01	Transformer				50.000	32.409	64.8	31.395	62.8
S002-TR-PSS1AC02	Transformer				50.000	30.986	62.0	30.107	60.2
S003-TMM1A [EXISTING]	Transformer				3.500	0.345	9.9	0.344	9.8
S003-TMM1B [EXISTING]	Transformer				3.500	1.169	33.4	1.149	32.8
S003-TR-11CE001	Transformer				1.600	0.577	36.1	0.567	35.4
S003-TR-11CE002	Transformer				1.600	0.626	39.1	0.615	38.4
S003-TR-11CN001	Transformer				2.500	0.817	32.7	0.804	32.2
S003-TR-11CN002	Transformer				2.500	0.919	36.8	0.903	36.1
T2	Transformer				1.000	0.774	77.4	0.757	75.7
TR-05CE001A	Transformer				2.500	1.046	41.9	1.021	40.8
TR-05CE001B	Transformer				2.500	1.346	53.8	1.302	52.1
TX-A	Transformer				3.150	2.681	85.1	2.578	81.8
TX-B	Transformer				3.150	2.218	70.4	2.192	69.6
S002-TR-CMP1_VFD	3W XFMR p				15.000	5.883	39.2		
	3W XFMR s				7.500	2.932	39.1		
	3W XFMR t				7.500	2.932	39.1		
S002-TR-CMP2_VFD	3W XFMR p				15.000	5.884	39.2		
	3W XFMR s				7.500	2.932	39.1		

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
S002-TR-CMP2_VFD	3W XFMR t				7.500	2.932	39.1		

* Indicates a branch with operating load exceeding the branch capability.

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
C_688-KM-01	-0.901	-0.410	0.906	0.411	5.0	0.5	98.5	103.5	0.55
Cable10	-0.040	-0.025	0.042	0.025	1.8	0.4	100.2	104.0	3.78
Cable11	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.71
Cable118	-2.253	-1.190	2.265	1.182	12.6	-7.4	100.3	100.9	0.54
Cable12	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.65
Cable123	-0.016	-0.012	0.016	0.012	0.2	0.0	107.5	102.9	0.77
Cable128	0.542	0.345	-0.541	-0.344	1.4	1.4	99.5	102.9	0.30
Cable13	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.71
Cable134	0.726	0.037	-0.726	-0.036	0.6	0.5	105.2	101.3	0.08
Cable136	-0.726	-0.037	0.728	0.039	1.6	1.6	105.2	105.4	0.24
Cable137	0.726	0.037	-0.726	-0.036	0.6	0.5	105.2	101.3	0.08
Cable144	0.724	0.462	-0.723	-0.460	1.7	1.7	103.6	103.4	0.28
Cable145	0.724	0.462	-0.723	-0.460	1.7	1.7	103.6	103.4	0.28
Cable146	0.724	0.462	-0.723	-0.460	1.7	1.7	103.6	103.4	0.28
Cable147	0.723	0.460	-0.722	-0.459	0.8	0.8	103.4	99.5	0.13
Cable148	0.723	0.460	-0.722	-0.459	0.8	0.8	103.4	99.5	0.13
Cable149	0.723	0.460	-0.722	-0.459	0.8	0.8	103.4	99.5	0.13
Cable150	-0.726	-0.037	0.728	0.039	1.6	1.6	105.2	105.4	0.24
Cable151	0.542	0.345	-0.540	-0.343	1.4	1.4	99.5	99.2	0.30
Cable152	0.710	0.029	-0.709	-0.029	0.5	0.5	101.4	101.3	0.08
Cable153	0.254	0.160	-0.253	-0.159	0.7	0.7	99.2	94.6	0.34
Cable154	0.254	0.160	-0.253	-0.159	0.9	0.9	99.2	94.5	0.39
Cable155	-0.016	-0.012	0.016	0.012	0.1	0.0	107.8	99.2	0.47
Cable156	-0.016	-0.012	0.016	0.012	0.1	0.0	107.7	99.2	0.54
Cable157	-0.732	-0.040	0.734	0.041	1.6	1.6	101.4	105.4	0.24
Cable158	0.540	0.025	-0.539	-0.024	1.0	1.0	101.3	101.1	0.19
Cable159	0.540	0.026	-0.540	-0.025	0.9	0.9	101.3	101.1	0.18
Cable160	0.541	0.026	-0.540	-0.025	0.9	0.9	101.3	101.1	0.17
Cable161	0.540	0.025	-0.539	-0.024	0.6	0.6	101.3	101.2	0.11
Cable162	0.541	0.344	-0.539	-0.342	1.4	1.4	99.5	102.9	0.30
Cable163	0.254	0.159	-0.253	-0.159	0.3	0.3	102.9	94.8	0.14
Cable164	0.254	0.159	-0.253	-0.159	0.4	0.4	102.9	94.7	0.21
Cable165	-0.016	-0.012	0.016	0.012	0.1	0.0	107.9	102.9	0.38
Cable166	-0.016	-0.012	0.016	0.012	0.1	0.0	107.9	102.9	0.45

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd
	MW	Mvar	MW	Mvar	kW	kvar	From	To	% Drop in Vmag
Cable167	0.541	0.345	-0.540	-0.343	1.4	1.4	99.5	102.9	0.30
Cable168	0.254	0.160	-0.253	-0.159	0.6	0.6	102.9	94.6	0.29
Cable169	0.254	0.160	-0.253	-0.159	0.9	0.9	102.9	94.5	0.42
Cable170	-0.016	-0.012	0.016	0.012	0.1	0.0	107.8	102.9	0.49
Cable171	-0.016	-0.012	0.016	0.012	0.1	0.0	107.7	102.9	0.65
Cable176	0.022	0.011	-0.022	-0.011	0.0	0.0	101.4	101.4	0.02
Cable188	0.254	0.000	-0.253	0.000	0.2	0.2	101.1	96.6	0.07
Cable189	0.254	0.000	-0.253	0.000	0.3	0.3	101.1	96.6	0.10
Cable19	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.65
Cable190	-0.016	-0.012	0.016	0.012	0.1	0.0	104.6	101.1	0.33
Cable191	-0.016	-0.012	0.016	0.012	0.1	0.0	104.5	101.1	0.40
Cable192	0.254	0.000	-0.253	0.000	0.4	0.4	101.1	96.6	0.17
Cable193	0.254	0.001	-0.253	0.000	0.6	0.6	101.1	96.5	0.24
Cable194	-0.016	-0.012	0.016	0.012	0.1	0.0	104.4	101.1	0.46
Cable195	-0.016	-0.012	0.016	0.012	0.1	0.0	104.3	101.1	0.60
Cable196	0.254	0.001	-0.253	0.000	0.7	0.7	101.1	96.5	0.29
Cable197	0.254	0.000	-0.253	0.000	0.4	0.4	101.1	96.6	0.18
Cable198	-0.016	-0.012	0.016	0.012	0.2	0.0	104.2	101.1	0.66
Cable199	-0.016	-0.012	0.016	0.012	0.1	0.0	104.4	101.1	0.46
Cable2	-0.040	-0.025	0.042	0.025	1.9	0.4	100.0	104.0	3.97
Cable20	-0.016	-0.007	0.017	0.007	0.5	0.1	100.0	102.7	2.72
Cable200	0.254	0.000	-0.253	0.000	0.3	0.3	101.2	96.7	0.11
Cable201	0.254	0.000	-0.253	0.000	0.4	0.4	101.2	96.6	0.16
Cable202	-0.016	-0.012	0.016	0.012	0.1	0.0	104.6	101.2	0.33
Cable203	-0.016	-0.012	0.016	0.012	0.1	0.0	104.5	101.2	0.42
Cable21	-0.040	-0.025	0.042	0.025	1.7	0.4	100.4	104.0	3.58
Cable22	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.65
Cable23	-0.001	-0.001	0.001	0.001	0.0	0.0	102.4	102.5	0.11
Cable24	-0.003	-0.002	0.003	0.002	0.0	0.0	99.6	99.9	0.33
Cable25	-0.040	-0.025	0.042	0.025	1.7	0.4	100.4	104.0	3.58
Cable26	-0.016	-0.007	0.017	0.007	0.5	0.1	97.1	99.9	2.80
Cable27	-0.040	-0.025	0.042	0.025	1.7	0.4	100.5	104.0	3.52
Cable28	-0.040	-0.025	0.042	0.025	1.7	0.4	98.7	102.1	3.39
Cable29	-0.040	-0.025	0.042	0.025	1.7	0.4	98.6	102.1	3.46
Cable3	-0.040	-0.025	0.042	0.025	1.9	0.4	100.1	104.0	3.91
Cable30	-0.040	-0.025	0.042	0.025	1.7	0.4	98.6	102.1	3.53
Cable31	-0.040	-0.025	0.042	0.025	1.7	0.4	98.6	102.1	3.46

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
Cable32	-0.040	-0.025	0.042	0.025	1.7	0.4	98.6	102.1	3.53
Cable33	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.59
Cable34	-0.040	-0.025	0.042	0.025	1.7	0.4	98.6	102.1	3.53
Cable35	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.59
Cable36	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.66
Cable37	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.59
Cable38	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.66
Cable39	-0.040	-0.025	0.042	0.025	1.8	0.4	98.4	102.1	3.72
Cable4	-0.040	-0.025	0.042	0.025	1.9	0.4	100.2	104.0	3.84
Cable40	0.254	0.160	-0.253	-0.159	1.1	1.1	102.9	94.4	0.52
Cable41	0.255	0.160	-0.253	-0.159	1.2	1.2	102.9	94.4	0.58
Cable42	-0.040	-0.025	0.042	0.025	1.8	0.4	98.5	102.1	3.66
Cable43	-0.040	-0.025	0.042	0.025	1.8	0.4	98.4	102.1	3.72
Cable44	-0.040	-0.025	0.042	0.025	1.9	0.4	98.3	102.1	3.79
Cable45	-0.025	-0.015	0.025	0.015	0.2	0.1	101.2	102.1	0.91
Cable48	-0.025	-0.015	0.025	0.015	0.1	0.1	101.7	102.1	0.43
Cable49	-0.025	-0.015	0.025	0.015	0.1	0.1	101.6	102.1	0.53
Cable5	-0.040	-0.025	0.042	0.025	1.9	0.4	100.2	104.0	3.84
Cable6	-0.040	-0.025	0.042	0.025	1.8	0.4	100.2	104.0	3.78
Cable7	-0.040	-0.025	0.042	0.025	1.8	0.4	100.3	104.0	3.71
Cable83	-0.016	-0.012	0.016	0.012	0.2	0.0	107.6	102.9	0.70
Cable84	0.000	0.000	0.000	0.000	0.0	0.0	100.0	100.0	0.00
FD Tranfo	-0.647	-0.485	0.659	0.529	12.6	44.1	97.4	101.9	4.46
S002-830-1063-N-P	0.894	0.655	-0.892	-0.651	2.0	4.3	101.2	100.8	0.38
S002-830-1066-N-P	0.513	0.379	-0.513	-0.378	0.7	1.4	101.4	101.2	0.22
S002-830-1069-N-P	0.491	0.336	-0.490	-0.333	1.0	2.2	104.7	104.4	0.37
S002-830-1072-E-P	0.000	0.000	0.000	0.000	0.0	0.0	100.0	100.0	0.00
S002-830-1072-N-P	0.560	0.422	-0.559	-0.419	1.5	3.2	103.3	102.8	0.46
S002-830-1420-N-P-1	0.795	0.436	-0.793	-0.434	1.1	2.3	102.0	101.7	0.24
S002-830-1500-N-P-1	0.791	0.457	-0.790	-0.454	1.4	2.9	100.9	100.6	0.29
S002-830-1601-N-P-1	0.000	0.000	0.000	0.000	0.0	0.0	100.6	91.4	0.00
S002-830-9001-A-P	27.937	12.242	-27.936	-13.407	1.3	-1164.5	100.0	100.0	0.01
S002-830-9001-C-P	-28.572	-13.018	28.574	13.005	2.6	-13.1	101.2	101.3	0.02
S002-830-9001-D-P-1	4.445	2.595	-4.442	-2.591	2.3	3.9	103.6	103.5	0.09
S002-830-9004-A-P	28.642	14.004	-28.641	-15.168	1.4	-1164.3	100.0	100.0	0.01
S002-830-9004-D-P	-0.742	-0.515	0.743	0.516	0.6	0.3	98.9	103.5	0.09
S002-830-9005-D-P	-0.949	-0.706	0.950	0.707	1.0	0.6	98.9	103.5	0.11

	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
Branch ID	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
S002-830-9006-D-P	-0.181	-0.105	0.183	0.105	2.0	-0.2	102.6	103.5	0.88
S002-830-9007-D-P	-0.018	-0.010	0.018	0.010	0.0	-0.4	103.4	103.5	0.09
S002-830-9008-C-P-1	-27.871	-11.397	27.874	11.380	3.2	-17.1	100.3	100.3	0.02
S002-830-9008-D-P	-0.018	-0.010	0.018	0.010	0.0	-0.3	103.4	103.5	0.06
S002-830-9009-D-P	-0.018	-0.010	0.018	0.010	0.0	-0.3	103.4	103.5	0.06
S002-830-9010-D-P	-0.187	-0.108	0.189	0.108	2.5	-0.2	102.4	103.5	1.09
S002-830-9011-D-P	-0.019	-0.011	0.019	0.010	0.0	-0.5	103.4	103.5	0.11
S002-830-9012-D-P	-0.019	-0.011	0.019	0.010	0.0	-0.6	103.4	103.5	0.14
S002-830-9013-D-P	-0.019	-0.011	0.019	0.010	0.0	-0.6	103.4	103.5	0.14
S002-830-9014-D-P	-0.020	-0.011	0.020	0.011	0.0	-0.4	103.4	103.5	0.09
S002-830-9015-D-P	-0.020	-0.011	0.020	0.011	0.0	-0.4	103.4	103.5	0.09
S002-830-9016-D-P	-0.087	-0.043	0.087	0.042	0.0	-1.0	103.5	103.5	0.05
S002-830-9017-D-P	-0.087	-0.043	0.087	0.042	0.0	-1.1	103.5	103.5	0.05
S002-830-9018-D-P	-0.871	-0.434	0.875	0.434	4.5	0.4	103.0	103.5	0.50
S002-830-9019-D-P	-0.038	-0.022	0.039	0.022	0.0	-0.2	103.5	103.5	0.02
S002-830-9020-D-P	-0.021	-0.011	0.021	0.010	0.0	-0.2	103.5	103.5	0.03
S002-830-9021-D-P-1	2.849	1.576	-2.848	-1.575	0.9	1.0	103.5	103.5	0.06
S002-830-9024-D-P	-0.331	-0.212	0.332	0.212	0.1	-0.1	98.9	103.5	0.04
S002-830-9025-D-P	-0.212	-0.145	0.212	0.145	0.0	-0.1	98.9	103.5	0.02
S002-830-9026-D-P	-0.181	-0.105	0.183	0.105	2.0	-0.2	102.6	103.5	0.88
S002-830-9027-D-P	-0.183	-0.105	0.185	0.105	2.0	-0.2	102.6	103.5	0.89
S002-830-9028-D-P	-0.018	-0.010	0.018	0.010	0.0	-0.3	103.4	103.5	0.06
S002-830-9029-D-P	-0.018	-0.010	0.018	0.010	0.0	-0.3	103.4	103.5	0.06
S002-830-9030-D-P	-0.190	-0.109	0.192	0.109	2.6	-0.2	102.4	103.5	1.10
S002-830-9031-D-P	-0.190	-0.109	0.192	0.109	2.6	-0.2	102.4	103.5	1.10
S002-830-9032-D-P	-0.019	-0.011	0.019	0.010	0.0	-0.6	103.3	103.5	0.14
S002-830-9033-D-P	-0.019	-0.011	0.019	0.010	0.0	-0.6	103.3	103.5	0.14
S002-830-9034-D-P	-0.198	-0.115	0.200	0.115	2.2	-0.1	102.5	103.5	0.92
S002-830-9035-D-P	-0.020	-0.011	0.020	0.011	0.0	-0.4	103.4	103.5	0.09
S002-830-9036-D-P	-0.087	-0.043	0.087	0.042	0.0	-1.0	103.4	103.5	0.05
S002-830-9037-D-P	-0.871	-0.434	0.875	0.434	4.3	0.4	103.0	103.5	0.48
S002-830-9038-D-P	-0.087	-0.043	0.087	0.042	0.0	-1.1	103.4	103.5	0.05
S002-830-9039-D-P	-0.208	-0.106	0.209	0.106	1.1	-0.1	103.0	103.5	0.47
S002-830-9051-C-P	-0.799	-0.464	0.799	0.462	0.1	-1.9	101.2	101.2	0.01
S002-830-9053-C-P	-0.321	-0.222	0.322	0.208	0.1	-14.1	101.2	101.2	0.04
S002-830-9054-C-P	-0.900	-0.695	0.901	0.680	1.2	-15.1	101.1	101.2	0.14
S002-830-9055-C-P	-0.494	-0.349	0.494	0.334	0.3	-15.8	101.2	101.2	0.07

Branch ID	From-To Bus Flow MW	From-To Bus Flow Mvar	To-From Bus Flow MW	To-From Bus Flow Mvar	Losses kW	Losses kvar	% Bus Voltage From	% Bus Voltage To	Vd % Drop in Vmag
S002-830-9056-C-P-1	-4.455	-2.794	4.455	2.792	0.3	-2.6	101.2	101.2	0.01
S002-830-9059-C-P	-0.973	-0.471	0.974	0.464	0.5	-7.5	101.2	101.2	0.06
S002-830-9061-C-P-1	-6.532	-3.057	6.536	3.048	4.2	-9.0	101.2	101.2	0.09
S002-830-9062-C-P	-0.531	-0.323	0.531	0.314	0.2	-9.9	101.2	101.2	0.05
S002-830-9063-C-P	-0.053	-0.032	0.053	0.022	0.0	-9.9	101.2	101.2	0.00
S002-830-9064-C-P	-0.796	-0.486	0.796	0.484	0.1	-1.4	100.3	100.3	0.01
S002-830-9066-C-P	-0.173	-0.118	0.173	0.104	0.0	-13.9	100.3	100.3	0.02
S002-830-9067-C-P	-0.515	-0.392	0.516	0.377	0.4	-15.5	100.2	100.3	0.08
S002-830-9068-C-P	-0.564	-0.442	0.565	0.427	0.5	-15.4	100.2	100.3	0.09
S002-830-9069-C-P-1	-2.853	-1.654	2.853	1.651	0.1	-2.9	100.3	100.3	0.01
S002-830-9072-C-P	-0.098	-0.052	0.098	0.044	0.0	-8.0	100.3	100.3	0.01
S002-830-9073-C-P-1	-6.527	-3.054	6.531	3.046	4.2	-8.4	100.2	100.3	0.09
S002-830-9074-C-P	-0.978	-0.516	0.978	0.508	0.6	-7.4	100.2	100.3	0.07
S002-830-9076-C-P	-0.531	-0.323	0.531	0.314	0.2	-9.5	100.3	100.3	0.05
S002-830-9077-C-P-1	3.733	2.335	-3.648	-2.491	85.2	-155.1	101.2	99.0	2.29
S002-830-9078-C-P-1	4.526	1.646	-4.423	-1.786	102.7	-139.7	100.3	97.8	2.46
S002-830-9079-C-P	-2.990	-1.455	3.002	1.453	11.7	-2.1	100.9	101.2	0.39
S002-C_TR-04DE001	0.213	0.110	-0.213	-0.111	0.3	-0.6	103.5	103.3	0.15
S002-C_TR-04DE002	0.000	-0.001	0.000	0.000	0.0	-0.8	103.5	103.5	0.00
S002-Cable115	-0.127	-0.086	0.127	0.086	0.2	0.4	102.7	103.0	0.24
S002-Cable119	-0.138	-0.090	0.138	0.091	0.2	0.5	99.9	100.2	0.28
S002-Cable124	-0.094	-0.058	0.094	0.058	0.1	0.2	100.9	101.1	0.16
S002-Cable125	-0.315	-0.195	0.316	0.198	0.9	2.3	102.1	102.6	0.54
S002-Cable126	-0.334	-0.241	0.334	0.244	0.9	2.6	99.6	100.2	0.54
S002-Cable127	-0.030	-0.022	0.030	0.022	0.0	0.0	102.9	103.0	0.05
S002-Cable131	-0.053	-0.038	0.054	0.038	0.5	0.1	101.0	101.7	0.71
S002-Cable132	-0.056	-0.041	0.057	0.041	0.8	0.2	99.5	100.6	1.08
S002-Cable138	-0.009	-0.007	0.009	0.007	0.1	0.0	103.1	103.9	0.89
S002-Cable181	-0.036	-0.018	0.037	0.018	0.8	0.1	98.2	99.9	1.74
S002-Cable20	0.094	0.027	-0.094	-0.026	0.4	0.9	101.7	101.0	0.70
S002-Cable21	0.014	0.010	-0.014	-0.010	0.0	0.0	102.9	102.6	0.32
S002-Cable22	0.095	0.060	-0.095	-0.059	0.6	1.2	100.6	99.6	1.04
S002-Cable23	0.000	0.000	0.000	0.000		0.0	100.0	100.0	0.00
S002-Cable26	-0.003	-0.002	0.003	0.002	0.0	0.0	103.6	103.7	0.10
S002-Cable29	0.000	0.000	0.000	0.000	0.0	0.0	102.6	102.6	0.00
S002-Cable34	0.095	0.103	-0.094	-0.103	1.5	0.9	101.1	99.9	1.18
S002-Cable35	0.320	0.216	-0.319	-0.215	0.1	0.2	104.0	103.9	0.06

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
S002-Cable85	-0.229	-0.135	0.230	0.137	0.5	1.3	100.7	101.1	0.40
S002-Cable87	-0.236	-0.122	0.237	0.123	0.4	1.1	100.8	101.1	0.32
S002-Cable88	-0.233	-0.156	0.233	0.158	0.6	1.7	99.7	100.2	0.52
S002-Cable89	0.172	0.116	-0.172	-0.116	0.0	0.1	103.8	103.7	0.03
S002-Cable90	0.014	0.009	-0.014	-0.009	0.0	0.0	102.6	102.5	0.14
S002-Cable95	-0.233	-0.156	0.233	0.158	0.6	1.8	99.6	100.2	0.54
S002-TML1 [EXISTING]	0.321	0.222	-0.320	-0.216	2.0	7.0	101.2	104.0	6.20
S002-TML1A-1 [EXISTING]	0.742	0.515	-0.737	-0.483	5.4	32.2	98.9	101.1	7.05
S002-TML1A-2 [EXISTING]	0.949	0.706	-0.939	-0.651	9.2	55.0	98.9	100.2	7.86
S002-TML1B-1 [EXISTING]	0.331	0.212	-0.330	-0.206	1.0	6.1	98.9	102.6	5.69
S002-TML1B-2 [EXISTING]	0.212	0.145	-0.212	-0.142	0.4	2.6	98.9	103.0	5.37
S002-TML2 [EXISTING]	-0.172	-0.116	0.173	0.118	0.6	2.0	103.8	100.3	5.49
S002-TMM-1A [EXISTING]	4.455	2.794	-4.445	-2.595	10.0	199.4	101.2	103.6	2.15
S002-TMM-1B [EXISTING]	2.853	1.654	-2.849	-1.576	3.9	78.1	100.3	103.5	1.27
S002-TR-01CE001	-0.795	-0.436	0.799	0.464	4.6	27.9	102.0	101.2	4.37
S002-TR-01CE002	-0.791	-0.457	0.796	0.486	4.8	28.9	100.9	100.3	4.43
S002-TR-03CE001	0.494	0.349	-0.491	-0.336	2.6	13.7	101.2	104.7	1.67
S002-TR-03CE002	0.564	0.442	-0.560	-0.422	3.7	19.7	100.2	103.3	2.09
S002-TR-03CN001	-0.894	-0.655	0.900	0.695	6.3	39.9	101.2	101.1	4.98
S002-TR-03CN002	0.515	0.392	-0.513	-0.379	2.1	13.1	100.2	101.4	3.88
S002-TR-04DE001	-0.209	-0.105	0.213	0.111	3.5	5.2	103.7	103.3	4.83
S002-TR-CMP1_VFD	5.872	0.371	-2.932	0.000	8.3	371.3	101.2	96.5	0.34
	0.000	0.000	-2.932	0.000			101.2	96.5	0.34
S002-TR-CMP2_VFD	5.872	0.378	-2.932	0.000	8.4	378.5	100.3	95.6	0.35
	0.000	0.000	-2.932	0.000			100.3	95.6	0.35
S002-TR-PSS1AC01	-28.574	-13.005	28.641	15.168	66.6	2163.0	101.3	100.0	3.13
S002-TR-PSS1AC02	-27.874	-11.380	27.936	13.407	62.4	2027.0	100.3	100.0	4.04
S003-830-1000-E-P	0.479	0.302	-0.478	-0.300	1.1	2.3	102.3	101.9	0.38
S003-830-1004-E-P	0.510	0.343	-0.509	-0.341	1.3	2.7	101.1	100.7	0.43
S003-830-2000-E-P	0.000	0.000	0.000	0.000	0.0	0.0	100.7	91.5	0.00
S003-830-9001-C-P	-0.657	-0.485	0.657	0.484	0.1	-1.3	99.0	99.0	0.01
S003-830-9001-D-P	0.304	0.161	-0.303	-0.161	0.0	-0.1	104.9	104.9	0.01
S003-830-9002-C-P	-0.758	-0.520	0.758	0.519	0.0	-0.8	97.8	97.8	0.01
S003-830-9003-C-P	-0.483	-0.315	0.483	0.315	0.0	-0.8	99.0	99.0	0.00
S003-830-9004-C-P	-0.513	-0.360	0.513	0.359	0.0	-0.7	97.8	97.8	0.00
S003-830-9005-C-P	-0.304	-0.164	0.304	0.163	0.0	-0.8	99.0	99.0	0.00
S003-830-9006-C-P	-0.942	-0.692	0.942	0.691	0.1	-1.0	97.8	97.8	0.01

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
S003-830-9019-D-P	0.938	0.662	-0.938	-0.662	0.2	0.4	102.3	102.3	0.04
S003-830-9021-C-P	-2.207	-0.223	2.210	0.216	3.9	-6.1	97.7	97.8	0.18
S003-830-9022-C-P	-2.198	-1.535	2.204	1.529	5.6	-5.3	98.7	99.0	0.24
S003-Cable105	-0.315	-0.237	0.318	0.242	2.5	5.4	101.2	102.6	1.36
S003-Cable113	-0.659	-0.529	0.662	0.527	2.7	-1.8	101.9	102.3	0.38
S003-Cable120	0.000	0.000	0.000	0.000		0.0	100.0	100.0	0.00
S003-Cable121	-0.023	-0.017	0.023	0.017	0.1	0.0	100.9	101.2	0.31
S003-Cable142	-0.088	-0.043	0.090	0.043	1.7	0.4	99.4	101.2	1.78
S003-Cable173	-0.036	-0.017	0.036	0.017	0.3	0.1	100.5	101.2	0.72
S003-Cable19	-0.020	-0.015	0.020	0.015	0.1	0.0	102.3	102.6	0.26
S003-Cable42	-0.136	-0.102	0.138	0.106	1.6	3.3	99.3	101.2	1.94
S003-Cable43	0.000	0.000	0.000	0.000	0.0	0.0	100.0	100.0	0.00
S003-Cable44	-0.021	-0.016	0.021	0.016	0.1	0.0	102.3	102.6	0.27
S003-Cable45	0.000	0.000	0.000	0.000	0.0	0.0	100.0	100.0	0.00
S003-Cable93	-0.028	-0.021	0.028	0.021	0.1	0.0	101.5	101.9	0.40
S003-Cable94	-0.028	-0.020	0.028	0.020	0.1	0.0	100.3	100.7	0.36
S003-Cable96	-0.025	-0.019	0.025	0.019	0.1	0.0	100.9	101.2	0.33
S003-TMM1A [EXISTING]	-0.304	-0.161	0.304	0.164	0.3	2.5	104.9	99.0	2.81
S003-TMM1B [EXISTING]	-0.938	-0.662	0.942	0.692	3.5	29.5	102.3	97.8	4.03
S003-TR-11CE001	0.483	0.315	-0.479	-0.302	3.8	13.2	99.0	102.3	1.76
S003-TR-11CE002	0.513	0.360	-0.510	-0.343	2.7	16.3	97.8	101.1	1.79
S003-TR-11CN001	0.657	0.485	-0.654	-0.468	2.9	17.3	99.0	102.6	1.52
S003-TR-11CN002	0.758	0.520	-0.754	-0.498	3.7	22.5	97.8	101.2	1.67
T2	0.725	0.273	-0.717	-0.245	8.1	28.3	100.9	102.6	2.24
TR-05CE001A	0.900	0.534	-0.893	-0.495	6.6	39.4	101.2	104.0	2.45
TR-05CE001B	1.133	0.726	-1.122	-0.660	11.1	66.4	100.3	102.1	3.29
TX-A	-2.173	-1.386	2.198	1.535	24.8	149.0	103.6	98.7	1.19
TX-B	-2.189	-0.118	2.207	0.223	17.4	104.2	105.4	97.7	3.96
					701.0	3238.6			

* This Transmission Line includes Series Capacitor.

Equipment Cable and Overload Heater Losses Summary Report

Connected Load		Cable /Overload Heater		Losses		% Voltage			% Vd	% Vst
							Terminal on			
ID	Type	ID	Library	kW	kvar	Bus	Bus kV	Load kV	Operating	Starting
614-KM-01A	Ind. Motor	Cable16	11NCUS3	3.2	4.4	100.00	99.92	99.92	0.08	99.50
614-KM-01B	Ind. Motor	Cable17	11NCUS3	3.2	4.4	100.00	99.92	99.92	0.08	99.50
S002-40-MP-5C	Ind. Motor	Cable47	11NCUS3	0.0	0.0	101.24	101.24	101.24	0.00	101.06
S002-40-MP-05A	Ind. Motor	Cable46	11NCUS3	0.2	0.2	101.20	101.15	101.15	0.05	101.01
S002-6100-MK-1D	Ind. Motor	MSB1A-MK1D-P	11NCUS3	4.2	5.9	101.16	101.07	101.07	0.09	100.69
S002-6100-MK-1C	Ind. Motor	MSB1B-MK1C-P	11NCUS3	4.2	5.8	100.21	100.12	100.12	0.09	99.79
S002-40-MP-05B	Ind. Motor	Cable50	11NCUS3	0.2	0.2	100.25	100.20	100.20	0.05	100.07
S002-6100-MP-3D	Ind. Motor	Cable54	3.3NCUN3	2.0	0.2	102.57	101.68	101.68	0.89	100.92
S002-6100-MP-4A	Ind. Motor	Cable55	3.3NCUN3	0.0	0.0	103.40	103.34	103.34	0.06	102.28
S002-6100-MP-4C	Ind. Motor	Cable57	3.3NCUN3	0.0	0.0	103.40	103.34	103.34	0.06	102.28
S002-6100-MP-6A	Ind. Motor	Cable63	3.3NCUN3	2.6	0.3	102.36	101.24	101.24	1.12	100.36
S002-6100-MP-6C	Ind. Motor	Cable65	3.3NCUN3	2.6	0.3	102.36	101.24	101.24	1.12	100.36
S002-40-MP-04B	Ind. Motor	Cable86	3.3NCUN3	1.1	0.1	103.00	102.53	102.53	0.47	101.66
614-KM-11	Ind. Motor	Cable9	11NCUS3	1.0	1.4	100.30	100.25	100.25	0.04	100.08
S003-7200-MP-03A	Ind. Motor	Cable97	3.3NCUN3	0.0	0.0	104.87	104.82	104.82	0.05	103.15
S003-7100-MK-2A	Ind. Motor	Cable98	3.3NCUN3	0.0	0.0	104.87	104.84	104.84	0.03	104.03
S003-7100-MK-1B	Ind. Motor	Cable100	3.3NCUN3	0.0	0.0	104.87	104.84	104.84	0.03	103.98
S003-7100-MK-1A	Ind. Motor	Cable99	3.3NCUN3	0.9	0.2	104.87	104.50	104.50	0.37	103.84
S003-7100-MK-1C	Ind. Motor	Cable101	3.3NCUN3	0.0	0.0	102.28	102.24	102.24	0.04	101.33
S003-7100-MK-2B	Ind. Motor	Cable102	3.3NCUN3	0.7	0.1	102.28	101.94	101.94	0.34	101.42
S003-7100-MK-2C	Ind. Motor	Cable103	3.3NCUN3	0.0	0.0	102.28	102.25	102.25	0.03	101.55
S003-7200-MP-02B	Ind. Motor	Cable104	3.3NCUN3	0.0	0.0	102.28	102.27	102.27	0.01	101.75
S002-MCC1-4_M	Ind. Motor	Cable135	1.0NCUN3	0.6	0.0	101.02	98.10	98.10	2.92	86.96
S002-6100-MP-1B	Ind. Motor	Cable130	1.0NCUN3	0.7	0.2	99.90	98.56	98.56	1.34	93.46
S002-6300-MK-1A	Ind. Motor	Cable108	1.0NCUN3	3.0	1.1	101.73	99.61	99.61	2.12	93.40
S003-7500-K-1A	Ind. Motor	Cable110	1.0NCUN3	2.1	1.6	100.66	99.15	99.15	1.51	94.93
S003-7400-MP-4A	Ind. Motor	Cable129	1.0NCUN3	0.2	0.0	102.56	101.64	101.64	0.93	97.84
S003-7400-P-21B	Ind. Motor	Cable133	1.0NCUN3	2.4	0.9	101.22	98.95	98.95	2.27	93.00

Alert Summary Report

	% Alert Settings	
	Critical	**Marginal**
Loading		
Bus	106.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
Bus Voltage		
OverVoltage	106.0	105.0
UnderVoltage	94.0	95.0
Generator Excitation		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Critical Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus14	Bus	Over Voltage	0.380	kV	0.409	107.6	3-Phase
Bus15	Bus	Over Voltage	0.380	kV	0.409	107.5	3-Phase
Bus2	Bus	Over Voltage	0.380	kV	0.410	107.8	3-Phase
Bus4	Bus	Over Voltage	0.380	kV	0.409	107.7	3-Phase
Bus57	Bus	Over Voltage	0.380	kV	0.410	108.0	3-Phase
Bus59	Bus	Over Voltage	0.380	kV	0.410	107.9	3-Phase
Bus60	Bus	Over Voltage	0.380	kV	0.410	107.8	3-Phase
Bus62	Bus	Over Voltage	0.380	kV	0.409	107.7	3-Phase
Bus68	Bus	Over Voltage	0.380	kV	0.411	108.2	3-Phase
Bus70	Bus	Over Voltage	0.380	kV	0.411	108.1	3-Phase
Bus72	Bus	Over Voltage	0.380	kV	0.410	107.9	3-Phase
Bus73	Bus	Over Voltage	0.380	kV	0.410	107.9	3-Phase
Bus74	Bus	Over Voltage	0.380	kV	0.411	108.0	3-Phase
Bus75	Bus	Over Voltage	0.380	kV	0.410	107.9	3-Phase
Bus76	Bus	Over Voltage	0.380	kV	0.410	107.8	3-Phase

Critical Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus77	Bus	Over Voltage	0.380	kV	0.409	107.7	3-Phase
DG-1 (TOFS)	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
FD Tranfo	Transformer	Overload	0.750	MVA	0.845	112.7	3-Phase
S002-EDG-01E001	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
S002-EDG-01E001	Generator	Under Power	0.000	MW	0.000	0.0	3-Phase
S002-EDG-03E001	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
S003-A-77301A	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
S003-DG-1 [EXISTING]	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
S003-EDG-11E001	Generator	Under Excited	0.000	Mvar	0.000	0.0	3-Phase
S003-EDG-11E001	Generator	Under Power	0.000	MW	0.000	0.0	3-Phase

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus52	Bus	Over Voltage	0.400	kV	0.421	105.2	3-Phase
Bus63	Bus	Over Voltage	0.400	kV	0.421	105.2	3-Phase
Bus65	Bus	Over Voltage	0.400	kV	0.422	105.4	3-Phase

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	56.580	26.246	62.371	90.72 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	56.580	26.246	62.371	90.72 Lagging
Total Motor Load:	50.343	27.926	57.570	87.45 Lagging
Total Static Load:	4.949	2.968	5.771	85.76 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	1.287	3.239		
System Mismatch:	0.000	0.000		

Number of Iterations: 1